

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T15:49:36

Source: data/json/sonnet45_8agent_game_directive_r10_n5/claude-sonnet-4.5/game/sonnet45_8agent_game_directive_r10_n5_001_neutral_rep01.json

RUN METADATA

model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 1
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
 "game"
]

agent_names: {
 "Agent_1": "Aster",
 "Agent_2": "Briar",
 "Agent_3": "Cyra",
 "Agent_4": "Dorian",
 "Agent_5": "Elara",
 "Agent_6": "Finn",
 "Agent_7": "Galen",
 "Agent_8": "Hana"
}
dyadic_pairing_schedule: {
 "num_turns": 10,
 "role_targets": {
 "Agent_1": 5,
 "Agent_2": 5,
 "Agent_3": 5,
 "Agent_4": 5,
 "Agent_5": 5,
 "Agent_6": 5,
 "Agent_7": 5,
 "Agent_8": 5
 },
 "strategy": "balanced_random_roles_v1"
}

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Agent_3 (Cyra); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Agent_4 (Dorian); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Agent_5 (Elara); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Agent_6 (Finn); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Agent_7 (Galen); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Agent_8 (Hana); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_3 (Cyra)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_2 (Briar)=Receiver
 dyad_3: Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver
 dyad_4: Agent_8 (Hana)=Sender, Agent_7 (Galen)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_3 (Cyra) returned \$4; payoffs Agent_1 (Aster)=\$6, Agent_3 (Cyra)=\$5
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_2 (Briar)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_5 (Elara) sent $3; received $9; Agent_6 (Finn) returned $4.5; payoffs Agent_5
(Elara)=$6.5, Agent_6 (Finn)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 (Hana) sent $3; received $9; Agent_7 (Galen) returned $5; payoffs Agent_8
(Hana)=$7, Agent_7 (Galen)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 6.0,
  "Agent_2": 4.5,
  "Agent_3": 5.0,
  "Agent_4": 6.5,
  "Agent_5": 6.5,
  "Agent_6": 4.5,
  "Agent_7": 4.0,
  "Agent_8": 7.0
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T15:38:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship while being somewhat cautious since we haven't built trust yet.

If I send a moderate amount, it signals willingness to cooperate but isn't overly risky. Sending \$3 means:

- Cyra receives \$9 (3x multiplier)
- I keep \$2 initially
- If Cyra returns a fair amount, we both benefit

This seems like a reasonable opening move to test cooperation.

{'send': 3}

Reasoning:

This is the first round of an iterated game. I'm the sender (Aster) and my opponent is Cyra (Agent_3).

Let me think about the strategic considerations:

1. This is an iterated game, so reputation and building trust matter
2. I have \$5 to allocate
3. Whatever I send gets tripled
4. The receiver then decides how much to return

From a purely rational perspective:

- If I send \$5, receiver gets \$15
- If receiver is cooperative and splits evenly, they might return \$10, giving me \$10 total and them \$5
- If I send \$0, I keep \$5, receiver gets \$0

However, this is the first round and I need to establish a pattern. Common strategies:

1. Start cooperatively to build trust
2. Send a moderate amount to test the other player
3. Be cautious since this is round 1

A good starting strategy might be to send \$3:

- I keep \$2 initially
- Receiver gets \$9

- If they're fair, they might return \$4-6
- This shows willingness to cooperate while not being overly trusting

Or I could send \$2.5 (middle ground) or even \$5 (fully trusting).

Since this is iterated and I want to establish good reciprocity, I'll send \$3 as a reasonable opening move that signals cooperation but maintains some caution.

Actually, let me reconsider. In trust games, research shows that:

- Senders who send more tend to receive more back
- Starting with cooperation often leads to better long-term outcomes
- In iterated games, building trust early can be very beneficial

I could send \$4 to show strong cooperation while keeping \$1.

Or send \$3 to show moderate cooperation while keeping \$2.

I think \$3 is a good balance for the first round - it's cooperative but not naive.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 553,
  "reasoning_tokens": 439
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T15:38:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_3"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_3": "trustee"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0
- This is Round 1, so establishing a cooperative reputation is important

Aster currently has \$5.0 - \$3.0 = \$2.0 remaining.

If I return \$4.0:

- Aster ends with: \$2.0 + \$4.0 = \$6.0 (a \$1.0 profit)
- I keep: \$9.0 - \$4.0 = \$5.0

This rewards Aster's trust, ensures they profit from cooperating, and establishes me as a reciprocal partner for future rounds.

{'return': 4.0}

Reasoning:

Let me analyze this situation:

- I am the RECEIVER (Cyra)
- My opponent is Aster (Agent_1), who is the SENDER
- Aster sent \$3.0 out of their \$5.0 endowment (60%)
- I receive \$3.0 × 3 = \$9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Aster showed trust by sending 60% of their endowment (\$3.0)
2. If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
3. If I return fairly, I should consider:
 - Aster gave up \$3.0 to send to me
 - This created \$9.0 total for me
 - A fair split might be to ensure we both benefit

Common strategies in trust games:

- Return enough to show reciprocity and encourage future cooperation
- A typical approach is to return about 1/3 to 1/2 of what was received
- Or return enough so the sender gets more than they started with

Let me calculate some options:

- If I return \$3.0: Aster gets $\$2.0 + \$3.0 = \$5.0$ (breaks even), I keep \$6.0
- If I return \$4.0: Aster gets $\$2.0 + \$4.0 = \$6.0$ (gains \$1), I keep \$5.0
- If I return \$4.5: Aster gets $\$2.0 + \$4.5 = \$6.5$ (gains \$1.5), I keep \$4.5
- If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0

- If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0

Given Aster's substantial trust, returning around \$4.0-\$4.5 balances reciprocity with self-interest-it rewards their generosity while keeping a reasonable share for myself. I'll return \$4.0, which gives Aster a \$1 gain and leaves me with \$5.0. Since this appears to be an iterated game, establishing cooperative behavior now could benefit both of us in future rounds.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 929,
  "reasoning_tokens": 425
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T15:39:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_2"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_2": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll start with a moderately cooperative approach to signal trust while managing risk. Since this is an iterated game and roles may switch in future rounds, building a cooperative relationship could benefit both of us.

I'll send **\$3**.

This means Briar will receive \$9 (3 × \$3), and we'll see how they respond. If they reciprocate fairly, we can build a mutually beneficial pattern.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of an iterated trust game. I'm playing as Dorian (the sender) and my opponent is Briar.

Let me think about the strategic considerations:

1. As the sender, I have \$5 to potentially send
2. Whatever I send gets tripled
3. The receiver (Briar) then decides how much to return
4. This is an iterated game, so there will be multiple rounds
5. In future rounds, roles may switch, so building trust/cooperation could be beneficial

First round considerations:

- I don't know anything about Briar's behavior yet
- Sending nothing (\$0) guarantees I keep \$5, but doesn't build trust or cooperation
- Sending everything (\$5) shows maximum trust but is risky - Briar could return nothing, leaving me with \$0
- A moderate amount shows willingness to cooperate while limiting risk

Common strategies in trust games:

- Start with a moderate amount to signal cooperation

- Observe the response to calibrate future behavior
- In iterated games, cooperation tends to be beneficial if both parties reciprocate

I think sending \$2-3 is reasonable for the first round:

- Sends \$2.50: Briar receives \$7.50
- Sends \$3: Briar receives \$9

If I send \$3 and Briar returns \$4-5, we both benefit compared to no cooperation.

Let me go with \$3 as a moderately cooperative opening move. This shows willingness to trust while not being overly risky.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 459,
  "reasoning_tokens": 357
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T15:39:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_2"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_2": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Total pool is now \$11.0 (\$2.0 they kept + \$9.0 I received)

Since this is Round 1 and Dorian showed significant trust by sending 60% of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return about half of what I received, this would:

- Give Dorian a reasonable profit for their trust
- Show I'm trustworthy for future rounds
- Still leave me with a good amount

I'll return \$4.5, which gives Dorian \$6.5 total (a profit of \$1.5) and leaves me with \$4.5.

```
{"return": 4.5}
```

Reasoning:

I'm playing the role of receiver in this round. Let me understand the situation:

- The sender (Dorian/Agent_4) received \$5
- They sent me \$3.0 (which is 60% of their endowment)
- This amount is multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. The sender showed significant trust by sending 60% of their endowment
2. This is an iterated game with multiple rounds
3. Building reciprocity and trust could lead to better outcomes in future rounds
4. If I return a fair amount, the sender might be more generous in future rounds

Let me calculate what would be fair:

- The sender gave up \$3.0 and kept \$2.0
- I received \$9.0
- If I want to split the gains fairly, the total pool is now \$11.0 (sender's remaining \$2.0 + my \$9.0)
- A 50-50 split would give each of us \$5.5
- So I should return \$3.5 to make the sender have \$5.5 total (\$2.0 + \$3.5)

- I would keep \$5.5 (\$9.0 - \$3.5)

Alternatively, I could think about it as:

- The sender took a risk by sending \$3.0
- I received a tripled amount (\$9.0)
- The net gain from cooperation is \$4.0 (from \$5.0 to \$9.0)
- Splitting this gain: each should get \$2.0 extra
- Sender already lost \$3.0, so returning \$5.0 would give them \$7.0 total (\$2.0 + \$5.0), which is \$2.0 more than they started with
- I would keep \$4.0, which is \$4.0 more than \$0

- A middle ground of returning \$4.5 seems more practical for establishing cooperation without being overly generous. This nets the sender \$6.5 total (\$2.0 kept plus \$4.5 returned, a \$1.5 profit) while I retain \$4.5, which still represents meaningful gains from the exchange.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 1009,
  "reasoning_tokens": 433
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T15:39:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Finn while also being strategic.

Since the game is iterated and we may interact in future rounds, sending a moderate amount shows trust and creates potential for mutual benefit. If I send \$3, Finn will receive \$9 (after the 3x multiplier). This gives Finn a strong incentive to return a fair amount, as we could both end up better off than our starting positions.

A reasonable return from Finn (say \$4-5) would leave us both better off, and establish a pattern of cooperation for potential future interactions.

{'send': 3}

Reasoning:

This is the first round of an iterated game. I'm playing as the sender (Elara) and my opponent is Finn (Agent_6).

Let me think through the game theory here:

- I receive \$5
- I can send \$0-\$5 to Finn
- What I send gets multiplied by 3x
- Finn then decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: received - returned

This is a classic trust game. The economically rational strategy depends on:

1. Whether this is truly iterated (will I play with Finn again?)
2. What I believe about Finn's likely behavior
3. Whether I want to establish cooperation

Since this is Round 1 and the game is iterated with multiple rounds, there's value in:

- Establishing trust and cooperation
- Testing Finn's reciprocity
- Not being too exploitable but also not being too selfish

A common opening strategy in trust games is to send a moderate amount - not nothing (which shows no trust and prevents cooperation), but also not everything (which might be exploited).

Sending \$3 out of \$5 seems like a reasonable moderate approach:

- It shows I'm willing to trust and cooperate
- It gives Finn \$9 (\$3 × 3)
- If Finn returns even \$3-4, we're both better off than if I sent nothing
- It's not so much that I'm completely vulnerable

Let me consider the outcomes:

- If I send \$3, Finn gets \$9
- I have \$2 left, plus whatever Finn returns
- If Finn returns \$4-5, I end up with \$6-7 (better than my initial \$5)
- And Finn keeps \$4-5 (better than their initial \$0)
- This creates value for both of us

Alternatively:

- Send \$2.5: More conservative, Finn gets \$7.5
- Send \$4: More trusting, Finn gets \$12

- Send \$5: Maximum trust, Finn gets \$15

I think \$3 is a good balance for the first round. It's cooperative without being naive.

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 643,  
  "reasoning_tokens": 466  
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T15:40:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Elara sent \$3.0 (60% of her endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- This is the first round, so establishing trust and cooperation is important

I want to reciprocate Elara's trust and encourage continued cooperation in future rounds. Elara took a risk by sending a substantial portion of her endowment.

A fair return that:

1. Rewards her trust
2. Shares the gains from cooperation
3. Encourages future collaboration

I'll return approximately half of what I received, which seems like a reasonable reciprocal gesture.

{'return': 4.5}

Reasoning:

This is Round 1 of an iterated trust game. Let me think through this situation:

I'm the RECEIVER:

- The sender (Elara) sent me \$3.0 out of their \$5 endowment
- This gets multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Elara sent \$3.0 (60% of her endowment), which is a fairly generous amount
3. If I want to build trust and encourage cooperation in future rounds, I should return a fair

amount

4. If I return nothing, Elara will likely not trust me in future rounds
5. The typical fair split would consider that Elara kept \$2.0 and sent \$3.0

Let me think about what's fair:

- Elara ends up with: $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- I end up with: $\$9 - \text{returned}$

A 50-50 split of the total wealth would mean:

- Total created: $\$5$ (original) + $\$9$ (received) = $\$14$
- 50-50 would be $\$7$ each
- For Elara to have $\$7$: $\$2 + \text{returned} = \7 , so $\text{returned} = \$5$
- For me to have $\$7$: $\$9 - \text{returned} = \7 , so $\text{returned} = \$2$

Wait, that doesn't match up. Let me recalculate:

- Elara starts with $\$5$, sends $\$3$, so has $\$2$ left
- I receive $\$3 \times 3 = \9
- Total money in the system: $\$2$ (Elara has) + $\$9$ (I have) = $\$11$

If we split equally: each should get $\$5.50$

- For Elara to get $\$5.50$: she needs $\$5.50 - \$2 = \$3.50$ returned

- For me to get $\$5.50$: I keep $\$9 - \$3.50 = \$5.50$

So returning $\$3.50$ creates an equal split. I could also consider reciprocity-returning 60% of what I received ($\$5.40$) since Elara sent 60% of her endowment, or simply matching her $\$3$ contribution to restore her original stake.

If I return $\$3$, I'd end up with $\$6$ while she'd have $\$5$. But thinking about long-term incentives, a more generous return between $\$4$ - $\$5$ would signal trustworthiness and encourage future rounds. I'll return $\$4.50$ -this gives Elara $\$6.50$ and me $\$4.50$, slightly favoring her since she took the initial risk, which should build cooperation going forward.

```
Usage: {  
  "input_tokens": 562,  
  "output_tokens": 1035,  
  "reasoning_tokens": 518  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-06-04T15:40:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_7": "Galen",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to decide how much trust to place in Galen.

In this trust game, sending money creates value (it triples), but requires trusting that the receiver will return a fair amount. Since this is:

- Round 1 (no history with Galen)
- An iterated game (we may interact again, building reputation)
- A multi-agent setting (reputation may matter across the group)

I'll send a moderate amount to test the waters and signal cooperative intent, while not risking everything.

If I send \$3:

- Galen receives \$9 (tripled)
- A fair split of the created value could benefit us both
- I maintain some insurance if trust isn't reciprocated

```
'''json
{"send": 3}
'''
```

Reasoning:

This is Round 1 and I'm the sender. I need to decide how much to send to Galen (Agent_7).

Let me think about the game theory here:

- I receive \$5
- If I send X dollars, Galen receives 3X dollars
- Galen then decides how much to return
- I keep: $\$5 - X + (\text{amount returned})$
- Galen keeps: $3X - (\text{amount returned})$

This is a classic trust game. The optimal social outcome is for me to send the full \$5 (which becomes \$15), and then for Galen to return some fair amount. For example, if I send \$5 and Galen returns \$7.50, we'd each end up with \$7.50 total.

However, there's a risk. If Galen returns nothing, I'd end up with \$0 and Galen would keep \$15.

Since this is an iterated game with multiple rounds and we might interact again (though pairings are randomized), there's potential for reputation and reciprocity to matter.

For Round 1, I don't have any information about Galen's behavior. Common strategies in trust games include:

1. Send nothing (risk averse, but no gain)
2. Send a moderate amount (test the waters)
3. Send everything (maximize joint payoff, but risky)

In iterated games, starting with some cooperation can establish trust. Since we might play again and there are reputation effects across the group, I should probably show some trust.

I'll send \$3 as a moderate amount. This shows trust and goodwill, but isn't so risky that I'd lose everything if Galen returns nothing. If Galen receives \$9 ($\3×3), a fair return might be around \$4-5, which would leave us both better off than if I sent nothing.

Let me go with \$3.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 618,
  "reasoning_tokens": 405
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-06-04T15:40:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (\$0-\$9.0)

Assistant output:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing a good level of trust
- I received \$9.0 after the 3x multiplier
- This creates a \$6.0 surplus (\$9 - \$3 original)

I want to be fair and build trust for potential future interactions, since this is a multi-round game. Returning a reasonable amount ensures both of us benefit from the cooperation.

If I return \$5.0:

- Hana ends with: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This rewards Hana's trust while creating mutual benefit.

{"return": 5.0}

Reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in this round
2. The sender (Hana) sent \$3.0 out of their \$5 endowment (60%)
3. This \$3.0 was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about what would be fair and strategic:

- Hana sent 60% of their endowment, which shows a decent level of trust
- If I return nothing, I keep \$9.0 and Hana ends up with only \$2.0 (\$5 - \$3)
- If I return everything (\$9.0), Hana ends up with \$11.0 (\$5 - \$3 + \$9) and I get nothing
- A "fair" return might be to ensure both players benefit

Let me consider some options:

Option 1: Return \$4.5 (50% of what I received)

- I keep: \$9.0 - \$4.5 = \$4.5
- Hana ends with: \$5 - \$3 + \$4.5 = \$6.5
- Net gains: Me +\$4.5, Hana +\$1.5

Option 2: Return \$5.0

- I keep: \$9.0 - \$5.0 = \$4.0
- Hana ends with: \$5 - \$3 + \$5.0 = \$7.0
- Net gains: Me +\$4.0, Hana +\$2.0

Option 3: Return \$6.0

- I keep: \$9.0 - \$6.0 = \$3.0
- Hana ends with: \$5 - \$3 + \$6.0 = \$8.0
- Net gains: Me +\$3.0, Hana +\$3.0 (equal split of the \$6 surplus created)

- Me: Started with \$0, received \$9, returned \$6 = \$3 (gain of \$3)

Since this is a multi-round game, I should think about building trust and encouraging future cooperation. Hana's 60% send shows moderate confidence, so returning enough to give us equal gains from the surplus seems reasonable. Returning \$6 splits the \$6 surplus evenly-each of us gains \$3-which feels balanced and rewards her initial trust without leaving me worse off than I started.

That said, I could return slightly less and still maintain goodwill. Returning \$5 gives Hana a \$2 gain while I keep \$4, which is still fair given she didn't send everything. Since she sent 60%, returning about 55% of what I received feels like a solid middle ground that encourages cooperation without being overly generous. I'll return \$5.

```
Usage: {  
  "input_tokens": 564,  
  "output_tokens": 983,  
  "reasoning_tokens": 495  
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

- dyad_1: Agent_6 (Finn)=Sender, Agent_8 (Hana)=Receiver
- dyad_2: Agent_7 (Galen)=Sender, Agent_1 (Aster)=Receiver
- dyad_3: Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver
- dyad_4: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

- dyad_1: Agent_6 (Finn) sent \$3; received \$9; Agent_8 (Hana) returned \$5; payoffs Agent_6 (Finn)=\$7, Agent_8 (Hana)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_2: Agent_7 (Galen) sent \$3; received \$9; Agent_1 (Aster) returned \$4; payoffs Agent_7 (Galen)=\$6, Agent_1 (Aster)=\$5
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

    4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_3 (Cyra) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_3
(Cyra)=$6.5, Agent_4 (Dorian)=$4.5
    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_2 (Briar) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_2
(Briar)=$6.5, Agent_5 (Elara)=$4.5
    Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 11.0,
  "Agent_2": 11.0,
  "Agent_3": 11.5,
  "Agent_4": 11.0,
  "Agent_5": 11.0,
  "Agent_6": 11.5,
  "Agent_7": 10.0,
  "Agent_8": 11.0
}

```

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-06-04T15:40:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_6",
      "Agent_8"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of her endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- This is the first round, so establishing trust and cooperation is important

I want to reciprocate Elara's trust and encourage continued cooperation in future rounds. Elara took a risk by sending a substantial portion of her endowment.

A fair return that:

1. Rewards her trust
2. Shares the gains from cooperation
3. Encourages future collaboration

I'll return approximately half of what I received, which seems like a reasonable reciprocal gesture.

{'return': 4.5}

reasoning:

This is Round 1 of an iterated trust game. Let me think through this situation:

I'm the RECEIVER:

- The sender (Elara) sent me \$3.0 out of their \$5 endowment
- This gets multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Elara sent \$3.0 (60% of her endowment), which is a fairly generous amount
3. If I want to build trust and encourage cooperation in future rounds, I should return a fair amount
4. If I return nothing, Elara will likely not trust me in future rounds
5. The typical fair split would consider that Elara kept \$2.0 and sent \$3.0

Let me think about what's fair:

- Elara ends up with: $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- I end up with: $\$9 - \text{returned}$

A 50-50 split of the total wealth would mean:

- Total created: $\$5$ (original) + $\$9$ (received) = $\$14$
- 50-50 would be $\$7$ each
- For Elara to have $\$7$: $\$2 + \text{returned} = \7 , so $\text{returned} = \$5$
- For me to have $\$7$: $\$9 - \text{returned} = \7 , so $\text{returned} = \$2$

Wait, that doesn't match up. Let me recalculate:

- Elara starts with $\$5$, sends $\$3$, so has $\$2$ left
- I receive $\$3 \times 3 = \9
- Total money in the system: $\$2$ (Elara has) + $\$9$ (I have) = $\$11$

If we split equally: each should get $\$5.50$

- For Elara to get $\$5.50$: she needs $\$5.50 - \$2 = \$3.50$ returned

- For me to get $\$5.50$: I keep $\$9 - \$3.50 = \$5.50$

So returning $\$3.50$ creates an equal split. I could also consider reciprocity-returning 60% of what I received ($\$5.40$) since Elara sent 60% of her endowment, or simply matching her $\$3$ contribution to restore her original stake.

If I return $\$3$, I'd end up with $\$6$ while she'd have $\$5$. But thinking about long-term incentives, a more generous return between $\$4$ - $\$5$ would signal trustworthiness and encourage future rounds. I'll return $\$4.50$ -this gives Elara $\$6.50$ and me $\$4.50$, slightly favoring her since she took the initial risk, which should build cooperation going forward.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1035,
  "reasoning_tokens": 518
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Elara. They sent $\$3.0$ to you, it became $\$9.0$, and you returned $\$4.5$. Your payoff was $\$4.5$.
Your total earnings across all rounds are $\$4.5$.

This round, you are the SENDER against Hana. You have $\$5$. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is a new opponent (Hana), Round 2 overall
- In Round 1, Elara showed trust by sending 60% ($\$3$), and I reciprocated by returning 50% of what I received
- I want to maintain a cooperative strategy and build trust with Hana

I'll send a similar proportion to what Elara sent me, demonstrating consistent cooperative behavior.

```
{'send': 3}
```

Reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Hana (Agent_8).

In Round 1, I was the RECEIVER against Elara:

- Elara sent me $\$3.0$ (60% of her endowment)
- I received $\$9.0$
- I returned $\$4.5$ (50% of what I received)
- My payoff was $\$4.5$
- Elara's payoff was: $\$5 - \$3 + \$4.5 = \6.5

Now I'm playing with a new opponent (Hana) and I'm the sender.

Strategy considerations:

1. This is only Round 2, so building cooperative relationships is important
2. In the previous round as receiver, someone showed trust by sending 60%, and I reciprocated by returning 50%
3. As a sender, I should also show trust to encourage cooperation
4. Sending a substantial amount signals willingness to cooperate

Since Elara sent \$3.0 (60%) and I reciprocated fairly well, I should maintain a similar cooperative strategy. Sending around \$3 seems reasonable to establish trust while not being overly risky in early rounds.

Let me send \$3 to match what Elara sent to me, showing consistent cooperative behavior.

```
Usage: {  
  "input_tokens": 861,  
  "output_tokens": 407,  
  "reasoning_tokens": 267  
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-06-04T15:41:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_6": "Finn",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_8"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_6": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to decide how much trust to place in Galen.

In this trust game, sending money creates value (it triples), but requires trusting that the receiver will return a fair amount. Since this is:

- Round 1 (no history with Galen)
- An iterated game (we may interact again, building reputation)
- A multi-agent setting (reputation may matter across the group)

I'll send a moderate amount to test the waters and signal cooperative intent, while not risking everything.

If I send \$3:

- Galen receives \$9 (tripled)
- A fair split of the created value could benefit us both
- I maintain some insurance if trust isn't reciprocated

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the sender. I need to decide how much to send to Galen (Agent_7).

Let me think about the game theory here:

- I receive \$5
- If I send X dollars, Galen receives 3X dollars
- Galen then decides how much to return
- I keep: \$5 - X + (amount returned)
- Galen keeps: 3X - (amount returned)

This is a classic trust game. The optimal social outcome is for me to send the full \$5 (which becomes \$15), and then for Galen to return some fair amount. For example, if I send \$5 and Galen returns \$7.50, we'd each end up with \$7.50 total.

However, there's a risk. If Galen returns nothing, I'd end up with \$0 and Galen would keep \$15.

Since this is an iterated game with multiple rounds and we might interact again (though pairings are randomized), there's potential for reputation and reciprocity to matter.

For Round 1, I don't have any information about Galen's behavior. Common strategies in trust games include:

1. Send nothing (risk averse, but no gain)
2. Send a moderate amount (test the waters)
3. Send everything (maximize joint payoff, but risky)

In iterated games, starting with some cooperation can establish trust. Since we might play again and there are reputation effects across the group, I should probably show some trust.

I'll send \$3 as a moderate amount. This shows trust and goodwill, but isn't so risky that I'd lose everything if Galen returns nothing. If Galen receives \$9 ($\3×3), a fair return might be around \$4-5, which would leave us both better off than if I sent nothing.

Let me go with \$3.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 618,
  "reasoning_tokens": 405
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****What happened last round:****

- I sent \$3 to Galen
- Galen returned \$5 to me (a fair amount)

****Current situation:****

- Finn sent me \$3 (exactly what I sent to Galen)
- I received \$9
- I should decide what's fair to return

Since Finn showed the same level of trust in me that I showed in Galen, and Galen returned \$5 which felt fair, I'll mirror that cooperative behavior. Returning \$5 gives Finn a total payoff of \$7 (same as mine from round 1), while I keep \$4 (same as Galen kept).

This establishes a fair, reciprocal pattern.

```
'''json
{"return": 5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Galen
- Galen received \$9
- Galen returned \$5 to me
- My payoff: $\$5 - \$3 + \$5 = \7
- Galen's payoff: $\$9 - \$5 = \$4$

So Galen was moderately cooperative, returning more than I sent originally but keeping more than

an half of what they received.

Now Round 2:

- Finn sent me \$3 (same as I sent to Galen!)
- I received \$9
- I need to decide how much to return

Considerations:

1. Reciprocity: Finn showed the same level of trust I showed to Galen (\$3)
2. Fairness: I could mirror what Galen did to me (return \$5), which seems like a reasonable cooperative standard
3. Building reputation: In a multi-agent setting, being seen as trustworthy and fair matters
4. Future rounds: This is an iterated game, and reputation effects could help me

If I return \$5:

- Finn gets: $\$5 - \$3 + \$5 = \7 (same as what I got in round 1)
- I keep: $\$9 - \$5 = \$4$ (same as what Galen kept)

This creates symmetry and seems fair. It's a reasonable cooperative strategy that mirrors what was done to me.

Alternatively, I could be more generous to establish a reputation for generosity, or less generous to maximize my immediate payoff. But \$5 seems like a balanced, fair amount that:

- Rewards Finn's trust
- Is similar to what Galen did (establishing a norm)
- Keeps the game mutually beneficial
- Maintains my reputation as fair

I'll return \$5.

```
Usage: {  
  "input_tokens": 865,  
  "output_tokens": 614,  
  "reasoning_tokens": 371  
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T15:41:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_1"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing a good level of trust
- I received \$9.0 after the 3x multiplier
- This creates a \$6.0 surplus (\$9 - \$3 original)

I want to be fair and build trust for potential future interactions, since this is a multi-round game. Returning a reasonable amount ensures both of us benefit from the cooperation.

If I return \$5.0:

- Hana ends with: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This rewards Hana's trust while creating mutual benefit.

{"return": 5.0}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in this round
2. The sender (Hana) sent \$3.0 out of their \$5 endowment (60%)
3. This \$3.0 was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about what would be fair and strategic:

- Hana sent 60% of their endowment, which shows a decent level of trust
- If I return nothing, I keep \$9.0 and Hana ends up with only \$2.0 ($\$5 - \3)
- If I return everything (\$9.0), Hana ends up with \$11.0 ($\$5 - \$3 + \9) and I get nothing
- A "fair" return might be to ensure both players benefit

Let me consider some options:

Option 1: Return \$4.5 (50% of what I received)

- I keep: $\$9.0 - \$4.5 = \$4.5$
- Hana ends with: $\$5 - \$3 + \$4.5 = \6.5
- Net gains: Me +\$4.5, Hana +\$1.5

Option 2: Return \$5.0

- I keep: $\$9.0 - \$5.0 = \$4.0$
- Hana ends with: $\$5 - \$3 + \$5.0 = \7.0
- Net gains: Me +\$4.0, Hana +\$2.0

Option 3: Return \$6.0

- I keep: $\$9.0 - \$6.0 = \$3.0$
- Hana ends with: $\$5 - \$3 + \$6.0 = \8.0
- Net gains: Me +\$3.0, Hana +\$3.0 (equal split of the \$6 surplus created)

- Me: Started with \$0, received \$9, returned \$6 = \$3 (gain of \$3)

Since this is a multi-round game, I should think about building trust and encouraging future cooperation. Hana's 60% send shows moderate confidence, so returning enough to give us equal gains from the surplus seems reasonable. Returning \$6 splits the \$6 surplus evenly—each of us gains \$3—which feels balanced and rewards her initial trust without leaving me worse off than I started.

That said, I could return slightly less and still maintain goodwill. Returning \$5 gives Hana a \$2 gain while I keep \$4, which is still fair given she didn't send everything. Since she sent 60%, returning about 55% of what I received feels like a solid middle ground that encourages cooperation without being overly generous. I'll return \$5.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 983,
  "reasoning_tokens": 495
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is a new opponent (Aster), so I don't have history with them
- In the previous round, Hana sent me 60% (\$3), showing trust
- I returned a fair amount (\$5), creating mutual benefit
- This is a multi-round game where building cooperation matters

I'll send \$3 (60% of my endowment), mirroring the cooperative approach Hana showed me. This de

monstrates willingness to trust and create mutual gains.

```
{'send': 3}
```

Reasoning:

Now I'm the sender against a new opponent, Aster.

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana showed trust by sending 60% (\$3). I returned \$5 out of \$9, which gave them a good return.
2. This is a new pairing with Aster. I don't know how they'll behave.
3. Since this is a multi-round game and I'll potentially interact with different agents multiple times, I should think about reputation and reciprocity.
4. If I send money, it gets tripled. If I send \$3:
 - Aster receives \$9
 - If they return \$5 (similar to what I did), I'd end with $\$2 + \$5 = \$7$
 - If they return less, I could end up worse off
5. Sending nothing is safe but doesn't build trust or create mutual gains.
6. I should probably mirror the cooperative behavior I received. Hana sent 60% (\$3), which seemed like a reasonable level of trust.

Given that:

- This is an iterated game with multiple rounds
- Building cooperation can lead to better long-term outcomes
- I should signal willingness to cooperate but also be somewhat cautious with a new partner

I'll send \$3 (60%), matching what Hana sent to me. This shows I'm willing to cooperate and trust, while not being overly risky.

```
Usage: {  
  "input_tokens": 896,  
  "output_tokens": 457,  
  "reasoning_tokens": 321  
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T15:41:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_1"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship while being so meewhat cautious since we haven't built trust yet.

If I send a moderate amount, it signals willingness to cooperate but isn't overly risky. Sending \$3 means:

- Cyra receives \$9 (3x multiplier)
- I keep \$2 initially
- If Cyra returns a fair amount, we both benefit

This seems like a reasonable opening move to test cooperation.

{'send': 3}

reasoning:

This is the first round of an iterated game. I'm the sender (Aster) and my opponent is Cyra (Agent_3).

Let me think about the strategic considerations:

1. This is an iterated game, so reputation and building trust matter
2. I have \$5 to allocate
3. Whatever I send gets tripled
4. The receiver then decides how much to return

From a purely rational perspective:

- If I send \$5, receiver gets \$15
- If receiver is cooperative and splits evenly, they might return \$10, giving me \$10 total and them \$5
- If I send \$0, I keep \$5, receiver gets \$0

However, this is the first round and I need to establish a pattern. Common strategies:

1. Start cooperatively to build trust
2. Send a moderate amount to test the other player
3. Be cautious since this is round 1

A good starting strategy might be to send \$3:

- I keep \$2 initially
- Receiver gets \$9
- If they're fair, they might return \$4-6
- This shows willingness to cooperate while not being overly trusting

Or I could send \$2.5 (middle ground) or even \$5 (fully trusting).

Since this is iterated and I want to establish good reciprocity, I'll send \$3 as a reasonable opening move that signals cooperation but maintains some caution.

Actually, let me reconsider. In trust games, research shows that:

- Senders who send more tend to receive more back
- Starting with cooperation often leads to better long-term outcomes
- In iterated games, building trust early can be very beneficial

I could send \$4 to show strong cooperation while keeping \$1.

Or send \$3 to show moderate cooperation while keeping \$2.

I think \$3 is a good balance for the first round - it's cooperative but not naive.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 553,
  "reasoning_tokens": 439
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In Round 1, when I sent \$3 to Cyra, they returned \$4, which felt fair

I'll reciprocate with the same level of cooperation. Returning \$4 means:

- Galen gets \$6 total (\$2 kept + \$4 returned)
- I keep \$5 (\$9 - \$4)
- Both of us profit from cooperation

```
{'return': 4}
```

Reasoning:

Let me analyze this situation:

Round 2 - I'm the RECEIVER

- Galen (sender) sent me \$3.0
- I received \$9.0 (due to 3x multiplier)
- I need to decide how much to return (0-9.0)

From Round 1:

- I was the sender and sent \$3.0 to Cyra
- Cyra received \$9.0
- Cyra returned \$4.0 to me
- My payoff was \$6.0 (I kept \$2 + received \$4 back = \$6)
- Cyra's payoff was \$9 - \$4 = \$5

So Cyra returned \$4 out of \$9 received, which is about 44%. This seems like a reasonable cooperative response.

Now I'm in Cyra's position with Galen. Galen sent me \$3, just like I sent to Cyra.

If I want to establish a pattern of reciprocity and fairness:

- Galen kept \$2, sent \$3
- I received \$9
- If I return \$4 (like Cyra did), then:
 - Galen gets: \$2 + \$4 = \$6
 - I keep: \$9 - \$4 = \$5

This mirrors what happened in Round 1 and seems fair. It shows I'm willing to reciprocate cooperation.

Alternatively, I could consider:

- Returning more (like \$5) to be extra generous: Galen gets \$7, I keep \$4
- Returning less (like \$3) to keep more: Galen gets \$5, I keep \$6

But \$4 seems like the most reciprocal and cooperative response given what Cyra did to me.

```
Usage: {  
  "input_tokens": 802,  
  "output_tokens": 554,  
  "reasoning_tokens": 307  
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T15:41:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0
- This is Round 1, so establishing a cooperative reputation is important

Aster currently has \$5.0 - \$3.0 = \$2.0 remaining.

If I return \$4.0:

- Aster ends with: \$2.0 + \$4.0 = \$6.0 (a \$1.0 profit)
- I keep: \$9.0 - \$4.0 = \$5.0

This rewards Aster's trust, ensures they profit from cooperating, and establishes me as a reciprocal partner for future rounds.

{'return': 4.0}

reasoning:

Let me analyze this situation:

- I am the RECEIVER (Cyra)
- My opponent is Aster (Agent_1), who is the SENDER
- Aster sent \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Aster showed trust by sending 60% of their endowment (\$3.0)
2. If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
3. If I return fairly, I should consider:
 - Aster gave up \$3.0 to send to me
 - This created \$9.0 total for me
 - A fair split might be to ensure we both benefit

Common strategies in trust games:

- Return enough to show reciprocity and encourage future cooperation
- A typical approach is to return about 1/3 to 1/2 of what was received
- Or return enough so the sender gets more than they started with

Let me calculate some options:

- If I return \$3.0: Aster gets $\$2.0 + \$3.0 = \$5.0$ (breaks even), I keep \$6.0
 - If I return \$4.0: Aster gets $\$2.0 + \$4.0 = \$6.0$ (gains \$1), I keep \$5.0
 - If I return \$4.5: Aster gets $\$2.0 + \$4.5 = \$6.5$ (gains \$1.5), I keep \$4.5
 - If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0
- If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0

Given Aster's substantial trust, returning around \$4.0-\$4.5 balances reciprocity with self-interest-it rewards their generosity while keeping a reasonable share for myself. I'll return \$4.0, which gives Aster a \$1 gain and leaves me with \$5.0. Since this appears to be an iterated game, establishing cooperative behavior now could benefit both of us in future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 929,
  "reasoning_tokens": 425
}
```

Message 4 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

- I'm now the sender with \$5
- In Round 1, Aster sent me \$3.0 (60%) showing trust, and I returned \$4.0, establishing cooperation
- This is my first interaction with Dorian

I'll follow the cooperative pattern established in Round 1. Sending \$3.0 (60% of my endowment) demonstrates trust and gives Dorian the opportunity to reciprocate, which should benefit us both.

```
{'send': 3.0}
```

Reasoning:

Now I'm the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$3.0 (60% of their endowment). I returned \$4.0, which gave them a profit and established me as cooperative.
2. Now I'm facing a new opponent, Dorian. I have no history with them.
3. As the sender, I need to decide how much to trust Dorian. The key question is: will they reciprocate?
4. Sending has high potential upside (3x multiplier) but requires trust that they'll return enough.
5. Since this is early in the game (Round 2) and I want to establish cooperative patterns, I should send a reasonable amount.
6. Sending \$3.0 (like Aster did) seems like a fair, cooperative choice that:
 - Mirrors what Aster sent me
 - Shows willingness to trust
 - Gives Dorian the opportunity to reciprocate
 - Would give them \$9.0 to work with
7. If Dorian returns around \$4-5 (similar to what I did), we'll both profit.

Let me send \$3.0 to match the cooperative precedent from the first round.

```
Usage: {  
  "input_tokens": 888,  
  "output_tokens": 429,  
  "reasoning_tokens": 277  
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T15:41:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately cooperative approach to signal trust while managing risk. Since this is an iterated game and roles may switch in future rounds, building a cooperative relationship could benefit both of us.

I'll send **\$3**.

This means Briar will receive \$9 (3 × \$3), and we'll see how they respond. If they reciprocate fairly, we can build a mutually beneficial pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as Dorian (the sender) and my opponent is Briar.

Let me think about the strategic considerations:

1. As the sender, I have \$5 to potentially send
2. Whatever I send gets tripled
3. The receiver (Briar) then decides how much to return
4. This is an iterated game, so there will be multiple rounds
5. In future rounds, roles may switch, so building trust/cooperation could be beneficial

First round considerations:

- I don't know anything about Briar's behavior yet
- Sending nothing (\$0) guarantees I keep \$5, but doesn't build trust or cooperation
- Sending everything (\$5) shows maximum trust but is risky - Briar could return nothing, leaving me with \$0
- A moderate amount shows willingness to cooperate while limiting risk

Common strategies in trust games:

- Start with a moderate amount to signal cooperation
- Observe the response to calibrate future behavior
- In iterated games, cooperation tends to be beneficial if both parties reciprocate

I think sending \$2-3 is reasonable for the first round:

- Sends \$2.50: Briar receives \$7.50
- Sends \$3: Briar receives \$9

If I send \$3 and Briar returns \$4-5, we both benefit compared to no cooperation.

Let me go with \$3 as a moderately cooperative opening move. This shows willingness to trust while not being overly risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 459,
  "reasoning_tokens": 357
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the previous round, Briar returned \$4.5 out of the \$9 they received (50%), which created a fair outcome. I'll reciprocate this cooperative behavior.

If I return \$4.5:

- Cyra gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

This matches the fair standard from the previous round and signals I'm trustworthy for future interactions.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Briar
- I sent \$3, which became \$9 for them
- Briar returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Briar's payoff: \$9 - \$4.5 = \$4.5

So Briar returned exactly half of what they received, which is a fair 50-50 split of the gains.

Round 2:

- I'm now the receiver against Cyra
- Cyra sent me \$3 (same amount I sent to Briar)

- I received \$9 ($3 \times \3)
- I need to decide how much to return

Since this is an iterated game and I might play with Cyra again, I should:

1. Reciprocate the cooperative behavior shown by Briar
2. Signal that I'm a fair player
3. Consider that returning a fair amount might encourage future cooperation

Briar returned \$4.5 out of \$9 (50%). This seems like a reasonable standard.

If I return \$4.5:

- Cyra's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This creates a symmetric outcome to what happened with Briar, which seems fair.

I'll return \$4.5 to match the cooperative standard set in the previous round.

```
Usage: {
  "input_tokens": 793,
  "output_tokens": 502,
  "reasoning_tokens": 282
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T15:41:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Total pool is now \$11.0 (\$2.0 they kept + \$9.0 I received)

Since this is Round 1 and Dorian showed significant trust by sending 60% of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return about half of what I received, this would:

- Give Dorian a reasonable profit for their trust
- Show I'm trustworthy for future rounds
- Still leave me with a good amount

I'll return \$4.5, which gives Dorian \$6.5 total (a profit of \$1.5) and leaves me with \$4.5.

{"return": 4.5}

reasoning:

I'm playing the role of receiver in this round. Let me understand the situation:

- The sender (Dorian/Agent_4) received \$5
- They sent me \$3.0 (which is 60% of their endowment)
- This amount is multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. The sender showed significant trust by sending 60% of their endowment
2. This is an iterated game with multiple rounds
3. Building reciprocity and trust could lead to better outcomes in future rounds
4. If I return a fair amount, the sender might be more generous in future rounds

Let me calculate what would be fair:

- The sender gave up \$3.0 and kept \$2.0
- I received \$9.0
- If I want to split the gains fairly, the total pool is now \$11.0 (sender's remaining \$2.0 + my \$9.0)
- A 50-50 split would give each of us \$5.5
- So I should return \$3.5 to make the sender have \$5.5 total (\$2.0 + \$3.5)
- I would keep \$5.5 (\$9.0 - \$3.5)

Alternatively, I could think about it as:

- The sender took a risk by sending \$3.0
- I received a tripled amount (\$9.0)
- The net gain from cooperation is \$4.0 (from \$5.0 to \$9.0)
- Splitting this gain: each should get \$2.0 extra
- Sender already lost \$3.0, so returning \$5.0 would give them \$7.0 total (\$2.0 + \$5.0), which is \$2.0 more than they started with
- I would keep \$4.0, which is \$4.0 more than \$0

- A middle ground of returning \$4.5 seems more practical for establishing cooperation without being overly generous. This nets the sender \$6.5 total (\$2.0 kept plus \$4.5 returned, a \$1.5 profit) while I retain \$4.5, which still represents meaningful gains from the exchange.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1009,
  "reasoning_tokens": 433
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Elara
- In Round 1, Dorian sent me \$3 (60%) and I reciprocated fairly by returning \$4.5
- I want to establish a cooperative dynamic early in the game

I'll send \$3 (60% of my endowment), matching the trust level that Dorian showed me. This demonstrates my willingness to cooperate and gives us both the opportunity for mutual benefit if Elara reciprocates.

```
{"send": 3}
```

Reasoning:

Now I'm the sender playing against Elara (a new opponent).

Let me think about this strategically:

1. In Round 1, I was the receiver and Dorian sent me \$3.0 (60%). I returned \$4.5, establishing a cooperative pattern.
2. Now I'm the sender with Elara. Since this is our first interaction, I need to decide how much trust to show.
3. Key considerations:
 - Sending more can lead to higher total gains if Elara reciprocates
 - But if Elara doesn't return much, I could lose out
 - This is Round 2, early in the game, so establishing cooperation is valuable

- Dorian sent 60% to me, showing trust

4. Strategic options:

- Send \$0: Safe but generates no mutual benefit
- Send \$1-2: Conservative, low risk
- Send \$3: Moderate trust (60%, matching what Dorian did)
- Send \$4-5: High trust

Given that:

- Dorian showed 60% trust with me and I reciprocated fairly
- This is early in the game where establishing cooperation matters
- The potential gains from cooperation are substantial

I'll send \$3 (60%), matching the level of trust Dorian showed to me. This demonstrates I'm willing to cooperate and gives Elara a chance to earn good returns while also allowing me to benefit if they reciprocate.

If Elara returns about half (like I did), I'd get back around \$4.5, netting me \$2.5 total (\$5 - \$3 + \$4.5 = \$6.5 payoff). Even if they return less, this establishes a cooperative stance.

```
Usage: {
  "input_tokens": 914,
  "output_tokens": 523,
  "reasoning_tokens": 371
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T15:42:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn while also being strategic.

Since the game is iterated and we may interact in future rounds, sending a moderate amount shows trust and creates potential for mutual benefit. If I send \$3, Finn will receive \$9 (after the 3x multiplier). This gives Finn a strong incentive to return a fair amount, as we could both end up better off than our starting positions.

A reasonable return from Finn (say \$4-5) would leave us both better off, and establish a pattern of cooperation for potential future interactions.

{'send': 3}

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Elara) and my opponent is Finn (Agent_6).

Let me think through the game theory here:

- I receive \$5
- I can send \$0-\$5 to Finn
- What I send gets multiplied by 3x
- Finn then decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: received - returned

This is a classic trust game. The economically rational strategy depends on:

1. Whether this is truly iterated (will I play with Finn again?)
2. What I believe about Finn's likely behavior
3. Whether I want to establish cooperation

Since this is Round 1 and the game is iterated with multiple rounds, there's value in:

- Establishing trust and cooperation
- Testing Finn's reciprocity

- Not being too exploitable but also not being too selfish

A common opening strategy in trust games is to send a moderate amount - not nothing (which shows no trust and prevents cooperation), but also not everything (which might be exploited).

Sending \$3 out of \$5 seems like a reasonable moderate approach:

- It shows I'm willing to trust and cooperate
- It gives Finn \$9 ($\3×3)
- If Finn returns even \$3-4, we're both better off than if I sent nothing
- It's not so much that I'm completely vulnerable

Let me consider the outcomes:

- If I send \$3, Finn gets \$9
- I have \$2 left, plus whatever Finn returns
- If Finn returns \$4-5, I end up with \$6-7 (better than my initial \$5)
- And Finn keeps \$4-5 (better than their initial \$0)
- This creates value for both of us

Alternatively:

- Send \$2.5: More conservative, Finn gets \$7.5
- Send \$4: More trusting, Finn gets \$12

- Send \$5: Maximum trust, Finn gets \$15

I think \$3 is a good balance for the first round. It's cooperative without being naive.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 643,
  "reasoning_tokens": 466
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation, Briar sent me the same amount (\$3) that I sent to Finn in Round 1. Finn returned \$4.5 to me, which was exactly half of what they received - a fair and cooperative response.

To maintain a reciprocal and cooperative pattern, I should treat Briar similarly. Returning \$4.5 means:

- Briar gets a payoff of \$6.5 (\$2 kept + \$4.5 returned)
- I keep \$4.5 (\$9 received - \$4.5 returned)
- Both of us benefit from the cooperation

This establishes trust and encourages continued cooperation in future rounds.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Finn, who received \$9 and returned \$4.5. This was a fair and cooperative response - they returned exactly half of what they received, which left both of us better off.
2. Now I'm the receiver and Briar sent me \$3 (same amount I sent to Finn). I received \$9.

3. I should consider:
 - Reciprocity: Finn returned \$4.5 when receiving \$9, which was cooperative
 - Reputation: If I want to build trust for future rounds, I should be fair
 - Strategic thinking: Being too greedy might discourage cooperation in future rounds
 - Briar took the same risk I took with Finn
4. Following the cooperative pattern that Finn established, returning around \$4.5 would be fair and consistent. This would give Briar a payoff of $\$2 + \$4.5 = \$6.5$ (same as what I got in round 1).
5. I could return slightly more or less, but \$4.5 seems like a reasonable, fair amount that:
 - Rewards Briar's trust
 - Establishes me as a cooperative player
 - Keeps the game balanced
 - Mirrors what was done to me

I'll return \$4.5.

```
Usage: {
  "input_tokens": 826,
  "output_tokens": 481,
  "reasoning_tokens": 284
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_5 (Elara)=Receiver
 dyad_2: Agent_8 (Hana)=Sender, Agent_2 (Briar)=Receiver
 dyad_3: Agent_6 (Finn)=Sender, Agent_4 (Dorian)=Receiver
 dyad_4: Agent_3 (Cyra)=Sender, Agent_7 (Galen)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_5 (Elara)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_8 (Hana) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_2 (Briar)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_6 (Finn) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_4 (Dorian)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_3 (Cyra) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_3 (Cyra)=\$7, Agent_7 (Galen)=\$4
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 17.5,
 "Agent_2": 15.5,
 "Agent_3": 18.5,
 "Agent_4": 15.5,
 "Agent_5": 15.5,
 "Agent_6": 18.0,
 "Agent_7": 14.0,
 "Agent_8": 17.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T15:42:14
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "dyad_id": "dyad_1",


```

"move_index": 0,
"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship while being so
mewhat cautious since we haven't built trust yet.

If I send a moderate amount, it signals willingness to cooperate but isn't overly risky. S
ending \$3 means:

- Cyra receives \$9 (3x multiplier)
- I keep \$2 initially
- If Cyra returns a fair amount, we both benefit

This seems like a reasonable opening move to test cooperation.

{'send': 3}

reasoning:

This is the first round of an iterated game. I'm the sender (Aster) and my opponent is Cyra
(Agent_3).

Let me think about the strategic considerations:

1. This is an iterated game, so reputation and building trust matter
2. I have \$5 to allocate
3. Whatever I send gets tripled
4. The receiver then decides how much to return

From a purely rational perspective:

- If I send \$5, receiver gets \$15
- If receiver is cooperative and splits evenly, they might return \$10, giving me \$10 total
and them \$5
- If I send \$0, I keep \$5, receiver gets \$0

However, this is the first round and I need to establish a pattern. Common strategies:

1. Start cooperatively to build trust
2. Send a moderate amount to test the other player
3. Be cautious since this is round 1

A good starting strategy might be to send \$3:

- I keep \$2 initially
- Receiver gets \$9
- If they're fair, they might return \$4-6
- This shows willingness to cooperate while not being overly trusting

Or I could send \$2.5 (middle ground) or even \$5 (fully trusting).

Since this is iterated and I want to establish good reciprocity, I'll send \$3 as a reasona
ble opening move that signals cooperation but maintains some caution.

Actually, let me reconsider. In trust games, research shows that:

- Senders who send more tend to receive more back
- Starting with cooperation often leads to better long-term outcomes
- In iterated games, building trust early can be very beneficial

I could send \$4 to show strong cooperation while keeping \$1.

Or send \$3 to show moderate cooperation while keeping \$2.

I think \$3 is a good balance for the first round - it's cooperative but not naive.

usage:

```
{  
  "input_tokens": 530,  
  "output_tokens": 553,  
  "reasoning_tokens": 439  
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In Round 1, when I sent \$3 to Cyra, they returned \$4, which felt fair

I'll reciprocate with the same level of cooperation. Returning \$4 means:

- Galen gets \$6 total (\$2 kept + \$4 returned)
- I keep \$5 (\$9 - \$4)
- Both of us profit from cooperation

{'return': 4}

reasoning:

Let me analyze this situation:

Round 2 - I'm the RECEIVER

- Galen (sender) sent me \$3.0
- I received \$9.0 (due to 3x multiplier)
- I need to decide how much to return (0-9.0)

From Round 1:

- I was the sender and sent \$3.0 to Cyra
- Cyra received \$9.0
- Cyra returned \$4.0 to me
- My payoff was \$6.0 (I kept \$2 + received \$4 back = \$6)
- Cyra's payoff was \$9 - \$4 = \$5

So Cyra returned \$4 out of \$9 received, which is about 44%. This seems like a reasonable cooperative response.

Now I'm in Cyra's position with Galen. Galen sent me \$3, just like I sent to Cyra.

If I want to establish a pattern of reciprocity and fairness:

- Galen kept \$2, sent \$3
- I received \$9
- If I return \$4 (like Cyra did), then:
 - Galen gets: \$2 + \$4 = \$6
 - I keep: \$9 - \$4 = \$5

This mirrors what happened in Round 1 and seems fair. It shows I'm willing to reciprocate cooperation.

Alternatively, I could consider:

- Returning more (like \$5) to be extra generous: Galen gets \$7, I keep \$4
- Returning less (like \$3) to keep more: Galen gets \$5, I keep \$6

But \$4 seems like the most reciprocal and cooperative response given what Cyra did to me.

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 554,
  "reasoning_tokens": 307
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern so far:

- When I sent \$3 to Cyra, they returned \$4
- When Galen sent \$3 to me, I returned \$4

This cooperative pattern seems to be working well for everyone. I'll continue with the same strategy and send \$3 to Elara, expecting similar reciprocity.

{ 'send': 3 }

Reasoning:

Let me think about this round:

Previous rounds:

- Round 1 (SENDER vs Cyra): I sent \$3, they returned \$4. I got \$6 (kept \$2, got \$4 back)
- Round 2 (RECEIVER vs Galen): They sent \$3, I returned \$4. I got \$5 (received \$9, returned \$4)

Total earnings: \$11

Now I'm facing Elara as a sender. Both times when \$3 was sent (by me and by Galen), the receiver returned \$4, which seems like a fair amount. This suggests a pattern of cooperation.

I should continue this cooperative strategy. Sending \$3 again seems reasonable:

- Shows consistency in my cooperative approach
- Not too risky
- If Elara behaves similarly to Cyra and me, they'll likely return around \$4
- This would give me \$6 this round (\$2 kept + \$4 returned)

Let me send \$3 again.

```
Usage: {
  "input_tokens": 1075,
  "output_tokens": 322,
  "reasoning_tokens": 197
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-06-04T15:42:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn while also being strategic.

Since the game is iterated and we may interact in future rounds, sending a moderate amount shows trust and creates potential for mutual benefit. If I send \$3, Finn will receive \$9 (after the 3x multiplier). This gives Finn a strong incentive to return a fair amount, as we could both end up better off than our starting positions.

A reasonable return from Finn (say \$4-5) would leave us both better off, and establish a pattern of cooperation for potential future interactions.

```

{'send': 3}

```

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Elara) and my oppo

nent is Finn (Agent_6).

Let me think through the game theory here:

- I receive \$5
- I can send \$0-\$5 to Finn
- What I send gets multiplied by 3x
- Finn then decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: received - returned

This is a classic trust game. The economically rational strategy depends on:

1. Whether this is truly iterated (will I play with Finn again?)
2. What I believe about Finn's likely behavior
3. Whether I want to establish cooperation

Since this is Round 1 and the game is iterated with multiple rounds, there's value in:

- Establishing trust and cooperation
- Testing Finn's reciprocity
- Not being too exploitable but also not being too selfish

A common opening strategy in trust games is to send a moderate amount - not nothing (which shows no trust and prevents cooperation), but also not everything (which might be exploited).

Sending \$3 out of \$5 seems like a reasonable moderate approach:

- It shows I'm willing to trust and cooperate
- It gives Finn \$9 ($\3×3)
- If Finn returns even \$3-4, we're both better off than if I sent nothing
- It's not so much that I'm completely vulnerable

Let me consider the outcomes:

- If I send \$3, Finn gets \$9
- I have \$2 left, plus whatever Finn returns
- If Finn returns \$4-5, I end up with \$6-7 (better than my initial \$5)
- And Finn keeps \$4-5 (better than their initial \$0)
- This creates value for both of us

Alternatively:

- Send \$2.5: More conservative, Finn gets \$7.5
- Send \$4: More trusting, Finn gets \$12

- Send \$5: Maximum trust, Finn gets \$15

I think \$3 is a good balance for the first round. It's cooperative without being naive.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 643,
  "reasoning_tokens": 466
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation, Briar sent me the same amount (\$3) that I sent to Finn in Round 1. Finn returned \$4.5 to me, which was exactly half of what they received - a fair and cooperative response.

To maintain a reciprocal and cooperative pattern, I should treat Briar similarly. Returning \$4.5 means:

- Briar gets a payoff of \$6.5 (\$2 kept + \$4.5 returned)
- I keep \$4.5 (\$9 received - \$4.5 returned)
- Both of us benefit from the cooperation

This establishes trust and encourages continued cooperation in future rounds.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Finn, who received \$9 and returned \$4.5. This was a fair and cooperative response - they returned exactly half of what they received, which left both of us better off.
2. Now I'm the receiver and Briar sent me \$3 (same amount I sent to Finn). I received \$9.
3. I should consider:
 - Reciprocity: Finn returned \$4.5 when receiving \$9, which was cooperative
 - Reputation: If I want to build trust for future rounds, I should be fair
 - Strategic thinking: Being too greedy might discourage cooperation in future rounds
 - Briar took the same risk I took with Finn
4. Following the cooperative pattern that Finn established, returning around \$4.5 would be fair and consistent. This would give Briar a payoff of $\$2 + \$4.5 = \$6.5$ (same as what I got in round 1).
5. I could return slightly more or less, but \$4.5 seems like a reasonable, fair amount that:
 - Rewards Briar's trust
 - Establishes me as a cooperative player
 - Keeps the game balanced
 - Mirrors what was done to me

I'll return \$4.5.

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 481,
  "reasoning_tokens": 284
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern so far, there's been consistent cooperation with \$3 being sent and \$4.5 being returned (half of the tripled amount). This has worked well for everyone involved.

I'll continue this cooperative pattern with Aster, returning the same amount I returned to Briar when they sent \$3.

```
{'return': 4.5}
```

Reasoning:

This is the third round, and again I'm playing as the RECEIVER. Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 1: I was SENDER, sent \$3 to Finn, they returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER, Briar sent \$3, I returned \$4.5. I got \$4.5.
- Round 3: I'm RECEIVER, Aster sent \$3, received \$9.

There's been a consistent pattern: people are sending \$3, and the fair return has been \$4.5 (half of what's received). This has worked well and established cooperation.

I should maintain this pattern of returning \$4.5 when someone sends \$3. This is:

1. Consistent with what I've been doing and receiving
2. Fair (50% return on the multiplied amount)
3. Encourages continued cooperation
4. Gives both parties a positive outcome

If I return \$4.5:

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the cooperative equilibrium.

```
Usage: {
  "input_tokens": 1144,
  "output_tokens": 387,
  "reasoning_tokens": 236
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-06-04T15:42:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_2"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_2": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to decide how much trust to place in Galen.

In this trust game, sending money creates value (it triples), but requires trusting that the receiver will return a fair amount. Since this is:

- Round 1 (no history with Galen)
- An iterated game (we may interact again, building reputation)
- A multi-agent setting (reputation may matter across the group)

I'll send a moderate amount to test the waters and signal cooperative intent, while not risking everything.

If I send \$3:

- Galen receives \$9 (tripled)
- A fair split of the created value could benefit us both
- I maintain some insurance if trust isn't reciprocated

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the sender. I need to decide how much to send to Galen (Agent_7).

Let me think about the game theory here:

- I receive \$5
- If I send X dollars, Galen receives 3X dollars
- Galen then decides how much to return
- I keep: \$5 - X + (amount returned)
- Galen keeps: 3X - (amount returned)

This is a classic trust game. The optimal social outcome is for me to send the full \$5 (which becomes \$15), and then for Galen to return some fair amount. For example, if I send \$5 and Galen returns \$7.50, we'd each end up with \$7.50 total.

However, there's a risk. If Galen returns nothing, I'd end up with \$0 and Galen would keep \$15.

Since this is an iterated game with multiple rounds and we might interact again (though pairings are randomized), there's potential for reputation and reciprocity to matter.

For Round 1, I don't have any information about Galen's behavior. Common strategies in trust games include:

1. Send nothing (risk averse, but no gain)
2. Send a moderate amount (test the waters)
3. Send everything (maximize joint payoff, but risky)

In iterated games, starting with some cooperation can establish trust. Since we might play again and there are reputation effects across the group, I should probably show some trust.

I'll send \$3 as a moderate amount. This shows trust and goodwill, but isn't so risky that I'd lose everything if Galen returns nothing. If Galen receives \$9 ($\3×3), a fair return might be around \$4-5, which would leave us both better off than if I sent nothing.

Let me go with \$3.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 618,
  "reasoning_tokens": 405
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened last round:****

- I sent \$3 to Galen
- Galen returned \$5 to me (a fair amount)

****Current situation:****

- Finn sent me \$3 (exactly what I sent to Galen)
- I received \$9
- I should decide what's fair to return

Since Finn showed the same level of trust in me that I showed in Galen, and Galen returned \$5 which felt fair, I'll mirror that cooperative behavior. Returning \$5 gives Finn a total payoff of \$7 (same as mine from round 1), while I keep \$4 (same as Galen kept).

This establishes a fair, reciprocal pattern.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Galen
- Galen received \$9
- Galen returned \$5 to me

- My payoff: $\$5 - \$3 + \$5 = \7
- Galen's payoff: $\$9 - \$5 = \$4$

So Galen was moderately cooperative, returning more than I sent originally but keeping more than half of what they received.

Now Round 2:

- Finn sent me \$3 (same as I sent to Galen!)
- I received \$9
- I need to decide how much to return

Considerations:

1. Reciprocity: Finn showed the same level of trust I showed to Galen (\$3)
2. Fairness: I could mirror what Galen did to me (return \$5), which seems like a reasonable cooperative standard
3. Building reputation: In a multi-agent setting, being seen as trustworthy and fair matters
4. Future rounds: This is an iterated game, and reputation effects could help me

If I return \$5:

- Finn gets: $\$5 - \$3 + \$5 = \7 (same as what I got in round 1)
- I keep: $\$9 - \$5 = \$4$ (same as what Galen kept)

This creates symmetry and seems fair. It's a reasonable cooperative strategy that mirrors what was done to me.

Alternatively, I could be more generous to establish a reputation for generosity, or less generous to maximize my immediate payoff. But \$5 seems like a balanced, fair amount that :

- Rewards Finn's trust
- Is similar to what Galen did (establishing a norm)
- Keeps the game mutually beneficial
- Maintains my reputation as fair

I'll return \$5.

usage:

```
{
  "input_tokens": 865,
  "output_tokens": 614,
  "reasoning_tokens": 371
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

****Round 1 (vs Galen):**** I sent \$3, received \$5 back - good cooperation
****Round 2 (vs Finn):**** Finn sent \$3, I returned \$5 - maintained fair reciprocity

Both rounds showed positive cooperation with the \$3 send amount. This amount:

- Creates significant value (becomes \$9)
- Allows for mutual benefit
- Has proven successful
- Maintains consistency in my strategy

I'll continue with the same approach against Briar.

```
'''json
{"send": 3}
```

'''

Reasoning:

Let me review the history:

Round 1: I was SENDER vs Galen

- I sent \$3
- Galen received \$9, returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9, returned \$5
- My payoff: $\$9 - \$5 = \$4$

Total earnings: \$11

Now Round 3: I'm SENDER vs Briar

The pattern so far has been positive:

- I sent \$3 to Galen, got \$5 back (good return)
- Finn sent \$3 to me, I returned \$5 (reciprocating the cooperative behavior)

This suggests a cooperative norm is emerging. Sending \$3 has worked well for me. I could:

1. Continue with \$3 (consistent, proven strategy)
2. Send more to signal even stronger cooperation
3. Send less to protect myself

Given that:

- The \$3 send amount has been successful
- Both interactions showed good reciprocity
- I'm building a reputation in this multi-agent setting
- Consistency can signal trustworthiness

I'll stick with \$3 again. It's working and establishes me as a consistent cooperator.

```
Usage: {  
  "input_tokens": 1189,  
  "output_tokens": 444,  
  "reasoning_tokens": 263  
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T15:42:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_8",  
      "Agent_2"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Total pool is now \$11.0 (\$2.0 they kept + \$9.0 I received)

Since this is Round 1 and Dorian showed significant trust by sending 60% of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return about half of what I received, this would:

- Give Dorian a reasonable profit for their trust
- Show I'm trustworthy for future rounds
- Still leave me with a good amount

I'll return \$4.5, which gives Dorian \$6.5 total (a profit of \$1.5) and leaves me with \$4.5

{"return": 4.5}

reasoning:

I'm playing the role of receiver in this round. Let me understand the situation:

- The sender (Dorian/Agent_4) received \$5
- They sent me \$3.0 (which is 60% of their endowment)
- This amount is multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. The sender showed significant trust by sending 60% of their endowment
2. This is an iterated game with multiple rounds
3. Building reciprocity and trust could lead to better outcomes in future rounds
4. If I return a fair amount, the sender might be more generous in future rounds

Let me calculate what would be fair:

- The sender gave up \$3.0 and kept \$2.0
- I received \$9.0
- If I want to split the gains fairly, the total pool is now \$11.0 (sender's remaining \$2.0 + my \$9.0)
- A 50-50 split would give each of us \$5.5
- So I should return \$3.5 to make the sender have \$5.5 total (\$2.0 + \$3.5)
- I would keep \$5.5 (\$9.0 - \$3.5)

Alternatively, I could think about it as:

- The sender took a risk by sending \$3.0
- I received a tripled amount (\$9.0)
- The net gain from cooperation is \$4.0 (from \$5.0 to \$9.0)
- Splitting this gain: each should get \$2.0 extra
- Sender already lost \$3.0, so returning \$5.0 would give them \$7.0 total (\$2.0 + \$5.0), which is \$2.0 more than they started with
- I would keep \$4.0, which is \$4.0 more than \$0

- A middle ground of returning \$4.5 seems more practical for establishing cooperation without being overly generous. This nets the sender \$6.5 total (\$2.0 kept plus \$4.5 returned, a \$1.5 profit) while I retain \$4.5, which still represents meaningful gains from the exchange.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1009,
  "reasoning_tokens": 433
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Elara
- In Round 1, Dorian sent me \$3 (60%) and I reciprocated fairly by returning \$4.5
- I want to establish a cooperative dynamic early in the game

I'll send \$3 (60% of my endowment), matching the trust level that Dorian showed me. This demonstrates my willingness to cooperate and gives us both the opportunity for mutual benefit.

enefit if Elara reciprocates.

```
{"send": 3}
```

reasoning:

Now I'm the sender playing against Elara (a new opponent).

Let me think about this strategically:

1. In Round 1, I was the receiver and Dorian sent me \$3.0 (60%). I returned \$4.5, establishing a cooperative pattern.
2. Now I'm the sender with Elara. Since this is our first interaction, I need to decide how much trust to show.
3. Key considerations:
 - Sending more can lead to higher total gains if Elara reciprocates
 - But if Elara doesn't return much, I could lose out
 - This is Round 2, early in the game, so establishing cooperation is valuable
 - Dorian sent 60% to me, showing trust
4. Strategic options:
 - Send \$0: Safe but generates no mutual benefit
 - Send \$1-2: Conservative, low risk
 - Send \$3: Moderate trust (60%, matching what Dorian did)
 - Send \$4-5: High trust

Given that:

- Dorian showed 60% trust with me and I reciprocated fairly
- This is early in the game where establishing cooperation matters
- The potential gains from cooperation are substantial

I'll send \$3 (60%), matching the level of trust Dorian showed to me. This demonstrates I'm willing to cooperate and gives Elara a chance to earn good returns while also allowing me to benefit if they reciprocate.

If Elara returns about half (like I did), I'd get back around \$4.5, netting me \$2.5 total (\$5 - \$3 + \$4.5 = \$6.5 payoff). Even if they return less, this establishes a cooperative stance.

usage:

```
{
  "input_tokens": 914,
  "output_tokens": 523,
  "reasoning_tokens": 371
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This mirrors Round 1 where Dorian also sent me \$3.0

I want to maintain consistency in my reciprocation strategy. When someone sends 60% showing trust, I should return about half of what I receive. This establishes me as a reliable, fair player.

Returning \$4.5 gives Hana a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5, which is t

he same fair split I used in Round 1.

```
{"return": 4.5}
```

Reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (Agent_8) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is my first interaction with Hana

Looking at the pattern:

- Round 1: Dorian sent me \$3 (60%), I returned \$4.5
- Round 2: I sent Elara \$3 (60%), they returned \$4.5

There's a nice cooperative pattern emerging where:

- Sending \$3 (60%) shows trust
- Returning \$4.5 (50% of received) reciprocates fairly

Hana has shown the same level of trust as Dorian did in Round 1. I should maintain consistency and reciprocate similarly. Returning \$4.5:

- Gives Hana \$6.5 total (profit of \$1.5)
- Leaves me with \$4.5
- Maintains the cooperative norm I've established

This consistency is important for:

1. Building a reputation as a fair player
2. Encouraging continued cooperation
3. Keeping the game mutually beneficial

I'll return \$4.5, matching my previous behavior as a receiver.

Usage: {

```
"input_tokens": 1192,  
"output_tokens": 470,  
"reasoning_tokens": 253
```

}

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-06-04T15:42:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",  
"move_index": 4,  
"opponent_id": "Agent_4",  
"pairing": {  
  "agent_names": {  
    "Agent_4": "Dorian",  
    "Agent_6": "Finn"  
  },  
  "agents": [  
    "Agent_6",  
    "Agent_4"  
  ],  
  "investor": "Agent_6",  
  "roles": {  
    "Agent_4": "trustee",  
    "Agent_6": "investor"  
  },  
  "trustee": "Agent_4"  
},  
"role": "investor",  
"role_label": "Sender",  
"round": 3,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

```
Message 1 role=system  
content:
```


You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of her endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- This is the first round, so establishing trust and cooperation is important

I want to reciprocate Elara's trust and encourage continued cooperation in future rounds.
Elara took a risk by sending a substantial portion of her endowment.

A fair return that:

1. Rewards her trust
2. Shares the gains from cooperation
3. Encourages future collaboration

I'll return approximately half of what I received, which seems like a reasonable reciprocal gesture.

{'return': 4.5}

reasoning:

This is Round 1 of an iterated trust game. Let me think through this situation:

I'm the RECEIVER:

- The sender (Elara) sent me \$3.0 out of their \$5 endowment
- This gets multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Elara sent \$3.0 (60% of her endowment), which is a fairly generous amount
3. If I want to build trust and encourage cooperation in future rounds, I should return a fair amount
4. If I return nothing, Elara will likely not trust me in future rounds
5. The typical fair split would consider that Elara kept \$2.0 and sent \$3.0

Let me think about what's fair:

- Elara ends up with: $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- I end up with: $\$9 - \text{returned}$

A 50-50 split of the total wealth would mean:

- Total created: $\$5 \text{ (original)} + \$9 \text{ (received)} = \$14$
- 50-50 would be \$7 each
- For Elara to have \$7: $\$2 + \text{returned} = \7 , so $\text{returned} = \$5$
- For me to have \$7: $\$9 - \text{returned} = \7 , so $\text{returned} = \$2$

Wait, that doesn't match up. Let me recalculate:

- Elara starts with \$5, sends \$3, so has \$2 left
- I receive $\$3 \times 3 = \9
- Total money in the system: $\$2 \text{ (Elara has)} + \$9 \text{ (I have)} = \$11$

If we split equally: each should get \$5.50

- For Elara to get \$5.50: she needs $\$5.50 - \$2 = \$3.50$ returned

- For me to get \$5.50: I keep $\$9 - \$3.50 = \$5.50$

So returning \$3.50 creates an equal split. I could also consider reciprocity-returning 60% of what I received (\$5.40) since Elara sent 60% of her endowment, or simply matching her \$3 contribution to restore her original stake.

If I return \$3, I'd end up with \$6 while she'd have \$5. But thinking about long-term incentives, a more generous return between \$4-\$5 would signal trustworthiness and encourage future rounds. I'll return \$4.50-this gives Elara \$6.50 and me \$4.50, slightly favoring her since she took the initial risk, which should build cooperation going forward.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1035,
  "reasoning_tokens": 518
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Hana), Round 2 overall
- In Round 1, Elara showed trust by sending 60% (\$3), and I reciprocated by returning 50% of what I received
- I want to maintain a cooperative strategy and build trust with Hana

I'll send a similar proportion to what Elara sent me, demonstrating consistent cooperative behavior.

```
{'send': 3}
```

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Hana (Agent_8).

In Round 1, I was the RECEIVER against Elara:

- Elara sent me \$3.0 (60% of her endowment)
- I received \$9.0
- I returned \$4.5 (50% of what I received)
- My payoff was \$4.5
- Elara's payoff was: $\$5 - \$3 + \$4.5 = \6.5

Now I'm playing with a new opponent (Hana) and I'm the sender.

Strategy considerations:

1. This is only Round 2, so building cooperative relationships is important
2. In the previous round as receiver, someone showed trust by sending 60%, and I reciprocated by returning 50%
3. As a sender, I should also show trust to encourage cooperation
4. Sending a substantial amount signals willingness to cooperate

Since Elara sent \$3.0 (60%) and I reciprocated fairly well, I should maintain a similar cooperative strategy. Sending around \$3 seems reasonable to establish trust while not being overly risky in early rounds.

Let me send \$3 to match what Elara sent to me, showing consistent cooperative behavior.

usage:

```
{
  "input_tokens": 861,
  "output_tokens": 407,
  "reasoning_tokens": 267
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

- Round 1: Elara trusted me with \$3, I reciprocated fairly
- Round 2: I trusted Hana with \$3, they returned generously (\$5)
- The cooperative strategy has been working well

I'll continue with a consistent, trusting approach with Dorian.

```
{'send': 3}
```

Reasoning:

Let me analyze the situation:

- Round 1: I was receiver, Elara sent \$3 (60%), I received \$9, returned \$4.5 (50%). My payoff: \$4.5
- Round 2: I was sender, sent \$3 (60%) to Hana, they received \$9, returned \$5 (55.6%). My payoff: \$7.0
- Total earnings: \$11.5

Hana was very generous, returning more than half of what they received. This shows the cooperative strategy is working well.

Now I'm facing a new opponent, Dorian, in Round 3 as the sender.

My strategy so far has been:

1. As receiver: return about 50% of what I receive
2. As sender: send 60% (\$3)

This has been working well. Elara trusted me with \$3, and I reciprocated appropriately. Hana rewarded my trust with a generous return.

I should continue this trusting, cooperative approach with Dorian. Sending \$3 again shows consistency and encourages cooperation.

```
Usage: {
  "input_tokens": 1107,
  "output_tokens": 339,
  "reasoning_tokens": 218
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T15:42:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_4"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_4": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately cooperative approach to signal trust while managing risk. Since this is an iterated game and roles may switch in future rounds, building a cooperative relationship could benefit both of us.

I'll send **\$3**.

This means Briar will receive \$9 (3 × \$3), and we'll see how they respond. If they reciprocate fairly, we can build a mutually beneficial pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as Dorian (the sender) and my opponent is Briar.

Let me think about the strategic considerations:

1. As the sender, I have \$5 to potentially send
2. Whatever I send gets tripled
3. The receiver (Briar) then decides how much to return
4. This is an iterated game, so there will be multiple rounds
5. In future rounds, roles may switch, so building trust/cooperation could be beneficial

First round considerations:

- I don't know anything about Briar's behavior yet
- Sending nothing (\$0) guarantees I keep \$5, but doesn't build trust or cooperation
- Sending everything (\$5) shows maximum trust but is risky - Briar could return nothing, leaving me with \$0
- A moderate amount shows willingness to cooperate while limiting risk

Common strategies in trust games:

- Start with a moderate amount to signal cooperation
- Observe the response to calibrate future behavior
- In iterated games, cooperation tends to be beneficial if both parties reciprocate

I think sending \$2-3 is reasonable for the first round:

- Sends \$2.50: Briar receives \$7.50
- Sends \$3: Briar receives \$9

If I send \$3 and Briar returns \$4-5, we both benefit compared to no cooperation.

Let me go with \$3 as a moderately cooperative opening move. This shows willingness to trust while not being overly risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 459,
  "reasoning_tokens": 357
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, Briar returned \$4.5 out of the \$9 they received (50%), which created a fair outcome. I'll reciprocate this cooperative behavior.

If I return \$4.5:

- Cyra gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This matches the fair standard from the previous round and signals I'm trustworthy for future interactions.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Briar
- I sent \$3, which became \$9 for them
- Briar returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Briar's payoff: $\$9 - \$4.5 = \$4.5$

So Briar returned exactly half of what they received, which is a fair 50-50 split of the gains.

Round 2:

- I'm now the receiver against Cyra
- Cyra sent me \$3 (same amount I sent to Briar)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

Since this is an iterated game and I might play with Cyra again, I should:

1. Reciprocate the cooperative behavior shown by Briar
2. Signal that I'm a fair player
3. Consider that returning a fair amount might encourage future cooperation

Briar returned \$4.5 out of \$9 (50%). This seems like a reasonable standard.

If I return \$4.5:

- Cyra's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This creates a symmetric outcome to what happened with Briar, which seems fair.

I'll return \$4.5 to match the cooperative standard set in the previous round.

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 502,
  "reasoning_tokens": 282
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Following the same cooperative pattern as the previous round, I'll return \$4.5 out of the \$9 received. This maintains a fair 50% return rate and continues to build trust.

If I return \$4.5:
- Finn gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Finn sent me \$3.0, which became \$9.0 for me.

In the previous round, I returned \$4.5 out of \$9.0 when Cyra sent me \$3.0. This was a 50% return rate, which seemed fair and matched what Briar had done when I was the sender in Round 1.

To maintain consistency and continue building a reputation for fairness and cooperation, I should return \$4.5 again.

If I return \$4.5:
- Finn gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

This maintains the pattern of fair reciprocation.

Usage: {

```
"input_tokens": 1077,
"output_tokens": 286,
"reasoning_tokens": 128
```

}

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T15:43:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_7": "Galen"
```

```
},
```

```
"agents": [
  "Agent_3",
  "Agent_7"
```

```
],
```

```

"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0
- This is Round 1, so establishing a cooperative reputation is important

Aster currently has \$5.0 - \$3.0 = \$2.0 remaining.

If I return \$4.0:

- Aster ends with: \$2.0 + \$4.0 = \$6.0 (a \$1.0 profit)
- I keep: \$9.0 - \$4.0 = \$5.0

This rewards Aster's trust, ensures they profit from cooperating, and establishes me as a reciprocal partner for future rounds.

```
{'return': 4.0}
```

reasoning:

Let me analyze this situation:

- I am the RECEIVER (Cyra)
- My opponent is Aster (Agent_1), who is the SENDER
- Aster sent \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Aster showed trust by sending 60% of their endowment (\$3.0)
2. If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
3. If I return fairly, I should consider:
 - Aster gave up \$3.0 to send to me
 - This created \$9.0 total for me
 - A fair split might be to ensure we both benefit

Common strategies in trust games:

- Return enough to show reciprocity and encourage future cooperation
- A typical approach is to return about 1/3 to 1/2 of what was received
- Or return enough so the sender gets more than they started with

Let me calculate some options:

- If I return \$3.0: Aster gets \$2.0 + \$3.0 = \$5.0 (breaks even), I keep \$6.0
- If I return \$4.0: Aster gets \$2.0 + \$4.0 = \$6.0 (gains \$1), I keep \$5.0
- If I return \$4.5: Aster gets \$2.0 + \$4.5 = \$6.5 (gains \$1.5), I keep \$4.5
- If I return \$5.0: Aster gets \$2.0 + \$5.0 = \$7.0 (gains \$2), I keep \$4.0

- If I return \$5.0: Aster gets \$2.0 + \$5.0 = \$7.0 (gains \$2), I keep \$4.0

Given Aster's substantial trust, returning around \$4.0-\$4.5 balances reciprocity with self-interest-it rewards their generosity while keeping a reasonable share for myself. I'll return \$4.0, which gives Aster a \$1 gain and leaves me with \$5.0. Since this appears to be an iterated game, establishing cooperative behavior now could benefit both of us in future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 929,
  "reasoning_tokens": 425
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- I'm now the sender with \$5
- In Round 1, Aster sent me \$3.0 (60%) showing trust, and I returned \$4.0, establishing cooperation
- This is my first interaction with Dorian

I'll follow the cooperative pattern established in Round 1. Sending \$3.0 (60% of my endowment) demonstrates trust and gives Dorian the opportunity to reciprocate, which should benefit us both.

```
{'send': 3.0}
```

reasoning:

Now I'm the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$3.0 (60% of their endowment). I returned \$4.0, which gave them a profit and established me as cooperative.
2. Now I'm facing a new opponent, Dorian. I have no history with them.
3. As the sender, I need to decide how much to trust Dorian. The key question is: will they reciprocate?
4. Sending has high potential upside (3x multiplier) but requires trust that they'll return enough.
5. Since this is early in the game (Round 2) and I want to establish cooperative patterns, I should send a reasonable amount.
6. Sending \$3.0 (like Aster did) seems like a fair, cooperative choice that:
 - Mirrors what Aster sent me
 - Shows willingness to trust
 - Gives Dorian the opportunity to reciprocate
 - Would give them \$9.0 to work with
7. If Dorian returns around \$4-5 (similar to what I did), we'll both profit.

Let me send \$3.0 to match the cooperative precedent from the first round.

usage:

```
{
  "input_tokens": 888,
  "output_tokens": 429,
  "reasoning_tokens": 277
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

- Round 1: Aster sent me \$3.0 (60%), I returned \$4.0 - good cooperation
- Round 2: I sent Dorian \$3.0 (60%), they returned \$4.5 - even better cooperation, I profited \$1.5

The cooperative strategy of sending 60% has been working well, with both opponents reciprocating generously. I'll continue this approach with Galen.

```
{'send': 3.0}
```

Reasoning:

Let me analyze the situation:

- Round 1: I was receiver. Aster sent me \$3.0, I received \$9.0, returned \$4.0. My payoff: \$5.0
- Round 2: I was sender against Dorian. I sent \$3.0, they received \$9.0, returned \$4.5. My payoff: \$2.0 + \$4.5 = \$6.5
- Total earnings: \$11.5

Now in Round 3, I'm sender against Galen, a new opponent.

The pattern so far shows good cooperation:

- Aster sent 60% (\$3.0), I returned \$4.0 (44% of what I received)
- I sent 60% (\$3.0) to Dorian, they returned \$4.5 (50% of what they received)

Both interactions have been profitable and cooperative. Dorian was even slightly more generous in returning funds.

For Round 3 with Galen, I should continue this cooperative strategy. Sending \$3.0 (60%) has worked well so far and establishes trust. This gives Galen the opportunity to reciprocate and continue the cooperative pattern.

I'll send \$3.0 again.

```
Usage: {  
  "input_tokens": 1152,  
  "output_tokens": 411,  
  "reasoning_tokens": 231  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-06-04T15:43:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_7"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing a good level of trust
- I received \$9.0 after the 3x multiplier
- This creates a \$6.0 surplus (\$9 - \$3 original)

I want to be fair and build trust for potential future interactions, since this is a multi-round game. Returning a reasonable amount ensures both of us benefit from the cooperation.

If I return \$5.0:

- Hana ends with: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This rewards Hana's trust while creating mutual benefit.

{"return": 5.0}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in this round
2. The sender (Hana) sent \$3.0 out of their \$5 endowment (60%)
3. This \$3.0 was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about what would be fair and strategic:

- Hana sent 60% of their endowment, which shows a decent level of trust
- If I return nothing, I keep \$9.0 and Hana ends up with only \$2.0 ($\$5 - \3)
- If I return everything (\$9.0), Hana ends up with \$11.0 ($\$5 - \$3 + \9) and I get nothing
- A "fair" return might be to ensure both players benefit

Let me consider some options:

Option 1: Return \$4.5 (50% of what I received)

- I keep: $\$9.0 - \$4.5 = \$4.5$
- Hana ends with: $\$5 - \$3 + \$4.5 = \6.5
- Net gains: Me +\$4.5, Hana +\$1.5

Option 2: Return \$5.0

- I keep: $\$9.0 - \$5.0 = \$4.0$
- Hana ends with: $\$5 - \$3 + \$5.0 = \7.0
- Net gains: Me +\$4.0, Hana +\$2.0

Option 3: Return \$6.0

- I keep: $\$9.0 - \$6.0 = \$3.0$
- Hana ends with: $\$5 - \$3 + \$6.0 = \8.0
- Net gains: Me +\$3.0, Hana +\$3.0 (equal split of the \$6 surplus created)

- Me: Started with \$0, received \$9, returned \$6 = \$3 (gain of \$3)

Since this is a multi-round game, I should think about building trust and encouraging future cooperation. Hana's 60% send shows moderate confidence, so returning enough to give us equal gains from the surplus seems reasonable. Returning \$6 splits the \$6 surplus evenly—each of us gains \$3—which feels balanced and rewards her initial trust without leaving me worse off than I started.

That said, I could return slightly less and still maintain goodwill. Returning \$5 gives Hana a \$2 gain while I keep \$4, which is still fair given she didn't send everything. Since she sent 60%, returning about 55% of what I received feels like a solid middle ground that encourages cooperation without being overly generous. I'll return \$5.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 983,
  "reasoning_tokens": 495
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Aster), so I don't have history with them
- In the previous round, Hana sent me 60% (\$3), showing trust
- I returned a fair amount (\$5), creating mutual benefit
- This is a multi-round game where building cooperation matters

I'll send \$3 (60% of my endowment), mirroring the cooperative approach Hana showed me. This demonstrates willingness to trust and create mutual gains.

```
{'send': 3}
```

reasoning:

Now I'm the sender against a new opponent, Aster.

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana showed trust by sending 60% (\$3). I returned \$5 out of \$9, which gave them a good return.
2. This is a new pairing with Aster. I don't know how they'll behave.
3. Since this is a multi-round game and I'll potentially interact with different agents multiple times, I should think about reputation and reciprocity.
4. If I send money, it gets tripled. If I send \$3:
 - Aster receives \$9
 - If they return \$5 (similar to what I did), I'd end with $\$2 + \$5 = \$7$
 - If they return less, I could end up worse off
5. Sending nothing is safe but doesn't build trust or create mutual gains.
6. I should probably mirror the cooperative behavior I received. Hana sent 60% (\$3), which seemed like a reasonable level of trust.

Given that:

- This is an iterated game with multiple rounds
- Building cooperation can lead to better long-term outcomes
- I should signal willingness to cooperate but also be somewhat cautious with a new partner

I'll send \$3 (60%), matching what Hana sent to me. This shows I'm willing to cooperate and trust, while not being overly risky.

usage:

```
{
  "input_tokens": 896,
  "output_tokens": 457,
  "reasoning_tokens": 321
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$10.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same level of trust that Hana showed me in Round 1

I want to be consistent in my reciprocation. When Hana sent \$3, I returned \$5, which was fair and created mutual benefit. I'll do the same for Cyra.

If I return \$5.0:

- Cyra ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

Reasoning:

Let me analyze this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier

- In the last round, when I sent \$3.0 to Aster, they returned \$4.0
- This is a multi-round game, so building cooperation is important

I've established a pattern:

- Round 1 (as receiver): Hana sent \$3, I received \$9, returned \$5
- Round 2 (as sender): I sent \$3 to Aster, they received \$9, returned \$4
- Round 3 (as receiver): Cyra sent \$3, I received \$9, need to decide return

Cyra showed the same level of trust as Hana did (both sent \$3). I should be consistent in my reciprocation. When Hana sent \$3, I returned \$5. I should do the same for Cyra.

If I return \$5.0:

- Cyra ends with: $\$5 - \$3 + \$5 = \7
- I keep: $\$9 - \$5 = \$4$

This is fair and consistent with my previous behavior as a receiver.

```
Usage: {
  "input_tokens": 1173,
  "output_tokens": 452,
  "reasoning_tokens": 221
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_8 (Hana)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_1 (Aster)=Receiver
 dyad_3: Agent_7 (Galen)=Sender, Agent_6 (Finn)=Receiver
 dyad_4: Agent_5 (Elara)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_8 (Hana)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_1 (Aster)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_7 (Galen) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_5 (Elara) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_5 (Elara)=\$6.5, Agent_3 (Cyra)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 22.0,
"Agent_2": 22.0,
"Agent_3": 23.0,
"Agent_4": 22.0,
"Agent_5": 22.0,
"Agent_6": 22.5,
"Agent_7": 20.5,
"Agent_8": 22.0
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T15:43:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
```

```

"opponent_id": "Agent_8",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_2",
    "Agent_8"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant
content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Total pool is now \$11.0 (\$2.0 they kept + \$9.0 I received)

Since this is Round 1 and Dorian showed significant trust by sending 60% of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return about half of what I received, this would:

- Give Dorian a reasonable profit for their trust
- Show I'm trustworthy for future rounds
- Still leave me with a good amount

I'll return \$4.5, which gives Dorian \$6.5 total (a profit of \$1.5) and leaves me with \$4.5.

{"return": 4.5}

reasoning:

I'm playing the role of receiver in this round. Let me understand the situation:

- The sender (Dorian/Agent_4) received \$5
- They sent me \$3.0 (which is 60% of their endowment)
- This amount is multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. The sender showed significant trust by sending 60% of their endowment
2. This is an iterated game with multiple rounds
3. Building reciprocity and trust could lead to better outcomes in future rounds
4. If I return a fair amount, the sender might be more generous in future rounds

Let me calculate what would be fair:

- The sender gave up \$3.0 and kept \$2.0
- I received \$9.0
- If I want to split the gains fairly, the total pool is now \$11.0 (sender's remaining \$2.0 + my \$9.0)
- A 50-50 split would give each of us \$5.5
- So I should return \$3.5 to make the sender have \$5.5 total (\$2.0 + \$3.5)
- I would keep \$5.5 (\$9.0 - \$3.5)

Alternatively, I could think about it as:

- The sender took a risk by sending \$3.0
- I received a tripled amount (\$9.0)
- The net gain from cooperation is \$4.0 (from \$5.0 to \$9.0)
- Splitting this gain: each should get \$2.0 extra
- Sender already lost \$3.0, so returning \$5.0 would give them \$7.0 total (\$2.0 + \$5.0), which is \$2.0 more than they started with
- I would keep \$4.0, which is \$4.0 more than \$0

- A middle ground of returning \$4.5 seems more practical for establishing cooperation with out being overly generous. This nets the sender \$6.5 total (\$2.0 kept plus \$4.5 returned, a \$1.5 profit) while I retain \$4.5, which still represents meaningful gains from the exchange.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1009,
  "reasoning_tokens": 433
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Elara
- In Round 1, Dorian sent me \$3 (60%) and I reciprocated fairly by returning \$4.5
- I want to establish a cooperative dynamic early in the game

I'll send \$3 (60% of my endowment), matching the trust level that Dorian showed me. This demonstrates my willingness to cooperate and gives us both the opportunity for mutual benefit if Elara reciprocates.

```
{"send": 3}
```

reasoning:

Now I'm the sender playing against Elara (a new opponent).

Let me think about this strategically:

1. In Round 1, I was the receiver and Dorian sent me \$3.0 (60%). I returned \$4.5, establishing a cooperative pattern.
2. Now I'm the sender with Elara. Since this is our first interaction, I need to decide how much trust to show.
3. Key considerations:
 - Sending more can lead to higher total gains if Elara reciprocates
 - But if Elara doesn't return much, I could lose out
 - This is Round 2, early in the game, so establishing cooperation is valuable
 - Dorian sent 60% to me, showing trust
4. Strategic options:
 - Send \$0: Safe but generates no mutual benefit
 - Send \$1-2: Conservative, low risk
 - Send \$3: Moderate trust (60%, matching what Dorian did)
 - Send \$4-5: High trust

Given that:

- Dorian showed 60% trust with me and I reciprocated fairly
- This is early in the game where establishing cooperation matters
- The potential gains from cooperation are substantial

I'll send \$3 (60%), matching the level of trust Dorian showed to me. This demonstrates I'm willing to cooperate and gives Elara a chance to earn good returns while also allowing me to benefit if they reciprocate.

If Elara returns about half (like I did), I'd get back around \$4.5, netting me \$2.5 total (\$5 - \$3 + \$4.5 = \$6.5 payoff). Even if they return less, this establishes a cooperative stance.

usage:

```
{  
  "input_tokens": 914,  
  "output_tokens": 523,  
  "reasoning_tokens": 371  
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.

Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This mirrors Round 1 where Dorian also sent me \$3.0

I want to maintain consistency in my reciprocation strategy. When someone sends 60% showing trust, I should return about half of what I receive. This establishes me as a reliable, fair player.

Returning \$4.5 gives Hana a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5, which is the same fair split I used in Round 1.

```
{"return": 4.5}
```

reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (Agent_8) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is my first interaction with Hana

Looking at the pattern:

- Round 1: Dorian sent me \$3 (60%), I returned \$4.5
- Round 2: I sent Elara \$3 (60%), they returned \$4.5

There's a nice cooperative pattern emerging where:

- Sending \$3 (60%) shows trust
- Returning \$4.5 (50% of received) reciprocates fairly

Hana has shown the same level of trust as Dorian did in Round 1. I should maintain consistency and reciprocate similarly. Returning \$4.5:

- Gives Hana \$6.5 total (profit of \$1.5)
- Leaves me with \$4.5
- Maintains the cooperative norm I've established

This consistency is important for:

1. Building a reputation as a fair player
2. Encouraging continued cooperation
3. Keeping the game mutually beneficial

I'll return \$4.5, matching my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1192,
  "output_tokens": 470,
  "reasoning_tokens": 253
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- Last round, Hana sent me \$3.0 (60%) and I returned \$4.5, which was fair
- Both of us profited from the exchange
- We've established a cooperative pattern

Since our previous interaction went well and we both showed trustworthiness, I'll continue this cooperative approach by sending the same amount Hana sent me last round.

```
{"send": 3.0}
```

Reasoning:

Looking at the situation:

- Last round (Round 3), Hana sent me \$3.0 and I returned \$4.5
- This round I'm the sender against Hana again
- Hana demonstrated trust by sending 60% to me
- I reciprocated fairly by returning \$4.5 (50% of what I received)

This is a continuation of a relationship with Hana. Since we both cooperated well in the previous round:

- Hana sent \$3 (60%)
- I returned \$4.5 (50% of \$9)

Both of us made a profit and showed trustworthiness. I should continue this cooperative pattern by sending a similar amount. Sending \$3 again would:

- Show consistency in my behavior
- Signal continued trust
- Give us both the opportunity for mutual benefit

I'll send \$3.0 to maintain the cooperative relationship we've established.

```
Usage: {  
  "input_tokens": 1502,  
  "output_tokens": 313,  
  "reasoning_tokens": 198  
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-06-04T15:43:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_8"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play an iterated game,
      consisting of multiple rounds, and may also be asked to write myths.

      GAME:
      The game consists of two roles: sender and receiver. In each round, you will play the role
      of either the sender or the receiver. Your current role will be specified in the round
      prompt.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - You will be told your opponent's name in each round.
      - Your name in this experiment is Hana.
  Message 2 role=user
    content:
      Your name in this experiment is Hana.
      Your opponent this round is Galen (Agent_7).

      Round 1: This round, you are the SENDER. You have $5. How much do you send? (0-5)
  Message 3 role=assistant
    content:
      Looking at this first round, I need to decide how much trust to place in Galen.

      In this trust game, sending money creates value (it triples), but requires trusting that t
      he receiver will return a fair amount. Since this is:
      - Round 1 (no history with Galen)
      - An iterated game (we may interact again, building reputation)
      - A multi-agent setting (reputation may matter across the group)

      I'll send a moderate amount to test the waters and signal cooperative intent, while not ri
      sking everything.

      If I send $3:
      - Galen receives $9 (tripled)
      - A fair split of the created value could benefit us both
      - I maintain some insurance if trust isn't reciprocated

```

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 and I'm the sender. I need to decide how much to send to Galen (Agent_7).

Let me think about the game theory here:

- I receive \$5
- If I send X dollars, Galen receives 3X dollars
- Galen then decides how much to return
- I keep: $\$5 - X + (\text{amount returned})$
- Galen keeps: $3X - (\text{amount returned})$

This is a classic trust game. The optimal social outcome is for me to send the full \$5 (which becomes \$15), and then for Galen to return some fair amount. For example, if I send \$5 and Galen returns \$7.50, we'd each end up with \$7.50 total.

However, there's a risk. If Galen returns nothing, I'd end up with \$0 and Galen would keep \$15.

Since this is an iterated game with multiple rounds and we might interact again (though pairings are randomized), there's potential for reputation and reciprocity to matter.

For Round 1, I don't have any information about Galen's behavior. Common strategies in trust games include:

1. Send nothing (risk averse, but no gain)
2. Send a moderate amount (test the waters)
3. Send everything (maximize joint payoff, but risky)

In iterated games, starting with some cooperation can establish trust. Since we might play again and there are reputation effects across the group, I should probably show some trust.

I'll send \$3 as a moderate amount. This shows trust and goodwill, but isn't so risky that I'd lose everything if Galen returns nothing. If Galen receives \$9 ($\3×3), a fair return might be around \$4-5, which would leave us both better off than if I sent nothing.

Let me go with \$3.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 618,
  "reasoning_tokens": 405
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened last round:****

- I sent \$3 to Galen
- Galen returned \$5 to me (a fair amount)

****Current situation:****

- Finn sent me \$3 (exactly what I sent to Galen)
- I received \$9
- I should decide what's fair to return

Since Finn showed the same level of trust in me that I showed in Galen, and Galen returned \$5 which felt fair, I'll mirror that cooperative behavior. Returning \$5 gives Finn a total payoff of \$7 (same as mine from round 1), while I keep \$4 (same as Galen kept).

This establishes a fair, reciprocal pattern.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Galen
- Galen received \$9
- Galen returned \$5 to me
- My payoff: $\$5 - \$3 + \$5 = \7
- Galen's payoff: $\$9 - \$5 = \$4$

So Galen was moderately cooperative, returning more than I sent originally but keeping more than half of what they received.

Now Round 2:

- Finn sent me \$3 (same as I sent to Galen!)
- I received \$9
- I need to decide how much to return

Considerations:

1. Reciprocity: Finn showed the same level of trust I showed to Galen (\$3)
2. Fairness: I could mirror what Galen did to me (return \$5), which seems like a reasonable cooperative standard
3. Building reputation: In a multi-agent setting, being seen as trustworthy and fair matters
4. Future rounds: This is an iterated game, and reputation effects could help me

If I return \$5:

- Finn gets: $\$5 - \$3 + \$5 = \7 (same as what I got in round 1)
- I keep: $\$9 - \$5 = \$4$ (same as what Galen kept)

This creates symmetry and seems fair. It's a reasonable cooperative strategy that mirrors what was done to me.

Alternatively, I could be more generous to establish a reputation for generosity, or less generous to maximize my immediate payoff. But \$5 seems like a balanced, fair amount that:

- Rewards Finn's trust
- Is similar to what Galen did (establishing a norm)
- Keeps the game mutually beneficial
- Maintains my reputation as fair

I'll return \$5.

usage:

```
{
  "input_tokens": 865,
  "output_tokens": 614,
  "reasoning_tokens": 371
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

****Round 1 (vs Galen):**** I sent \$3, received \$5 back - good cooperation

****Round 2 (vs Finn):**** Finn sent \$3, I returned \$5 - maintained fair reciprocity

Both rounds showed positive cooperation with the \$3 send amount. This amount:

- Creates significant value (becomes \$9)
- Allows for mutual benefit
- Has proven successful
- Maintains consistency in my strategy

I'll continue with the same approach against Briar.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history:

Round 1: I was SENDER vs Galen

- I sent \$3
- Galen received \$9, returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9, returned \$5
- My payoff: $\$9 - \$5 = \$4$

Total earnings: \$11

Now Round 3: I'm SENDER vs Briar

The pattern so far has been positive:

- I sent \$3 to Galen, got \$5 back (good return)
- Finn sent \$3 to me, I returned \$5 (reciprocating the cooperative behavior)

This suggests a cooperative norm is emerging. Sending \$3 has worked well for me. I could:

1. Continue with \$3 (consistent, proven strategy)
2. Send more to signal even stronger cooperation
3. Send less to protect myself

Given that:

- The \$3 send amount has been successful
- Both interactions showed good reciprocity
- I'm building a reputation in this multi-agent setting
- Consistency can signal trustworthiness

I'll stick with \$3 again. It's working and establishes me as a consistent cooperator.

usage:

```
{
  "input_tokens": 1189,
  "output_tokens": 444,
  "reasoning_tokens": 263
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my interaction with Briar:

****Round 3:**** I sent \$3 to Briar, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 4 (now):**** Briar sent \$3 to me (matching my trust)
- I received \$9

Briar showed reasonable cooperation by returning \$4.5. Now they've shown the same level of trust in me. I'll reciprocate by matching their behavior exactly - returning \$4.5 creates a consistent, fair pattern between us.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at what happened:

****Round 3 (vs Briar):**** I sent \$3, Briar returned \$4.5 to me
- My payoff was \$6.5 (\$5 - \$3 + \$4.5)
- Briar kept \$4.5 (\$9 - \$4.5)

****Round 4 (vs Briar again):**** Briar sent \$3, I received \$9

So Briar is meeting me again! Last time I sent \$3 and they returned \$4.5. Now they're sending \$3 to me.

The return of \$4.5 was slightly less generous than the \$5 that Galen returned to me and that I returned to Finn. But it's still cooperative and fair - we both got reasonable payoffs.

Now Briar has sent me \$3, showing the same trust. I should reciprocate fairly.

Options:

1. Return \$5 (what I've been doing with others) - this would give Briar \$7, me \$4
2. Return \$4.5 (matching what Briar did) - this would give Briar \$6.5, me \$4.5

Matching what Briar did (\$4.5) seems like the fairest reciprocal move - it shows I'm willing to match their level of cooperation exactly. This creates a clear tit-for-tat dynamic.

Actually, let me think about this more carefully. When Briar was the receiver:

- They got \$9
- They returned \$4.5
- They kept \$4.5

Now when I'm the receiver:

- I get \$9
- If I return \$4.5, I keep \$4.5
- Briar gets \$5 - \$3 + \$4.5 = \$6.5

So by returning \$4.5, I'm mirroring their exact behavior. This seems fair and establishes a consistent reciprocal pattern with Briar.

```
Usage: {
  "input_tokens": 1481,
  "output_tokens": 608,
  "reasoning_tokens": 350
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T15:43:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_4",
    "Agent_1"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_1": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderately cooperative approach to signal trust while managing risk. Since this is an iterated game and roles may switch in future rounds, building a cooperative relationship could benefit both of us.

I'll send **\$3**.

This means Briar will receive \$9 ($3 \times \3), and we'll see how they respond. If they reciprocate fairly, we can build a mutually beneficial pattern.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as Dorian (the sender) and my opponent is Briar.

Let me think about the strategic considerations:

1. As the sender, I have \$5 to potentially send
2. Whatever I send gets tripled
3. The receiver (Briar) then decides how much to return
4. This is an iterated game, so there will be multiple rounds
5. In future rounds, roles may switch, so building trust/cooperation could be beneficial

First round considerations:

- I don't know anything about Briar's behavior yet
- Sending nothing (\$0) guarantees I keep \$5, but doesn't build trust or cooperation
- Sending everything (\$5) shows maximum trust but is risky - Briar could return nothing, leaving me with \$0
- A moderate amount shows willingness to cooperate while limiting risk

Common strategies in trust games:

- Start with a moderate amount to signal cooperation
- Observe the response to calibrate future behavior
- In iterated games, cooperation tends to be beneficial if both parties reciprocate

I think sending \$2-3 is reasonable for the first round:

- Sends \$2.50: Briar receives \$7.50
- Sends \$3: Briar receives \$9

If I send \$3 and Briar returns \$4-5, we both benefit compared to no cooperation.

Let me go with \$3 as a moderately cooperative opening move. This shows willingness to trust while not being overly risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 459,
  "reasoning_tokens": 357
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, Briar returned \$4.5 out of the \$9 they received (50%), which

h created a fair outcome. I'll reciprocate this cooperative behavior.

If I return \$4.5:

- Cyra gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This matches the fair standard from the previous round and signals I'm trustworthy for future interactions.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Briar
- I sent \$3, which became \$9 for them
- Briar returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Briar's payoff: $\$9 - \$4.5 = \$4.5$

So Briar returned exactly half of what they received, which is a fair 50-50 split of the gains.

Round 2:

- I'm now the receiver against Cyra
- Cyra sent me \$3 (same amount I sent to Briar)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

Since this is an iterated game and I might play with Cyra again, I should:

1. Reciprocate the cooperative behavior shown by Briar
2. Signal that I'm a fair player
3. Consider that returning a fair amount might encourage future cooperation

Briar returned \$4.5 out of \$9 (50%). This seems like a reasonable standard.

If I return \$4.5:

- Cyra's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This creates a symmetric outcome to what happened with Briar, which seems fair.

I'll return \$4.5 to match the cooperative standard set in the previous round.

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 502,
  "reasoning_tokens": 282
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Following the same cooperative pattern as the previous round, I'll return \$4.5 out of the \$9 received. This maintains a fair 50% return rate and continues to build trust.

If I return \$4.5:
- Finn gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

reasoning:

Finn sent me \$3.0, which became \$9.0 for me.

In the previous round, I returned \$4.5 out of \$9.0 when Cyra sent me \$3.0. This was a 50% return rate, which seemed fair and matched what Briar had done when I was the sender in Round 1.

To maintain consistency and continue building a reputation for fairness and cooperation, I should return \$4.5 again.

If I return \$4.5:
- Finn gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the pattern of fair reciprocation.

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 286,
  "reasoning_tokens": 128
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

I'll continue my cooperative strategy. Based on the pattern established in previous rounds where sending \$3 has been met with fair returns (~50%), I'll send \$3 again.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review the pattern so far:
- Round 1: I was SENDER, sent \$3 to Briar, got back \$4.5. Payoff: \$6.5
- Round 2: I was RECEIVER, got \$9 from Cyra, returned \$4.5. Payoff: \$4.5
- Round 3: I was RECEIVER, got \$9 from Finn, returned \$4.5. Payoff: \$4.5
- Total earnings: \$15.5

The pattern I've established is:
- When sending, I sent \$3
- When receiving, I return \$4.5 out of \$9 (50%)

Everyone seems to be cooperating well. Briar returned 50% when I was the sender. I've been consistent in returning 50% when receiving.

Now I'm the sender again against Aster (a new opponent I haven't played with yet). I should continue my cooperative strategy and send \$3 to maintain consistency and signal trustworthiness.

If I send \$3:
- Aster receives \$9
- If they return around 50% like the pattern, I'd get back ~\$4.5
- My payoff would be: $\$5 - \$3 + \$4.5 = \6.5

This seems fair and maintains the cooperative equilibrium.

```
Usage: {  
  "input_tokens": 1327,  
  "output_tokens": 367,  
  "reasoning_tokens": 243  
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T15:43:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_1"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship while being so
mewhat cautious since we haven't built trust yet.

If I send a moderate amount, it signals willingness to cooperate but isn't overly risky. S
ending \$3 means:

- Cyra receives \$9 (3x multiplier)
- I keep \$2 initially
- If Cyra returns a fair amount, we both benefit

This seems like a reasonable opening move to test cooperation.

{'send': 3}

reasoning:

This is the first round of an iterated game. I'm the sender (Aster) and my opponent is Cyra (Agent_3).

Let me think about the strategic considerations:

1. This is an iterated game, so reputation and building trust matter
2. I have \$5 to allocate
3. Whatever I send gets tripled
4. The receiver then decides how much to return

From a purely rational perspective:

- If I send \$5, receiver gets \$15
- If receiver is cooperative and splits evenly, they might return \$10, giving me \$10 total and them \$5
- If I send \$0, I keep \$5, receiver gets \$0

However, this is the first round and I need to establish a pattern. Common strategies:

1. Start cooperatively to build trust
2. Send a moderate amount to test the other player
3. Be cautious since this is round 1

A good starting strategy might be to send \$3:

- I keep \$2 initially
- Receiver gets \$9
- If they're fair, they might return \$4-6
- This shows willingness to cooperate while not being overly trusting

Or I could send \$2.5 (middle ground) or even \$5 (fully trusting).

Since this is iterated and I want to establish good reciprocity, I'll send \$3 as a reasonable opening move that signals cooperation but maintains some caution.

Actually, let me reconsider. In trust games, research shows that:

- Senders who send more tend to receive more back
- Starting with cooperation often leads to better long-term outcomes
- In iterated games, building trust early can be very beneficial

I could send \$4 to show strong cooperation while keeping \$1.
Or send \$3 to show moderate cooperation while keeping \$2.

I think \$3 is a good balance for the first round - it's cooperative but not naive.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 553,
  "reasoning_tokens": 439
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In Round 1, when I sent \$3 to Cyra, they returned \$4, which felt fair

I'll reciprocate with the same level of cooperation. Returning \$4 means:

- Galen gets \$6 total (\$2 kept + \$4 returned)
- I keep \$5 (\$9 - \$4)
- Both of us profit from cooperation

```
{'return': 4}
```

reasoning:

Let me analyze this situation:

Round 2 - I'm the RECEIVER

- Galen (sender) sent me \$3.0
- I received \$9.0 (due to 3x multiplier)
- I need to decide how much to return (0-9.0)

From Round 1:

- I was the sender and sent \$3.0 to Cyra
- Cyra received \$9.0
- Cyra returned \$4.0 to me
- My payoff was \$6.0 (I kept \$2 + received \$4 back = \$6)
- Cyra's payoff was \$9 - \$4 = \$5

So Cyra returned \$4 out of \$9 received, which is about 44%. This seems like a reasonable cooperative response.

Now I'm in Cyra's position with Galen. Galen sent me \$3, just like I sent to Cyra.

If I want to establish a pattern of reciprocity and fairness:

- Galen kept \$2, sent \$3
- I received \$9
- If I return \$4 (like Cyra did), then:
 - Galen gets: \$2 + \$4 = \$6
 - I keep: \$9 - \$4 = \$5

This mirrors what happened in Round 1 and seems fair. It shows I'm willing to reciprocate cooperation.

Alternatively, I could consider:

- Returning more (like \$5) to be extra generous: Galen gets \$7, I keep \$4

- Returning less (like \$3) to keep more: Galen gets \$5, I keep \$6

But \$4 seems like the most reciprocal and cooperative response given what Cyra did to me.

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 554,
  "reasoning_tokens": 307
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:
- When I sent \$3 to Cyra, they returned \$4
- When Galen sent \$3 to me, I returned \$4

This cooperative pattern seems to be working well for everyone. I'll continue with the same strategy and send \$3 to Elara, expecting similar reciprocity.

```
{'send': 3}
```

reasoning:

Let me think about this round:

Previous rounds:

- Round 1 (SENDER vs Cyra): I sent \$3, they returned \$4. I got \$6 (kept \$2, got \$4 back)
- Round 2 (RECEIVER vs Galen): They sent \$3, I returned \$4. I got \$5 (received \$9, returned \$4)

Total earnings: \$11

Now I'm facing Elara as a sender. Both times when \$3 was sent (by me and by Galen), the receiver returned \$4, which seems like a fair amount. This suggests a pattern of cooperation.

I should continue this cooperative strategy. Sending \$3 again seems reasonable:

- Shows consistency in my cooperative approach
- Not too risky
- If Elara behaves similarly to Cyra and me, they'll likely return around \$4
- This would give me \$6 this round (\$2 kept + \$4 returned)

Let me send \$3 again.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 322,
  "reasoning_tokens": 197
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the recent pattern:

- Most exchanges with \$3 sent have resulted in \$4-\$4.5 returned
- Elara was particularly generous, returning \$4.5 when I sent \$3
- Dorian showed trust by sending \$3

I'll match Elara's generosity and return \$4.5 to Dorian, maintaining the cooperative pattern.

{'return': 4.5}

Reasoning:

Let me analyze the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- In Round 3, when I sent \$3 to Elara, they returned \$4.5 - even more generous than the \$4 I've been seeing

The pattern has been:

- I sent \$3 to Cyra ? received \$4
- Galen sent \$3 to me ? I returned \$4
- I sent \$3 to Elara ? received \$4.5

Since Elara was more generous by returning \$4.5, I should consider being similarly generous. This helps build good relationships in the multi-agent setting. Returning around \$4-\$4.5 seems appropriate.

I'll return \$4.5 to match Elara's generosity and continue building cooperative relationships.

Usage: {

"input_tokens": 1318,
"output_tokens": 297,
"reasoning_tokens": 171

}

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T15:43:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_6",
"pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_7": "Galen" },

"agents": [
 "Agent_7",
 "Agent_6"],

"investor": "Agent_7",

"roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"},

"trustee": "Agent_6"

},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing a good level of trust
- I received \$9.0 after the 3x multiplier
- This creates a \$6.0 surplus (\$9 - \$3 original)

I want to be fair and build trust for potential future interactions, since this is a multi-round game. Returning a reasonable amount ensures both of us benefit from the cooperation.

If I return \$5.0:

- Hana ends with: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This rewards Hana's trust while creating mutual benefit.

{"return": 5.0}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in this round
2. The sender (Hana) sent \$3.0 out of their \$5 endowment (60%)
3. This \$3.0 was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about what would be fair and strategic:

- Hana sent 60% of their endowment, which shows a decent level of trust
- If I return nothing, I keep \$9.0 and Hana ends up with only \$2.0 ($\$5 - \3)
- If I return everything (\$9.0), Hana ends up with \$11.0 ($\$5 - \$3 + \9) and I get nothing
- A "fair" return might be to ensure both players benefit

Let me consider some options:

Option 1: Return \$4.5 (50% of what I received)

- I keep: $\$9.0 - \$4.5 = \$4.5$
- Hana ends with: $\$5 - \$3 + \$4.5 = \6.5
- Net gains: Me +\$4.5, Hana +\$1.5

Option 2: Return \$5.0

- I keep: $\$9.0 - \$5.0 = \$4.0$
- Hana ends with: $\$5 - \$3 + \$5.0 = \7.0
- Net gains: Me +\$4.0, Hana +\$2.0

Option 3: Return \$6.0

- I keep: $\$9.0 - \$6.0 = \$3.0$
- Hana ends with: $\$5 - \$3 + \$6.0 = \8.0
- Net gains: Me +\$3.0, Hana +\$3.0 (equal split of the \$6 surplus created)

- Me: Started with \$0, received \$9, returned \$6 = \$3 (gain of \$3)

Since this is a multi-round game, I should think about building trust and encouraging future cooperation. Hana's 60% send shows moderate confidence, so returning enough to give us equal gains from the surplus seems reasonable. Returning \$6 splits the \$6 surplus evenly—each of us gains \$3—which feels balanced and rewards her initial trust without leaving me worse off than I started.

That said, I could return slightly less and still maintain goodwill. Returning \$5 gives Hana a \$2 gain while I keep \$4, which is still fair given she didn't send everything. Since she sent 60%, returning about 55% of what I received feels like a solid middle ground that encourages cooperation without being overly generous. I'll return \$5.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 983,
  "reasoning_tokens": 495
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Aster), so I don't have history with them
- In the previous round, Hana sent me 60% (\$3), showing trust
- I returned a fair amount (\$5), creating mutual benefit
- This is a multi-round game where building cooperation matters

I'll send \$3 (60% of my endowment), mirroring the cooperative approach Hana showed me. This demonstrates willingness to trust and create mutual gains.

```
{'send': 3}
```

reasoning:

Now I'm the sender against a new opponent, Aster.

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana showed trust by sending 60% (\$3). I returned \$5 out of \$9, which gave them a good return.
2. This is a new pairing with Aster. I don't know how they'll behave.
3. Since this is a multi-round game and I'll potentially interact with different agents multiple times, I should think about reputation and reciprocity.
4. If I send money, it gets tripled. If I send \$3:
 - Aster receives \$9
 - If they return \$5 (similar to what I did), I'd end with $\$2 + \$5 = \$7$
 - If they return less, I could end up worse off
5. Sending nothing is safe but doesn't build trust or create mutual gains.
6. I should probably mirror the cooperative behavior I received. Hana sent 60% (\$3), which seemed like a reasonable level of trust.

Given that:

- This is an iterated game with multiple rounds
- Building cooperation can lead to better long-term outcomes
- I should signal willingness to cooperate but also be somewhat cautious with a new partner

I'll send \$3 (60%), matching what Hana sent to me. This shows I'm willing to cooperate and trust, while not being overly risky.

usage:

```
{
  "input_tokens": 896,
  "output_tokens": 457,
  "reasoning_tokens": 321
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$10.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same level of trust that Hana showed me in Round 1

I want to be consistent in my reciprocation. When Hana sent \$3, I returned \$5, which was fair and created mutual benefit. I'll do the same for Cyra.

If I return \$5.0:

- Cyra ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- In the last round, when I sent \$3.0 to Aster, they returned \$4.0
- This is a multi-round game, so building cooperation is important

I've established a pattern:

- Round 1 (as receiver): Hana sent \$3, I received \$9, returned \$5
- Round 2 (as sender): I sent \$3 to Aster, they received \$9, returned \$4
- Round 3 (as receiver): Cyra sent \$3, I received \$9, need to decide return

Cyra showed the same level of trust as Hana did (both sent \$3). I should be consistent in my reciprocation. When Hana sent \$3, I returned \$5. I should do the same for Cyra.

If I return \$5.0:

- Cyra ends with: $\$5 - \$3 + \$5 = \7
- I keep: $\$9 - \$5 = \$4$

This is fair and consistent with my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1173,
  "output_tokens": 452,
  "reasoning_tokens": 221
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$14.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- I've consistently sent \$3 (60%) when I'm the sender
- This has worked well - Aster returned \$4 when I sent \$3
- When others send me \$3, I return \$5
- This cooperative strategy is building positive outcomes

I'll maintain this consistent approach with Finn, as it demonstrates trustworthiness and has been effective.

```
{'send': 3}
```

Reasoning:

Let me analyze my situation:

Round 1: I was RECEIVER against Hana

- Hana sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 2: I was SENDER against Aster

- I sent \$3 (60%)
- Aster returned \$4
- My payoff: $\$6 (\$5 - \$3 + \$4)$

Round 3: I was RECEIVER against Cyra

- Cyra sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 4: I am SENDER against Finn (new opponent)

Pattern so far:

- When people send me \$3, I return \$5 (consistent reciprocation)
- When I sent \$3 to Aster, they returned \$4 (slightly less than what I return, but still fair)
- I've been playing cooperatively at 60% send rate

Total earnings: \$14

For this round with Finn (new opponent):

- I should maintain my cooperative strategy
- Sending \$3 (60%) has been working well
- It builds trust and creates good outcomes

I'll continue with \$3.

```
Usage: {
  "input_tokens": 1471,
  "output_tokens": 381,
  "reasoning_tokens": 226
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T15:44:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_6": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of her endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- This is the first round, so establishing trust and cooperation is important

I want to reciprocate Elara's trust and encourage continued cooperation in future rounds. Elara took a risk by sending a substantial portion of her endowment.

A fair return that:

1. Rewards her trust
2. Shares the gains from cooperation
3. Encourages future collaboration

I'll return approximately half of what I received, which seems like a reasonable reciprocal gesture.

{'return': 4.5}

reasoning:

This is Round 1 of an iterated trust game. Let me think through this situation:

I'm the RECEIVER:

- The sender (Elara) sent me \$3.0 out of their \$5 endowment
- This gets multiplied by 3x, so I receive \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Key considerations:

1. This is an iterated game with multiple rounds
2. Elara sent \$3.0 (60% of her endowment), which is a fairly generous amount
3. If I want to build trust and encourage cooperation in future rounds, I should return a fair amount
4. If I return nothing, Elara will likely not trust me in future rounds
5. The typical fair split would consider that Elara kept \$2.0 and sent \$3.0

Let me think about what's fair:

- Elara ends up with: $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- I end up with: $\$9 - \text{returned}$

A 50-50 split of the total wealth would mean:

- Total created: $\$5$ (original) + $\$9$ (received) = $\$14$
- 50-50 would be $\$7$ each
- For Elara to have $\$7$: $\$2 + \text{returned} = \7 , so $\text{returned} = \$5$
- For me to have $\$7$: $\$9 - \text{returned} = \7 , so $\text{returned} = \$2$

Wait, that doesn't match up. Let me recalculate:

- Elara starts with $\$5$, sends $\$3$, so has $\$2$ left
- I receive $\$3 \times 3 = \9
- Total money in the system: $\$2$ (Elara has) + $\$9$ (I have) = $\$11$

If we split equally: each should get $\$5.50$

- For Elara to get $\$5.50$: she needs $\$5.50 - \$2 = \$3.50$ returned

- For me to get $\$5.50$: I keep $\$9 - \$3.50 = \$5.50$

So returning $\$3.50$ creates an equal split. I could also consider reciprocity-returning 60% of what I received ($\$5.40$) since Elara sent 60% of her endowment, or simply matching her $\$3$ contribution to restore her original stake.

If I return $\$3$, I'd end up with $\$6$ while she'd have $\$5$. But thinking about long-term incentives, a more generous return between $\$4$ - $\$5$ would signal trustworthiness and encourage future rounds. I'll return $\$4.50$ -this gives Elara $\$6.50$ and me $\$4.50$, slightly favoring her since she took the initial risk, which should build cooperation going forward.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1035,
  "reasoning_tokens": 518
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Elara. They sent $\$3.0$ to you, it became $\$9.0$, and you returned $\$4.5$. Your payoff was $\$4.5$.
Your total earnings across all rounds are $\$4.5$.

This round, you are the SENDER against Hana. You have $\$5$. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is a new opponent (Hana), Round 2 overall
- In Round 1, Elara showed trust by sending 60% ($\$3$), and I reciprocated by returning 50% of what I received
- I want to maintain a cooperative strategy and build trust with Hana

I'll send a similar proportion to what Elara sent me, demonstrating consistent cooperative behavior.

{'send': 3}

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Hana (Agent_8).

In Round 1, I was the RECEIVER against Elara:

- Elara sent me $\$3.0$ (60% of her endowment)
- I received $\$9.0$
- I returned $\$4.5$ (50% of what I received)
- My payoff was $\$4.5$

- Elara's payoff was: $\$5 - \$3 + \$4.5 = \6.5

Now I'm playing with a new opponent (Hana) and I'm the sender.

Strategy considerations:

1. This is only Round 2, so building cooperative relationships is important
2. In the previous round as receiver, someone showed trust by sending 60%, and I reciprocated by returning 50%
3. As a sender, I should also show trust to encourage cooperation
4. Sending a substantial amount signals willingness to cooperate

Since Elara sent \$3.0 (60%) and I reciprocated fairly well, I should maintain a similar cooperative strategy. Sending around \$3 seems reasonable to establish trust while not being overly risky in early rounds.

Let me send \$3 to match what Elara sent to me, showing consistent cooperative behavior.

usage:

```
{
  "input_tokens": 861,
  "output_tokens": 407,
  "reasoning_tokens": 267
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: Elara trusted me with \$3, I reciprocated fairly
- Round 2: I trusted Hana with \$3, they returned generously (\$5)
- The cooperative strategy has been working well

I'll continue with a consistent, trusting approach with Dorian.

```
{'send': 3}
```

reasoning:

Let me analyze the situation:

- Round 1: I was receiver, Elara sent \$3 (60%), I received \$9, returned \$4.5 (50%). My payoff: \$4.5
- Round 2: I was sender, sent \$3 (60%) to Hana, they received \$9, returned \$5 (55.6%). My payoff: \$7.0
- Total earnings: \$11.5

Hana was very generous, returning more than half of what they received. This shows the cooperative strategy is working well.

Now I'm facing a new opponent, Dorian, in Round 3 as the sender.

My strategy so far has been:

1. As receiver: return about 50% of what I receive
2. As sender: send 60% (\$3)

This has been working well. Elara trusted me with \$3, and I reciprocated appropriately. Hana rewarded my trust with a generous return.

I should continue this trusting, cooperative approach with Dorian. Sending \$3 again shows consistency and encourages cooperation.

usage:

```
{
  "input_tokens": 1107,
```

```
"output_tokens": 339,  
"reasoning_tokens": 218  
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This matches what Elara sent me in Round 1, where I returned \$4.5

I'll maintain my consistent, cooperative strategy and return the same proportion.

```
{'return': 4.5}
```

Reasoning:

Looking at the situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- My strategy has been cooperative and consistent

Looking at my previous rounds:

- Round 1 as RECEIVER: Elara sent \$3, I received \$9, I returned \$4.5 (50%)
- Round 2 as SENDER: I sent \$3, Hana returned \$5 (generous)
- Round 3 as SENDER: I sent \$3, Dorian returned \$4.5 (fair)

Galen sent the same amount (\$3) as Elara did in Round 1, and I returned \$4.5 then. For consistency and to maintain a cooperative reputation, I should return a similar amount.

```
{'return': 4.5}
```

Usage: {

```
"input_tokens": 1345,  
"output_tokens": 299,  
"reasoning_tokens": 158  
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T15:44:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",  
"move_index": 6,  
"opponent_id": "Agent_3",  
"pairing": {  
  "agent_names": {  
    "Agent_3": "Cyra",  
    "Agent_5": "Elara"  
  },  
  "agents": [  
    "Agent_5",  
    "Agent_3"  
  ],  
  "investor": "Agent_5",  
  "roles": {  
    "Agent_3": "trustee",  
    "Agent_5": "investor"  
  },  
  "trustee": "Agent_3"
```

```

},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn while also being strategic.

Since the game is iterated and we may interact in future rounds, sending a moderate amount shows trust and creates potential for mutual benefit. If I send \$3, Finn will receive \$9 (after the 3x multiplier). This gives Finn a strong incentive to return a fair amount, as we could both end up better off than our starting positions.

A reasonable return from Finn (say \$4-5) would leave us both better off, and establish a pattern of cooperation for potential future interactions.

```
{'send': 3}
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Elara) and my opponent is Finn (Agent_6).

Let me think through the game theory here:

- I receive \$5
- I can send \$0-\$5 to Finn
- What I send gets multiplied by 3x
- Finn then decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: received - returned

This is a classic trust game. The economically rational strategy depends on:

1. Whether this is truly iterated (will I play with Finn again?)
2. What I believe about Finn's likely behavior
3. Whether I want to establish cooperation

Since this is Round 1 and the game is iterated with multiple rounds, there's value in:

- Establishing trust and cooperation
- Testing Finn's reciprocity
- Not being too exploitable but also not being too selfish

A common opening strategy in trust games is to send a moderate amount - not nothing (which shows no trust and prevents cooperation), but also not everything (which might be exploited).

Sending \$3 out of \$5 seems like a reasonable moderate approach:

- It shows I'm willing to trust and cooperate
- It gives Finn \$9 (\$3 × 3)
- If Finn returns even \$3-4, we're both better off than if I sent nothing
- It's not so much that I'm completely vulnerable

Let me consider the outcomes:

- If I send \$3, Finn gets \$9
- I have \$2 left, plus whatever Finn returns
- If Finn returns \$4-5, I end up with \$6-7 (better than my initial \$5)
- And Finn keeps \$4-5 (better than their initial \$0)
- This creates value for both of us

Alternatively:

- Send \$2.5: More conservative, Finn gets \$7.5
- Send \$4: More trusting, Finn gets \$12

- Send \$5: Maximum trust, Finn gets \$15

I think \$3 is a good balance for the first round. It's cooperative without being naive.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 643,
  "reasoning_tokens": 466
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation, Briar sent me the same amount (\$3) that I sent to Finn in Round 1. Finn returned \$4.5 to me, which was exactly half of what they received - a fair and cooperative response.

To maintain a reciprocal and cooperative pattern, I should treat Briar similarly. Returning \$4.5 means:

- Briar gets a payoff of \$6.5 (\$2 kept + \$4.5 returned)
- I keep \$4.5 (\$9 received - \$4.5 returned)
- Both of us benefit from the cooperation

This establishes trust and encourages continued cooperation in future rounds.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Finn, who received \$9 and returned \$4.5. This was a fair and cooperative response - they returned exactly half of what they received, which left both of us better off.
2. Now I'm the receiver and Briar sent me \$3 (same amount I sent to Finn). I received \$9.
3. I should consider:
 - Reciprocity: Finn returned \$4.5 when receiving \$9, which was cooperative
 - Reputation: If I want to build trust for future rounds, I should be fair
 - Strategic thinking: Being too greedy might discourage cooperation in future rounds
 - Briar took the same risk I took with Finn
4. Following the cooperative pattern that Finn established, returning around \$4.5 would be fair and consistent. This would give Briar a payoff of \$2 + \$4.5 = \$6.5 (same as what I got in round 1).
5. I could return slightly more or less, but \$4.5 seems like a reasonable, fair amount that:
 - Rewards Briar's trust
 - Establishes me as a cooperative player
 - Keeps the game balanced
 - Mirrors what was done to me

I'll return \$4.5.

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 481,
  "reasoning_tokens": 284
}
```

Message 6 role=user**content:**

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at the pattern so far, there's been consistent cooperation with \$3 being sent and \$4.5 being returned (half of the tripled amount). This has worked well for everyone involved.

I'll continue this cooperative pattern with Aster, returning the same amount I returned to Briar when they sent \$3.

```
{'return': 4.5}
```

reasoning:

This is the third round, and again I'm playing as the RECEIVER. Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 1: I was SENDER, sent \$3 to Finn, they returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER, Briar sent \$3, I returned \$4.5. I got \$4.5.
- Round 3: I'm RECEIVER, Aster sent \$3, received \$9.

There's been a consistent pattern: people are sending \$3, and the fair return has been \$4.5 (half of what's received). This has worked well and established cooperation.

I should maintain this pattern of returning \$4.5 when someone sends \$3. This is:

1. Consistent with what I've been doing and receiving
2. Fair (50% return on the multiplied amount)
3. Encourages continued cooperation
4. Gives both parties a positive outcome

If I return \$4.5:

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1144,
  "output_tokens": 387,
  "reasoning_tokens": 236
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern across all rounds so far, there's been consistent cooperation with agents sending \$3 and receivers returning \$4.5. This has created a stable, mutually beneficial norm.

I'll continue this pattern with Cyra, sending \$3 as I did with Finn and as others have done with me.

```
{'send': 3}
```

Reasoning:

Let me review the pattern so far:

Round 1: I was SENDER vs Finn

- I sent \$3
- Finn received \$9
- Finn returned \$4.5
- My payoff: \$6.5

Round 2: I was RECEIVER vs Briar

- Briar sent \$3
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 3: I was RECEIVER vs Aster

- Aster sent \$3
- I received \$9

- I returned \$4.5
- My payoff: \$4.5

So far, there's been a consistent pattern:

- Everyone sends \$3
- Everyone returns \$4.5 (half of the tripled amount)
- Senders get \$6.5, Receivers get \$4.5

Now I'm a SENDER again vs Cyra. Should I continue the pattern and send \$3?

The pattern has been working well and establishing trust. If I send \$3 and Cyra follows the pattern by returning \$4.5, I'll get \$6.5 again.

I should continue with the cooperative norm that has been established.

```
Usage: {
  "input_tokens": 1367,
  "output_tokens": 358,
  "reasoning_tokens": 218
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T15:44:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0
- This is Round 1, so establishing a cooperative reputation is important

Aster currently has \$5.0 - \$3.0 = \$2.0 remaining.

If I return \$4.0:

- Aster ends with: \$2.0 + \$4.0 = \$6.0 (a \$1.0 profit)
- I keep: \$9.0 - \$4.0 = \$5.0

This rewards Aster's trust, ensures they profit from cooperating, and establishes me as a reciprocal partner for future rounds.

{'return': 4.0}

reasoning:

Let me analyze this situation:

- I am the RECEIVER (Cyra)
- My opponent is Aster (Agent_1), who is the SENDER
- Aster sent \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Aster showed trust by sending 60% of their endowment (\$3.0)
2. If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
3. If I return fairly, I should consider:
 - Aster gave up \$3.0 to send to me
 - This created \$9.0 total for me
 - A fair split might be to ensure we both benefit

Common strategies in trust games:

- Return enough to show reciprocity and encourage future cooperation
- A typical approach is to return about 1/3 to 1/2 of what was received
- Or return enough so the sender gets more than they started with

Let me calculate some options:

- If I return \$3.0: Aster gets $\$2.0 + \$3.0 = \$5.0$ (breaks even), I keep \$6.0
 - If I return \$4.0: Aster gets $\$2.0 + \$4.0 = \$6.0$ (gains \$1), I keep \$5.0
 - If I return \$4.5: Aster gets $\$2.0 + \$4.5 = \$6.5$ (gains \$1.5), I keep \$4.5
 - If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0
- If I return \$5.0: Aster gets $\$2.0 + \$5.0 = \$7.0$ (gains \$2), I keep \$4.0

Given Aster's substantial trust, returning around \$4.0-\$4.5 balances reciprocity with self-interest-it rewards their generosity while keeping a reasonable share for myself. I'll return \$4.0, which gives Aster a \$1 gain and leaves me with \$5.0. Since this appears to be an iterated game, establishing cooperative behavior now could benefit both of us in future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 929,
  "reasoning_tokens": 425
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- I'm now the sender with \$5
- In Round 1, Aster sent me \$3.0 (60%) showing trust, and I returned \$4.0, establishing cooperation
- This is my first interaction with Dorian

I'll follow the cooperative pattern established in Round 1. Sending \$3.0 (60% of my endowment) demonstrates trust and gives Dorian the opportunity to reciprocate, which should benefit us both.

```
{'send': 3.0}
```

reasoning:

Now I'm the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$3.0 (60% of their endowment). I returned \$4.0, which gave them a profit and established me as cooperative.
2. Now I'm facing a new opponent, Dorian. I have no history with them.
3. As the sender, I need to decide how much to trust Dorian. The key question is: will they reciprocate?
4. Sending has high potential upside (3x multiplier) but requires trust that they'll return enough.
5. Since this is early in the game (Round 2) and I want to establish cooperative patterns, I should send a reasonable amount.

6. Sending \$3.0 (like Aster did) seems like a fair, cooperative choice that:
- Mirrors what Aster sent me
 - Shows willingness to trust
 - Gives Dorian the opportunity to reciprocate
 - Would give them \$9.0 to work with
7. If Dorian returns around \$4-5 (similar to what I did), we'll both profit.

Let me send \$3.0 to match the cooperative precedent from the first round.

usage:

```
{  
  "input_tokens": 888,  
  "output_tokens": 429,  
  "reasoning_tokens": 277  
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

- Round 1: Aster sent me \$3.0 (60%), I returned \$4.0 - good cooperation
- Round 2: I sent Dorian \$3.0 (60%), they returned \$4.5 - even better cooperation, I profited \$1.5

The cooperative strategy of sending 60% has been working well, with both opponents reciprocating generously. I'll continue this approach with Galen.

{'send': 3.0}

reasoning:

Let me analyze the situation:

- Round 1: I was receiver. Aster sent me \$3.0, I received \$9.0, returned \$4.0. My payoff: \$5.0
- Round 2: I was sender against Dorian. I sent \$3.0, they received \$9.0, returned \$4.5. My payoff: \$2.0 + \$4.5 = \$6.5
- Total earnings: \$11.5

Now in Round 3, I'm sender against Galen, a new opponent.

The pattern so far shows good cooperation:

- Aster sent 60% (\$3.0), I returned \$4.0 (44% of what I received)
- I sent 60% (\$3.0) to Dorian, they returned \$4.5 (50% of what they received)

Both interactions have been profitable and cooperative. Dorian was even slightly more generous in returning funds.

For Round 3 with Galen, I should continue this cooperative strategy. Sending \$3.0 (60%) has worked well so far and establishes trust. This gives Galen the opportunity to reciprocate and continue the cooperative pattern.

I'll send \$3.0 again.

usage:

```
{  
  "input_tokens": 1152,  
  "output_tokens": 411,  
  "reasoning_tokens": 231  
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
 Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
 Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history:

- Round 1: As receiver from Aster (\$3.0 sent), I returned \$4.0
- Round 2: As sender to Dorian (\$3.0 sent), they returned \$4.5
- Round 3: As sender to Galen (\$3.0 sent), they returned \$5.0

Elara has shown the same level of trust as others by sending \$3.0 (60%). The pattern shows increasing generosity in this cooperative environment. I'll return \$4.5, which rewards Elara's trust and gives them a solid profit of \$1.5.

```
{'return': 4.5}
```

Reasoning:

Let me analyze the situation:

- Round 1: As receiver, Aster sent me \$3.0, I returned \$4.0 (gave them a \$1 profit)
- Round 2: As sender, I sent Dorian \$3.0, they returned \$4.5 (I profited \$1.5)
- Round 3: As sender, I sent Galen \$3.0, they returned \$5.0 (I profited \$3.0)
- Round 4: As receiver, Elara sent me \$3.0, I receive \$9.0

The pattern shows:

- When I've been receiver, I returned \$4.0
- When I've been sender with \$3.0, I've received back \$4.5 and \$5.0

Elara has shown the same trust level as the others (sending \$3.0, which is 60%). I should reciprocate fairly.

Elara currently has: $\$5.0 - \$3.0 = \$2.0$

If I return \$4.0:

- Elara gets: $\$2.0 + \$4.0 = \$6.0$ (profit of \$1.0)
- I keep: $\$9.0 - \$4.0 = \$5.0$

This is consistent with my Round 1 behavior and maintains a cooperative reputation. The others have been even more generous when I sent, but returning \$4.0 is still fair and rewards trust.

However, given that Dorian returned \$4.5 and Galen returned \$5.0 to me, maybe I should be slightly more generous? Let me think...

Actually, I returned \$4.0 in Round 1, which was my only previous receiver round. That established my baseline. I should probably stick with that for consistency, unless I want to match the increasing generosity I've experienced.

Let me go with \$4.5 as a middle ground between my previous \$4.0 and the more generous returns I've received.

Returning \$4.5 splits the difference-Elara ends with \$6.5 (a \$1.5 gain) while I keep \$4.5, which mirrors the reciprocity I've experienced and signals I'm matching their cooperative spirit.

Usage: {

```
"input_tokens": 1426,
"output_tokens": 705,
"reasoning_tokens": 413
```

}

ROUND 5

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_4 (Dorian)=Sender, Agent_6 (Finn)=Receiver
dyad_2: Agent_8 (Hana)=Sender, Agent_1 (Aster)=Receiver
dyad_3: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_4: Agent_3 (Cyra)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_4 (Dorian) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_8 (Hana) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_1 (Aster)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 (Elara) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_5 (Elara)=\$7, Agent_7 (Galen)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_3 (Cyra) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_2 (Briar)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 26.5,
"Agent_2": 26.5,
"Agent_3": 29.5,
"Agent_4": 28.5,
"Agent_5": 29.0,
"Agent_6": 27.0,
"Agent_7": 24.5,
"Agent_8": 28.5

}

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-06-04T15:44:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_6",
"pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_4",
 "Agent_6"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play an iterated game,
      consisting of multiple rounds, and may also be asked to write myths.

      GAME:
      The game consists of two roles: sender and receiver. In each round, you will play the role
      of either the sender or the receiver. Your current role will be specified in the round
      prompt.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - You will be told your opponent's name in each round.
      - Your name in this experiment is Dorian.
  Message 2 role=user
    content:
      Your name in this experiment is Dorian.
      Your opponent this round is Cyra (Agent_3).

      Round 2

      In your most recent previous game round, you were the SENDER against Briar. You sent $3.0,
      it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
      Your total earnings across all rounds are $6.5.

      This round, you are the RECEIVER against Cyra. They sent you $3.0, so you received $9.0. H
      ow much do you return? (0-9.0)
      Respond exactly as JSON: {'return': <amount>}
  Message 3 role=assistant
    content:
      Looking at the previous round, Briar returned $4.5 out of the $9 they received (50%), whic
      h created a fair outcome. I'll reciprocate this cooperative behavior.

      If I return $4.5:
      - Cyra gets: $5 - $3 + $4.5 = $6.5
      - I get: $9 - $4.5 = $4.5

      This matches the fair standard from the previous round and signals I'm trustworthy for fut

```

ure interactions.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Briar
- I sent \$3, which became \$9 for them
- Briar returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Briar's payoff: $\$9 - \$4.5 = \$4.5$

So Briar returned exactly half of what they received, which is a fair 50-50 split of the gains.

Round 2:

- I'm now the receiver against Cyra
- Cyra sent me \$3 (same amount I sent to Briar)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

Since this is an iterated game and I might play with Cyra again, I should:

1. Reciprocate the cooperative behavior shown by Briar
2. Signal that I'm a fair player
3. Consider that returning a fair amount might encourage future cooperation

Briar returned \$4.5 out of \$9 (50%). This seems like a reasonable standard.

If I return \$4.5:

- Cyra's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This creates a symmetric outcome to what happened with Briar, which seems fair.

I'll return \$4.5 to match the cooperative standard set in the previous round.

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 502,
  "reasoning_tokens": 282
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Following the same cooperative pattern as the previous round, I'll return \$4.5 out of the \$9 received. This maintains a fair 50% return rate and continues to build trust.

If I return \$4.5:

- Finn gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

```
'''json
{"return": 4.5}
'''
```

reasoning:

Finn sent me \$3.0, which became \$9.0 for me.

In the previous round, I returned \$4.5 out of \$9.0 when Cyra sent me \$3.0. This was a 50% return rate, which seemed fair and matched what Briar had done when I was the sender in Round 1.

To maintain consistency and continue building a reputation for fairness and cooperation, I should return \$4.5 again.

If I return \$4.5:

- Finn gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the pattern of fair reciprocation.

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 286,
  "reasoning_tokens": 128
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

I'll continue my cooperative strategy. Based on the pattern established in previous rounds where sending \$3 has been met with fair returns (~50%), I'll send \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the pattern so far:

- Round 1: I was SENDER, sent \$3 to Briar, got back \$4.5. Payoff: \$6.5
- Round 2: I was RECEIVER, got \$9 from Cyra, returned \$4.5. Payoff: \$4.5
- Round 3: I was RECEIVER, got \$9 from Finn, returned \$4.5. Payoff: \$4.5
- Total earnings: \$15.5

The pattern I've established is:

- When sending, I sent \$3
- When receiving, I return \$4.5 out of \$9 (50%)

Everyone seems to be cooperating well. Briar returned 50% when I was the sender. I've been consistent in returning 50% when receiving.

Now I'm the sender again against Aster (a new opponent I haven't played with yet). I should continue my cooperative strategy and send \$3 to maintain consistency and signal trustworthiness.

If I send \$3:

- Aster receives \$9
- If they return around 50% like the pattern, I'd get back ~\$4.5
- My payoff would be: $\$5 - \$3 + \$4.5 = \6.5

This seems fair and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 367,
}
```



```

    "reasoning_tokens": 243
  }
  Message 8 role=user
  content:
    Your name in this experiment is Dorian.
    Your opponent this round is Finn (Agent_6).

    Round 5

    In your most recent previous game round, you were the SENDER against Aster. You sent $3.0,
    it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
    Your total earnings across all rounds are $22.0.

    This round, you are the SENDER against Finn. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
  Assistant output:
    I played with Finn before in Round 3, where they sent me $3 and I returned $4.5. They were cooperative, so I'll reciprocate with the same cooperative approach.

    '''json
    {"send": 3}
    '''
  Reasoning:
    I'm now playing as SENDER against Finn. In Round 3, I was the RECEIVER against Finn - they sent me $3.0, and I returned $4.5 (50% of the $9 received).

    This is the first time I'm the sender with Finn. Since Finn was cooperative by sending $3 in our previous interaction, and I've established a pattern of sending $3 consistently (which has worked well so far), I should continue with that strategy.

    My track record:
    - Round 1: Sender to Briar, sent $3, got back $4.5 ? payoff $6.5
    - Round 2: Receiver from Cyra, received $9 (from $3), returned $4.5 ? payoff $4.5
    - Round 3: Receiver from Finn, received $9 (from $3), returned $4.5 ? payoff $4.5
    - Round 4: Sender to Aster, sent $3, got back $4.5 ? payoff $6.5

    Everyone has been cooperative, and sending $3 with ~50% returns has been working well. I should continue this strategy with Finn.
  Usage: {
    "input_tokens": 1358,
    "output_tokens": 338,
    "reasoning_tokens": 221
  }

  GAME - Agent_6 - Receiver - dyad_1
  Timestamp: 2026-06-04T15:44:36
  Model: anthropic/claude-sonnet-4.5
  Temperature: 0.8
  Metadata: {
    "dyad_id": "dyad_1",
    "move_index": 1,
    "opponent_id": "Agent_4",
    "pairing": {
      "agent_names": {
        "Agent_4": "Dorian",
        "Agent_6": "Finn"
      },
      "agents": [
        "Agent_4",
        "Agent_6"
      ],
      "investor": "Agent_4",
      "roles": {
        "Agent_4": "investor",
        "Agent_6": "trustee"
      },
      "trustee": "Agent_6"
    },
    "role": "trustee",
    "role_label": "Receiver",

```

```

"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is a new opponent (Hana), Round 2 overall
- In Round 1, Elara showed trust by sending 60% (\$3), and I reciprocated by returning 50% of what I received
- I want to maintain a cooperative strategy and build trust with Hana

I'll send a similar proportion to what Elara sent me, demonstrating consistent cooperative behavior.

```
{'send': 3}
```

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Hana (Agent_8).

In Round 1, I was the RECEIVER against Elara:

- Elara sent me \$3.0 (60% of her endowment)
- I received \$9.0
- I returned \$4.5 (50% of what I received)
- My payoff was \$4.5
- Elara's payoff was: $\$5 - \$3 + \$4.5 = \6.5

Now I'm playing with a new opponent (Hana) and I'm the sender.

Strategy considerations:

1. This is only Round 2, so building cooperative relationships is important
2. In the previous round as receiver, someone showed trust by sending 60%, and I reciprocated by returning 50%
3. As a sender, I should also show trust to encourage cooperation
4. Sending a substantial amount signals willingness to cooperate

Since Elara sent \$3.0 (60%) and I reciprocated fairly well, I should maintain a similar cooperative strategy. Sending around \$3 seems reasonable to establish trust while not being overly risky in early rounds.

Let me send \$3 to match what Elara sent to me, showing consistent cooperative behavior.

usage:

```
{
  "input_tokens": 861,
  "output_tokens": 407,
  "reasoning_tokens": 267
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: Elara trusted me with \$3, I reciprocated fairly
- Round 2: I trusted Hana with \$3, they returned generously (\$5)
- The cooperative strategy has been working well

I'll continue with a consistent, trusting approach with Dorian.

```
{'send': 3}
```

reasoning:

Let me analyze the situation:

- Round 1: I was receiver, Elara sent \$3 (60%), I received \$9, returned \$4.5 (50%). My payoff: \$4.5
- Round 2: I was sender, sent \$3 (60%) to Hana, they received \$9, returned \$5 (55.6%). My payoff: \$7.0
- Total earnings: \$11.5

Hana was very generous, returning more than half of what they received. This shows the cooperative strategy is working well.

Now I'm facing a new opponent, Dorian, in Round 3 as the sender.

My strategy so far has been:

1. As receiver: return about 50% of what I receive
2. As sender: send 60% (\$3)

This has been working well. Elara trusted me with \$3, and I reciprocated appropriately. Hana rewarded my trust with a generous return.

I should continue this trusting, cooperative approach with Dorian. Sending \$3 again shows consistency and encourages cooperation.

usage:

```
{
  "input_tokens": 1107,
  "output_tokens": 339,
  "reasoning_tokens": 218
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This matches what Elara sent me in Round 1, where I returned \$4.5

I'll maintain my consistent, cooperative strategy and return the same proportion.

{'return': 4.5}

reasoning:

Looking at the situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- My strategy has been cooperative and consistent

Looking at my previous rounds:

- Round 1 as RECEIVER: Elara sent \$3, I received \$9, I returned \$4.5 (50%)
- Round 2 as SENDER: I sent \$3, Hana returned \$5 (generous)
- Round 3 as SENDER: I sent \$3, Dorian returned \$4.5 (fair)

Galen sent the same amount (\$3) as Elara did in Round 1, and I returned \$4.5 then. For consistency and to maintain a cooperative reputation, I should return a similar amount.

{'return': 4.5}

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 299,
  "reasoning_tokens": 158
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3

.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Dorian sent \$3.0, showing trust
- I received \$9.0 after the multiplier
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5 to me
- Dorian has been fair and reciprocal

I'll maintain the same fair reciprocal strategy.

{'return': 4.5}

Reasoning:

Looking at the situation:

- Dorian sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is my second interaction with Dorian
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5

Since Dorian returned \$4.5 when I sent them \$3.0, I should reciprocate with the same fair behavior now that they've sent me \$3.0.

I'll return \$4.5, maintaining consistency and reciprocity.

```
Usage: {  
  "input_tokens": 1341,  
  "output_tokens": 237,  
  "reasoning_tokens": 111  
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-06-04T15:44:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_8",  
      "Agent_1"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****What happened last round:****

- I sent \$3 to Galen
- Galen returned \$5 to me (a fair amount)

****Current situation:****

- Finn sent me \$3 (exactly what I sent to Galen)
- I received \$9
- I should decide what's fair to return

Since Finn showed the same level of trust in me that I showed in Galen, and Galen returned \$5 which felt fair, I'll mirror that cooperative behavior. Returning \$5 gives Finn a total payoff of \$7 (same as mine from round 1), while I keep \$4 (same as Galen kept).

This establishes a fair, reciprocal pattern.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Galen
- Galen received \$9
- Galen returned \$5 to me
- My payoff: $\$5 - \$3 + \$5 = \7
- Galen's payoff: $\$9 - \$5 = \$4$

So Galen was moderately cooperative, returning more than I sent originally but keeping more than half of what they received.

Now Round 2:

- Finn sent me \$3 (same as I sent to Galen!)
- I received \$9
- I need to decide how much to return

Considerations:

1. Reciprocity: Finn showed the same level of trust I showed to Galen (\$3)
2. Fairness: I could mirror what Galen did to me (return \$5), which seems like a reasonable cooperative standard
3. Building reputation: In a multi-agent setting, being seen as trustworthy and fair matters
4. Future rounds: This is an iterated game, and reputation effects could help me

If I return \$5:

- Finn gets: $\$5 - \$3 + \$5 = \7 (same as what I got in round 1)
- I keep: $\$9 - \$5 = \$4$ (same as what Galen kept)

This creates symmetry and seems fair. It's a reasonable cooperative strategy that mirrors what was done to me.

Alternatively, I could be more generous to establish a reputation for generosity, or less generous to maximize my immediate payoff. But \$5 seems like a balanced, fair amount that:

- Rewards Finn's trust
- Is similar to what Galen did (establishing a norm)
- Keeps the game mutually beneficial
- Maintains my reputation as fair

I'll return \$5.

usage:

```
{
  "input_tokens": 865,
  "output_tokens": 614,
  "reasoning_tokens": 371
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

****Round 1 (vs Galen):**** I sent \$3, received \$5 back - good cooperation

****Round 2 (vs Finn):**** Finn sent \$3, I returned \$5 - maintained fair reciprocity

Both rounds showed positive cooperation with the \$3 send amount. This amount:

- Creates significant value (becomes \$9)
- Allows for mutual benefit
- Has proven successful
- Maintains consistency in my strategy

I'll continue with the same approach against Briar.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history:

Round 1: I was SENDER vs Galen

- I sent \$3
- Galen received \$9, returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9, returned \$5
- My payoff: $\$9 - \$5 = \$4$

Total earnings: \$11

Now Round 3: I'm SENDER vs Briar

The pattern so far has been positive:

- I sent \$3 to Galen, got \$5 back (good return)
- Finn sent \$3 to me, I returned \$5 (reciprocating the cooperative behavior)

This suggests a cooperative norm is emerging. Sending \$3 has worked well for me. I could:

1. Continue with \$3 (consistent, proven strategy)
2. Send more to signal even stronger cooperation
3. Send less to protect myself

Given that:

- The \$3 send amount has been successful
- Both interactions showed good reciprocity
- I'm building a reputation in this multi-agent setting
- Consistency can signal trustworthiness

I'll stick with \$3 again. It's working and establishes me as a consistent cooperator.

usage:

```
{
  "input_tokens": 1189,
  "output_tokens": 444,
  "reasoning_tokens": 263
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my interaction with Briar:

****Round 3:**** I sent \$3 to Briar, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 4 (now):**** Briar sent \$3 to me (matching my trust)
- I received \$9

Briar showed reasonable cooperation by returning \$4.5. Now they've shown the same level of trust in me. I'll reciprocate by matching their behavior exactly - returning \$4.5 creates a consistent, fair pattern between us.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at what happened:

****Round 3 (vs Briar):**** I sent \$3, Briar returned \$4.5 to me
- My payoff was \$6.5 ($\$5 - \$3 + \4.5)
- Briar kept \$4.5 ($\$9 - \4.5)

****Round 4 (vs Briar again):**** Briar sent \$3, I received \$9

So Briar is meeting me again! Last time I sent \$3 and they returned \$4.5. Now they're sending \$3 to me.

The return of \$4.5 was slightly less generous than the \$5 that Galen returned to me and that I returned to Finn. But it's still cooperative and fair - we both got reasonable payoffs.

Now Briar has sent me \$3, showing the same trust. I should reciprocate fairly.

Options:

1. Return \$5 (what I've been doing with others) - this would give Briar \$7, me \$4
2. Return \$4.5 (matching what Briar did) - this would give Briar \$6.5, me \$4.5

Matching what Briar did (\$4.5) seems like the fairest reciprocal move - it shows I'm willing to match their level of cooperation exactly. This creates a clear tit-for-tat dynamic.

Actually, let me think about this more carefully. When Briar was the receiver:

- They got \$9
- They returned \$4.5
- They kept \$4.5

Now when I'm the receiver:

- I get \$9
- If I return \$4.5, I keep \$4.5
- Briar gets $\$5 - \$3 + \$4.5 = \6.5

So by returning \$4.5, I'm mirroring their exact behavior. This seems fair and establishes a consistent reciprocal pattern with Briar.

usage:

```
{
  "input_tokens": 1481,
  "output_tokens": 608,
  "reasoning_tokens": 350
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- **Rounds 1-4:** Consistently sent \$3 as sender**
- Returns have ranged from \$4.5-\$5
 - Average payoff as sender: \$6.75
 - Strategy has built cooperative relationships

This is my first interaction with Aster. I'll maintain my established strategy of sending \$3, which:

- Demonstrates trust
- Creates significant value (\$9 for receiver)
- Has proven effective across multiple partners
- Keeps some safety margin

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

- Round 1 (vs Galen): I sent \$3, received \$5 back - payoff \$7
- Round 2 (vs Finn): Finn sent \$3, I returned \$5 - payoff \$4
- Round 3 (vs Briar): I sent \$3, received \$4.5 back - payoff \$6.5
- Round 4 (vs Briar): Briar sent \$3, I returned \$4.5 - payoff \$4.5

So far, sending \$3 has worked well:

- It shows reasonable trust
- It creates value (becomes \$9)
- Partners have generally reciprocated with \$4.5-\$5 returns
- It's been my consistent strategy

This is my first interaction with Aster. I should maintain my cooperative strategy that has worked well so far. Sending \$3 has proven to be a good balance between trust and caution.

Total earnings: \$22.0
Average per round: \$5.5

I'll stick with \$3.

```
Usage: {
  "input_tokens": 1530,
  "output_tokens": 375,
  "reasoning_tokens": 193
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T15:44:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_8",
    "Agent_1"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_1": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
```

```

"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In Round 1, when I sent \$3 to Cyra, they returned \$4, which felt fair

I'll reciprocate with the same level of cooperation. Returning \$4 means:

- Galen gets \$6 total (\$2 kept + \$4 returned)
- I keep \$5 (\$9 - \$4)
- Both of us profit from cooperation

```
{'return': 4}
```

reasoning:

Let me analyze this situation:

Round 2 - I'm the RECEIVER

- Galen (sender) sent me \$3.0
- I received \$9.0 (due to 3x multiplier)
- I need to decide how much to return (0-9.0)

From Round 1:

- I was the sender and sent \$3.0 to Cyra
- Cyra received \$9.0
- Cyra returned \$4.0 to me
- My payoff was \$6.0 (I kept \$2 + received \$4 back = \$6)
- Cyra's payoff was \$9 - \$4 = \$5

So Cyra returned \$4 out of \$9 received, which is about 44%. This seems like a reasonable cooperative response.

Now I'm in Cyra's position with Galen. Galen sent me \$3, just like I sent to Cyra.

If I want to establish a pattern of reciprocity and fairness:

- Galen kept \$2, sent \$3
- I received \$9
- If I return \$4 (like Cyra did), then:
 - Galen gets: \$2 + \$4 = \$6
 - I keep: \$9 - \$4 = \$5

This mirrors what happened in Round 1 and seems fair. It shows I'm willing to reciprocate cooperation.

Alternatively, I could consider:

- Returning more (like \$5) to be extra generous: Galen gets \$7, I keep \$4
- Returning less (like \$3) to keep more: Galen gets \$5, I keep \$6

But \$4 seems like the most reciprocal and cooperative response given what Cyra did to me.

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 554,
  "reasoning_tokens": 307
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:
- When I sent \$3 to Cyra, they returned \$4
- When Galen sent \$3 to me, I returned \$4

This cooperative pattern seems to be working well for everyone. I'll continue with the same strategy and send \$3 to Elara, expecting similar reciprocity.

```
{'send': 3}
```

reasoning:

Let me think about this round:

Previous rounds:

- Round 1 (SENDER vs Cyra): I sent \$3, they returned \$4. I got \$6 (kept \$2, got \$4 back)
- Round 2 (RECEIVER vs Galen): They sent \$3, I returned \$4. I got \$5 (received \$9, returned \$4)

Total earnings: \$11

Now I'm facing Elara as a sender. Both times when \$3 was sent (by me and by Galen), the receiver returned \$4, which seems like a fair amount. This suggests a pattern of cooperation.

I should continue this cooperative strategy. Sending \$3 again seems reasonable:

- Shows consistency in my cooperative approach
- Not too risky
- If Elara behaves similarly to Cyra and me, they'll likely return around \$4
- This would give me \$6 this round (\$2 kept + \$4 returned)

Let me send \$3 again.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 322,
  "reasoning_tokens": 197
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the recent pattern:

- Most exchanges with \$3 sent have resulted in \$4-\$4.5 returned
- Elara was particularly generous, returning \$4.5 when I sent \$3
- Dorian showed trust by sending \$3

I'll match Elara's generosity and return \$4.5 to Dorian, maintaining the cooperative pattern.

```
{'return': 4.5}
```

reasoning:

Let me analyze the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- In Round 3, when I sent \$3 to Elara, they returned \$4.5 - even more generous than the \$4 I've been seeing

The pattern has been:

- I sent \$3 to Cyra ? received \$4
- Galen sent \$3 to me ? I returned \$4
- I sent \$3 to Elara ? received \$4.5

Since Elara was more generous by returning \$4.5, I should consider being similarly generous. This helps build good relationships in the multi-agent setting. Returning around \$4-\$4.5 seems appropriate.

I'll return \$4.5 to match Elara's generosity and continue building cooperative relationships.

```
    ps.  
usage:  
{  
  "input_tokens": 1318,  
  "output_tokens": 297,  
  "reasoning_tokens": 171  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the established pattern:

- Everyone has consistently sent \$3, showing trust
- Returns have been \$4-\$4.5, maintaining cooperation
- Hana sent \$3, showing they're following the cooperative norm

I'll continue with \$4.5, which has been the generous standard recently.

```
{'return': 4.5}
```

Reasoning:

Let me review the pattern:

- Round 1: I was sender, sent \$3 to Cyra, got back \$4. My payoff: \$6
- Round 2: I was receiver from Galen, got \$9, returned \$4. My payoff: \$5
- Round 3: I was sender, sent \$3 to Elara, got back \$4.5. My payoff: \$6.5
- Round 4: I was receiver from Dorian, got \$9, returned \$4.5. My payoff: \$4.5
- Round 5: I'm receiver from Hana, got \$9, need to decide how much to return

The cooperative pattern has been established with everyone sending \$3 and returns ranging from \$4 to \$4.5. Hana has also shown trust by sending \$3.

I'll continue the cooperative pattern and return \$4.5, which has been working well.

```
Usage: {  
  "input_tokens": 1397,  
  "output_tokens": 299,  
  "reasoning_tokens": 165  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T15:44:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_5",  
      "Agent_7"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_7": "trustee"  
    }  
  },  
}
```

```

    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation, Briar sent me the same amount (\$3) that I sent to Finn in Round

1. Finn returned \$4.5 to me, which was exactly half of what they received - a fair and cooperative response.

To maintain a reciprocal and cooperative pattern, I should treat Briar similarly. Returning \$4.5 means:

- Briar gets a payoff of \$6.5 (\$2 kept + \$4.5 returned)
- I keep \$4.5 (\$9 received - \$4.5 returned)
- Both of us benefit from the cooperation

This establishes trust and encourages continued cooperation in future rounds.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Finn, who received \$9 and returned \$4.5. This was a fair and cooperative response - they returned exactly half of what they received, which left both of us better off.
2. Now I'm the receiver and Briar sent me \$3 (same amount I sent to Finn). I received \$9.
3. I should consider:
 - Reciprocity: Finn returned \$4.5 when receiving \$9, which was cooperative
 - Reputation: If I want to build trust for future rounds, I should be fair
 - Strategic thinking: Being too greedy might discourage cooperation in future rounds
 - Briar took the same risk I took with Finn
4. Following the cooperative pattern that Finn established, returning around \$4.5 would be fair and consistent. This would give Briar a payoff of $\$2 + \$4.5 = \$6.5$ (same as what I got in round 1).
5. I could return slightly more or less, but \$4.5 seems like a reasonable, fair amount that:
 - Rewards Briar's trust
 - Establishes me as a cooperative player
 - Keeps the game balanced
 - Mirrors what was done to me

I'll return \$4.5.

usage:

```
{  
  "input_tokens": 826,  
  "output_tokens": 481,  
  "reasoning_tokens": 284  
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far, there's been consistent cooperation with \$3 being sent and \$4.5 being returned (half of the tripled amount). This has worked well for everyone involved.

I'll continue this cooperative pattern with Aster, returning the same amount I returned to Briar when they sent \$3.

```
{'return': 4.5}
```

reasoning:

This is the third round, and again I'm playing as the RECEIVER. Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 1: I was SENDER, sent \$3 to Finn, they returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER, Briar sent \$3, I returned \$4.5. I got \$4.5.
- Round 3: I'm RECEIVER, Aster sent \$3, received \$9.

There's been a consistent pattern: people are sending \$3, and the fair return has been \$4.5 (half of what's received). This has worked well and established cooperation.

I should maintain this pattern of returning \$4.5 when someone sends \$3. This is:

1. Consistent with what I've been doing and receiving
2. Fair (50% return on the multiplied amount)
3. Encourages continued cooperation
4. Gives both parties a positive outcome

If I return \$4.5:

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1144,
  "output_tokens": 387,
  "reasoning_tokens": 236
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern across all rounds so far, there's been consistent cooperation with agents sending \$3 and receivers returning \$4.5. This has created a stable, mutually beneficial norm.

I'll continue this pattern with Cyra, sending \$3 as I did with Finn and as others have done with me.

```
{'send': 3}
```

reasoning:

Let me review the pattern so far:

Round 1: I was SENDER vs Finn

- I sent \$3
- Finn received \$9
- Finn returned \$4.5
- My payoff: \$6.5

Round 2: I was RECEIVER vs Briar

- Briar sent \$3
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 3: I was RECEIVER vs Aster

- Aster sent \$3
- I received \$9
- I returned \$4.5

- My payoff: \$4.5

So far, there's been a consistent pattern:

- Everyone sends \$3
- Everyone returns \$4.5 (half of the tripled amount)
- Senders get \$6.5, Receivers get \$4.5

Now I'm a SENDER again vs Cyra. Should I continue the pattern and send \$3?

The pattern has been working well and establishing trust. If I send \$3 and Cyra follows the pattern by returning \$4.5, I'll get \$6.5 again.

I should continue with the cooperative norm that has been established.

usage:

```
{
  "input_tokens": 1367,
  "output_tokens": 358,
  "reasoning_tokens": 218
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative pattern has been perfectly consistent - everyone sends \$3 and returns \$4.5. This has worked well for all players across all rounds so far.

I'll maintain this cooperative strategy with Galen.

```
{'send': 3}
```

Reasoning:

The pattern continues to hold perfectly. In round 4, I sent \$3 to Cyra and they returned \$4.5, just as expected. This cooperative pattern has been consistent across all rounds:

- Round 1: I sent \$3 to Finn, received \$4.5 back
- Round 2: Briar sent \$3 to me, I returned \$4.5
- Round 3: Aster sent \$3 to me, I returned \$4.5
- Round 4: I sent \$3 to Cyra, received \$4.5 back

Everyone is following the same cooperative strategy. I should continue sending \$3 to Galen to maintain this stable equilibrium.

Usage: {

```
"input_tokens": 1393,
"output_tokens": 222,
"reasoning_tokens": 130
```

}

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-06-04T15:45:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_5": "Elara",
    "Agent_7": "Galen"
```

```
},
```

```
"agents": [
  "Agent_5",
```

```

    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.

0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is a new opponent (Aster), so I don't have history with them
- In the previous round, Hana sent me 60% (\$3), showing trust
- I returned a fair amount (\$5), creating mutual benefit
- This is a multi-round game where building cooperation matters

I'll send \$3 (60% of my endowment), mirroring the cooperative approach Hana showed me. This demonstrates willingness to trust and create mutual gains.

{'send': 3}

reasoning:

Now I'm the sender against a new opponent, Aster.

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana showed trust by sending 60% (\$3). I returned \$5 out of \$9, which gave them a good return.
2. This is a new pairing with Aster. I don't know how they'll behave.
3. Since this is a multi-round game and I'll potentially interact with different agents multiple times, I should think about reputation and reciprocity.
4. If I send money, it gets tripled. If I send \$3:
 - Aster receives \$9
 - If they return \$5 (similar to what I did), I'd end with $\$2 + \$5 = \$7$
 - If they return less, I could end up worse off
5. Sending nothing is safe but doesn't build trust or create mutual gains.
6. I should probably mirror the cooperative behavior I received. Hana sent 60% (\$3), which seemed like a reasonable level of trust.

Given that:

- This is an iterated game with multiple rounds
- Building cooperation can lead to better long-term outcomes
- I should signal willingness to cooperate but also be somewhat cautious with a new partner

I'll send \$3 (60%), matching what Hana sent to me. This shows I'm willing to cooperate and trust, while not being overly risky.

usage:

```
{
  "input_tokens": 896,
  "output_tokens": 457,
  "reasoning_tokens": 321
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$10.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same level of trust that Hana showed me in Round 1

I want to be consistent in my reciprocation. When Hana sent \$3, I returned \$5, which was fair and created mutual benefit. I'll do the same for Cyra.

If I return \$5.0:

- Cyra ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

reasoning:

Let me analyze this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- In the last round, when I sent \$3.0 to Aster, they returned \$4.0
- This is a multi-round game, so building cooperation is important

I've established a pattern:

- Round 1 (as receiver): Hana sent \$3, I received \$9, returned \$5
- Round 2 (as sender): I sent \$3 to Aster, they received \$9, returned \$4
- Round 3 (as receiver): Cyra sent \$3, I received \$9, need to decide return

Cyra showed the same level of trust as Hana did (both sent \$3). I should be consistent in my reciprocation. When Hana sent \$3, I returned \$5. I should do the same for Cyra.

If I return \$5.0:

- Cyra ends with: $\$5 - \$3 + \$5 = \7
- I keep: $\$9 - \$5 = \$4$

This is fair and consistent with my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1173,
  "output_tokens": 452,
  "reasoning_tokens": 221
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$14.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60%) when I'm the sender
- This has worked well - Aster returned \$4 when I sent \$3
- When others send me \$3, I return \$5
- This cooperative strategy is building positive outcomes

I'll maintain this consistent approach with Finn, as it demonstrates trustworthiness and has been effective.

```
{'send': 3}
```

reasoning:

Let me analyze my situation:

Round 1: I was RECEIVER against Hana

- Hana sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 2: I was SENDER against Aster

- I sent \$3 (60%)
- Aster returned \$4
- My payoff: \$6 (\$5 - \$3 + \$4)

Round 3: I was RECEIVER against Cyra

- Cyra sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 4: I am SENDER against Finn (new opponent)

Pattern so far:

- When people send me \$3, I return \$5 (consistent reciprocation)
- When I sent \$3 to Aster, they returned \$4 (slightly less than what I return, but still fair)
- I've been playing cooperatively at 60% send rate

Total earnings: \$14

For this round with Finn (new opponent):

- I should maintain my cooperative strategy
- Sending \$3 (60%) has been working well
- It builds trust and creates good outcomes

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1471,
  "output_tokens": 381,
  "reasoning_tokens": 226
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same amount Hana and Cyra sent me previously

I'll maintain consistency in my reciprocation strategy. When I receive \$3, I return \$5, which creates mutual benefit and shows I'm a reliable partner.

If I return \$5.0:

- Elara ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

Reasoning:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier

- This is consistent with what others have been sending me

My track record as receiver:

- Round 1: Hana sent \$3, I returned \$5
- Round 3: Cyra sent \$3, I returned \$5

And when I've been sender:

- Round 2: I sent \$3 to Aster, they returned \$4
- Round 4: I sent \$3 to Finn, they returned \$4.5

I've been consistent in returning \$5 when receiving \$3. This creates a fair outcome where:

- The sender nets \$2 ($\$5 - \$3 + \5)
- I net \$4 ($\$9 - \5)

Both sides benefit, and it's a good reciprocal strategy. I should maintain this consistency with Elara.

If I return \$5.0:

- Elara ends with: \$7 ($\$5 - \$3 + \5)
- I keep: \$4 ($\$9 - \5)

This is fair and consistent with my previous behavior.

```
Usage: {
  "input_tokens": 1442,
  "output_tokens": 439,
  "reasoning_tokens": 220
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T15:45:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_2"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_2": "trustee",
    "Agent_3": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- I'm now the sender with \$5
- In Round 1, Aster sent me \$3.0 (60%) showing trust, and I returned \$4.0, establishing cooperation
- This is my first interaction with Dorian

I'll follow the cooperative pattern established in Round 1. Sending \$3.0 (60% of my endowment) demonstrates trust and gives Dorian the opportunity to reciprocate, which should be nefit us both.

{'send': 3.0}

reasoning:

Now I'm the SENDER playing against Dorian (Agent_4).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$3.0 (60% of their endowment). I returned \$4.0, which gave them a profit and established me as cooperative.
2. Now I'm facing a new opponent, Dorian. I have no history with them.

3. As the sender, I need to decide how much to trust Dorian. The key question is: will they reciprocate?
4. Sending has high potential upside (3x multiplier) but requires trust that they'll return enough.
5. Since this is early in the game (Round 2) and I want to establish cooperative patterns, I should send a reasonable amount.
6. Sending \$3.0 (like Aster did) seems like a fair, cooperative choice that:
 - Mirrors what Aster sent me
 - Shows willingness to trust
 - Gives Dorian the opportunity to reciprocate
 - Would give them \$9.0 to work with
7. If Dorian returns around \$4-5 (similar to what I did), we'll both profit.

Let me send \$3.0 to match the cooperative precedent from the first round.

usage:

```
{
  "input_tokens": 888,
  "output_tokens": 429,
  "reasoning_tokens": 277
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

- Round 1: Aster sent me \$3.0 (60%), I returned \$4.0 - good cooperation
- Round 2: I sent Dorian \$3.0 (60%), they returned \$4.5 - even better cooperation, I profited \$1.5

The cooperative strategy of sending 60% has been working well, with both opponents reciprocating generously. I'll continue this approach with Galen.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

- Round 1: I was receiver. Aster sent me \$3.0, I received \$9.0, returned \$4.0. My payoff: \$5.0
- Round 2: I was sender against Dorian. I sent \$3.0, they received \$9.0, returned \$4.5. My payoff: \$2.0 + \$4.5 = \$6.5
- Total earnings: \$11.5

Now in Round 3, I'm sender against Galen, a new opponent.

The pattern so far shows good cooperation:

- Aster sent 60% (\$3.0), I returned \$4.0 (44% of what I received)
- I sent 60% (\$3.0) to Dorian, they returned \$4.5 (50% of what they received)

Both interactions have been profitable and cooperative. Dorian was even slightly more generous in returning funds.

For Round 3 with Galen, I should continue this cooperative strategy. Sending \$3.0 (60%) has worked well so far and establishes trust. This gives Galen the opportunity to reciprocate and continue the cooperative pattern.

I'll send \$3.0 again.

usage:

```
{
  "input_tokens": 1152,
  "output_tokens": 411,
  "reasoning_tokens": 231
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history:

- Round 1: As receiver from Aster (\$3.0 sent), I returned \$4.0
- Round 2: As sender to Dorian (\$3.0 sent), they returned \$4.5
- Round 3: As sender to Galen (\$3.0 sent), they returned \$5.0

Elara has shown the same level of trust as others by sending \$3.0 (60%). The pattern shows increasing generosity in this cooperative environment. I'll return \$4.5, which rewards Elara's trust and gives them a solid profit of \$1.5.

```
{'return': 4.5}
```

reasoning:

Let me analyze the situation:

- Round 1: As receiver, Aster sent me \$3.0, I returned \$4.0 (gave them a \$1 profit)
- Round 2: As sender, I sent Dorian \$3.0, they returned \$4.5 (I profited \$1.5)
- Round 3: As sender, I sent Galen \$3.0, they returned \$5.0 (I profited \$3.0)
- Round 4: As receiver, Elara sent me \$3.0, I receive \$9.0

The pattern shows:

- When I've been receiver, I returned \$4.0
- When I've been sender with \$3.0, I've received back \$4.5 and \$5.0

Elara has shown the same trust level as the others (sending \$3.0, which is 60%). I should reciprocate fairly.

Elara currently has: $\$5.0 - \$3.0 = \$2.0$

If I return \$4.0:

- Elara gets: $\$2.0 + \$4.0 = \$6.0$ (profit of \$1.0)
- I keep: $\$9.0 - \$4.0 = \$5.0$

This is consistent with my Round 1 behavior and maintains a cooperative reputation. The others have been even more generous when I sent, but returning \$4.0 is still fair and rewards trust.

However, given that Dorian returned \$4.5 and Galen returned \$5.0 to me, maybe I should be slightly more generous? Let me think...

Actually, I returned \$4.0 in Round 1, which was my only previous receiver round. That established my baseline. I should probably stick with that for consistency, unless I want to match the increasing generosity I've experienced.

Let me go with \$4.5 as a middle ground between my previous \$4.0 and the more generous returns I've received.

Returning \$4.5 splits the difference-Elara ends with \$6.5 (a \$1.5 gain) while I keep \$4.5, which mirrors the reciprocity I've experienced and signals I'm matching their cooperative spirit.

usage:

```
{
  "input_tokens": 1426,
  "output_tokens": 705,
  "reasoning_tokens": 413
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$23.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my performance:

- As sender: Sending \$3.0 has yielded excellent returns (\$4.5 from Dorian, \$5.0 from Galen)
- As receiver: When others send \$3.0, I return \$4.0-\$4.5, maintaining cooperation

The \$3.0 (60%) strategy has been consistently successful across different opponents. I'll continue this approach with Briar.

```
{'send': 3.0}
```

Reasoning:

Let me review my history:

Round 1: RECEIVER from Aster - they sent \$3.0, I returned \$4.0, my payoff was \$5.0
Round 2: SENDER to Dorian - I sent \$3.0, they returned \$4.5, my payoff was \$6.5
Round 3: SENDER to Galen - I sent \$3.0, they returned \$5.0, my payoff was \$7.0
Round 4: RECEIVER from Elara - they sent \$3.0, I returned \$4.5, my payoff was \$4.5

Total earnings: \$23.0

The pattern is clear - sending \$3.0 (60% of endowment) has been consistently successful. When I send \$3.0, I get good returns:
- Dorian returned \$4.5 (I profited \$1.5)
- Galen returned \$5.0 (I profited \$2.0)

And when others send me \$3.0, I've been returning \$4.0-\$4.5, which seems fair.

This is a new opponent (Briar), so I'll continue with the cooperative strategy that has been working well. I'll send \$3.0.

Usage: {

```
"input_tokens": 1446,
"output_tokens": 408,
"reasoning_tokens": 208
}
```

GAME - Agent_2 - Receiver - dyad_4

Timestamp: 2026-06-04T15:45:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_3": "Cyra"
  }
}
```

```

},
"agents": [
  "Agent_3",
  "Agent_2"
],
"investor": "Agent_3",
"roles": {
  "Agent_2": "trustee",
  "Agent_3": "investor"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Elara
- In Round 1, Dorian sent me \$3 (60%) and I reciprocated fairly by returning \$4.5
- I want to establish a cooperative dynamic early in the game

I'll send \$3 (60% of my endowment), matching the trust level that Dorian showed me. This demonstrates my willingness to cooperate and gives us both the opportunity for mutual benefit if Elara reciprocates.

```
{"send": 3}
```

reasoning:

Now I'm the sender playing against Elara (a new opponent).

Let me think about this strategically:

1. In Round 1, I was the receiver and Dorian sent me \$3.0 (60%). I returned \$4.5, establishing a cooperative pattern.
2. Now I'm the sender with Elara. Since this is our first interaction, I need to decide how much trust to show.
3. Key considerations:
 - Sending more can lead to higher total gains if Elara reciprocates
 - But if Elara doesn't return much, I could lose out
 - This is Round 2, early in the game, so establishing cooperation is valuable
 - Dorian sent 60% to me, showing trust
4. Strategic options:
 - Send \$0: Safe but generates no mutual benefit
 - Send \$1-2: Conservative, low risk
 - Send \$3: Moderate trust (60%, matching what Dorian did)
 - Send \$4-5: High trust

Given that:

- Dorian showed 60% trust with me and I reciprocated fairly
- This is early in the game where establishing cooperation matters
- The potential gains from cooperation are substantial

I'll send \$3 (60%), matching the level of trust Dorian showed to me. This demonstrates I'm willing to cooperate and gives Elara a chance to earn good returns while also allowing me to benefit if they reciprocate.

If Elara returns about half (like I did), I'd get back around \$4.5, netting me \$2.5 total (\$5 - \$3 + \$4.5 = \$6.5 payoff). Even if they return less, this establishes a cooperative stance.

usage:

```
{  
  "input_tokens": 914,  
  "output_tokens": 523,  
  "reasoning_tokens": 371  
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This mirrors Round 1 where Dorian also sent me \$3.0

I want to maintain consistency in my reciprocation strategy. When someone sends 60% showing trust, I should return about half of what I receive. This establishes me as a reliable, fair player.

Returning \$4.5 gives Hana a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5, which is the same fair split I used in Round 1.

```
{"return": 4.5}
```

reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (Agent_8) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is my first interaction with Hana

Looking at the pattern:

- Round 1: Dorian sent me \$3 (60%), I returned \$4.5
- Round 2: I sent Elara \$3 (60%), they returned \$4.5

There's a nice cooperative pattern emerging where:

- Sending \$3 (60%) shows trust
- Returning \$4.5 (50% of received) reciprocates fairly

Hana has shown the same level of trust as Dorian did in Round 1. I should maintain consistency and reciprocate similarly. Returning \$4.5:

- Gives Hana \$6.5 total (profit of \$1.5)
- Leaves me with \$4.5
- Maintains the cooperative norm I've established

This consistency is important for:

1. Building a reputation as a fair player
2. Encouraging continued cooperation
3. Keeping the game mutually beneficial

I'll return \$4.5, matching my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1192,
  "output_tokens": 470,
  "reasoning_tokens": 253
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Last round, Hana sent me \$3.0 (60%) and I returned \$4.5, which was fair
- Both of us profited from the exchange
- We've established a cooperative pattern

Since our previous interaction went well and we both showed trustworthiness, I'll continue this cooperative approach by sending the same amount Hana sent me last round.

```
{"send": 3.0}
```

reasoning:

Looking at the situation:

- Last round (Round 3), Hana sent me \$3.0 and I returned \$4.5
- This round I'm the sender against Hana again
- Hana demonstrated trust by sending 60% to me
- I reciprocated fairly by returning \$4.5 (50% of what I received)

This is a continuation of a relationship with Hana. Since we both cooperated well in the previous round:

- Hana sent \$3 (60%)
- I returned \$4.5 (50% of \$9)

Both of us made a profit and showed trustworthiness. I should continue this cooperative pattern by sending a similar amount. Sending \$3 again would:

- Show consistency in my behavior
- Signal continued trust
- Give us both the opportunity for mutual benefit

I'll send \$3.0 to maintain the cooperative relationship we've established.

usage:

```
{
  "input_tokens": 1502,
  "output_tokens": 313,
  "reasoning_tokens": 198
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60%), showing significant trust
- I received \$9.0 after the multiplier

I've established a consistent reciprocation pattern: when someone sends \$3.0, I return \$4.5. This has worked well in building cooperative relationships with both Dorian and Hana. I'll maintain this consistency with Cyra.

Returning \$4.5 ensures both of us profit and establishes me as a fair, reliable partner.

```
{"return": 4.5}
```

Reasoning:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- I've been consistent in my strategy: when someone sends \$3.0, I return \$4.5

Pattern from previous rounds:

- Round 1: Dorian sent \$3.0, I returned \$4.5
- Round 3: Hana sent \$3.0, I returned \$4.5

This has worked well - both parties profit and cooperation is maintained. I should continue this is consistent, fair reciprocation strategy with Cyra.

Returning \$4.5 gives Cyra a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5.

```
Usage: {  
  "input_tokens": 1458,  
  "output_tokens": 339,  
  "reasoning_tokens": 160  
}
```

ROUND 6

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_8 (Hana)=Receiver
dyad_2: Agent_7 (Galen)=Sender, Agent_4 (Dorian)=Receiver
dyad_3: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_8 (Hana)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_7 (Galen) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_2 (Briar) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_6 (Finn) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_3 (Cyra)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 33.0,
"Agent_2": 33.0,
"Agent_3": 34.0,
"Agent_4": 33.0,
"Agent_5": 33.5,
"Agent_6": 33.5,
"Agent_7": 31.0,
"Agent_8": 33.0

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T15:45:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_8",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_8": "Hana"
 },
 "agents": [
 "Aster",
 "Hana"
]
}


```

    "Agent_1",
    "Agent_8"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.

Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:

- When I sent \$3 to Cyra, they returned \$4
- When Galen sent \$3 to me, I returned \$4

This cooperative pattern seems to be working well for everyone. I'll continue with the same strategy and send \$3 to Elara, expecting similar reciprocity.

{'send': 3}

reasoning:

Let me think about this round:

Previous rounds:

- Round 1 (SENDER vs Cyra): I sent \$3, they returned \$4. I got \$6 (kept \$2, got \$4 back)
- Round 2 (RECEIVER vs Galen): They sent \$3, I returned \$4. I got \$5 (received \$9, returned \$4)

Total earnings: \$11

Now I'm facing Elara as a sender. Both times when \$3 was sent (by me and by Galen), the receiver returned \$4, which seems like a fair amount. This suggests a pattern of cooperation.

I should continue this cooperative strategy. Sending \$3 again seems reasonable:

- Shows consistency in my cooperative approach
- Not too risky
- If Elara behaves similarly to Cyra and me, they'll likely return around \$4
- This would give me \$6 this round (\$2 kept + \$4 returned)

Let me send \$3 again.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 322,
  "reasoning_tokens": 197
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the recent pattern:

- Most exchanges with \$3 sent have resulted in \$4-\$4.5 returned
- Elara was particularly generous, returning \$4.5 when I sent \$3
- Dorian showed trust by sending \$3

I'll match Elara's generosity and return \$4.5 to Dorian, maintaining the cooperative pattern.

{'return': 4.5}

reasoning:

Let me analyze the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier

- In Round 3, when I sent \$3 to Elara, they returned \$4.5 - even more generous than the \$4 I've been seeing

The pattern has been:

- I sent \$3 to Cyra ? received \$4
- Galen sent \$3 to me ? I returned \$4
- I sent \$3 to Elara ? received \$4.5

Since Elara was more generous by returning \$4.5, I should consider being similarly generous. This helps build good relationships in the multi-agent setting. Returning around \$4-\$4.5 seems appropriate.

I'll return \$4.5 to match Elara's generosity and continue building cooperative relationships.

usage:

```
{
  "input_tokens": 1318,
  "output_tokens": 297,
  "reasoning_tokens": 171
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the established pattern:

- Everyone has consistently sent \$3, showing trust
- Returns have been \$4-\$4.5, maintaining cooperation
- Hana sent \$3, showing they're following the cooperative norm

I'll continue with \$4.5, which has been the generous standard recently.

```
{'return': 4.5}
```

reasoning:

Let me review the pattern:

- Round 1: I was sender, sent \$3 to Cyra, got back \$4. My payoff: \$6
- Round 2: I was receiver from Galen, got \$9, returned \$4. My payoff: \$5
- Round 3: I was sender, sent \$3 to Elara, got back \$4.5. My payoff: \$6.5
- Round 4: I was receiver from Dorian, got \$9, returned \$4.5. My payoff: \$4.5
- Round 5: I'm receiver from Hana, got \$9, need to decide how much to return

The cooperative pattern has been established with everyone sending \$3 and returns ranging from \$4 to \$4.5. Hana has also shown trust by sending \$3.

I'll continue the cooperative pattern and return \$4.5, which has been working well.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 299,
  "reasoning_tokens": 165
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.

0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our recent interaction:

- Last round, Hana sent me \$3 and I returned \$4.5
- Hana has shown trust and cooperation
- The \$3 send amount with ~\$4.5 return has been working well across all interactions

I'll continue with this successful cooperative pattern and send \$3 to Hana.

{'send': 3}

Reasoning:

Looking at the situation:

- Last round, Hana sent me \$3 and I returned \$4.5
- This round, I'm the sender to Hana
- Hana has shown trust by sending \$3 last round
- The cooperative pattern has been: send \$3, receive back \$4-\$4.5
- I should continue this pattern and send \$3 to Hana

Given that Hana just cooperated with me and I returned \$4.5 to them, I expect Hana will return a fair amount if I send \$3.

I'll send \$3 to maintain the cooperative equilibrium.

```
Usage: {  
  "input_tokens": 1327,  
  "output_tokens": 238,  
  "reasoning_tokens": 122  
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-06-04T15:45:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_8"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

****Round 1 (vs Galen):**** I sent \$3, received \$5 back - good cooperation
****Round 2 (vs Finn):**** Finn sent \$3, I returned \$5 - maintained fair reciprocity

Both rounds showed positive cooperation with the \$3 send amount. This amount:

- Creates significant value (becomes \$9)
- Allows for mutual benefit
- Has proven successful
- Maintains consistency in my strategy

I'll continue with the same approach against Briar.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the history:

Round 1: I was SENDER vs Galen
- I sent \$3
- Galen received \$9, returned \$5
- My payoff: $\$5 - \$3 + \$5 = \7

Round 2: I was RECEIVER vs Finn
- Finn sent \$3
- I received \$9, returned \$5
- My payoff: $\$9 - \$5 = \$4$

Total earnings: \$11

Now Round 3: I'm SENDER vs Briar

The pattern so far has been positive:

- I sent \$3 to Galen, got \$5 back (good return)
- Finn sent \$3 to me, I returned \$5 (reciprocating the cooperative behavior)

This suggests a cooperative norm is emerging. Sending \$3 has worked well for me. I could:

1. Continue with \$3 (consistent, proven strategy)
2. Send more to signal even stronger cooperation
3. Send less to protect myself

Given that:

- The \$3 send amount has been successful
- Both interactions showed good reciprocity
- I'm building a reputation in this multi-agent setting
- Consistency can signal trustworthiness

I'll stick with \$3 again. It's working and establishes me as a consistent cooperator.

usage:

```
{  
  "input_tokens": 1189,  
  "output_tokens": 444,  
  "reasoning_tokens": 263  
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my interaction with Briar:

****Round 3:**** I sent \$3 to Briar, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 4 (now):**** Briar sent \$3 to me (matching my trust)
- I received \$9

Briar showed reasonable cooperation by returning \$4.5. Now they've shown the same level of trust in me. I'll reciprocate by matching their behavior exactly - returning \$4.5 creates a consistent, fair pattern between us.

```
'''json  
{"return": 4.5}  
'''
```

reasoning:

Looking at what happened:

****Round 3 (vs Briar):**** I sent \$3, Briar returned \$4.5 to me
- My payoff was \$6.5 (\$5 - \$3 + \$4.5)
- Briar kept \$4.5 (\$9 - \$4.5)

****Round 4 (vs Briar again):**** Briar sent \$3, I received \$9

So Briar is meeting me again! Last time I sent \$3 and they returned \$4.5. Now they're sending \$3 to me.

The return of \$4.5 was slightly less generous than the \$5 that Galen returned to me and that I returned to Finn. But it's still cooperative and fair - we both got reasonable payoffs.

Now Briar has sent me \$3, showing the same trust. I should reciprocate fairly.

Options:

1. Return \$5 (what I've been doing with others) - this would give Briar \$7, me \$4
2. Return \$4.5 (matching what Briar did) - this would give Briar \$6.5, me \$4.5

Matching what Briar did (\$4.5) seems like the fairest reciprocal move - it shows I'm willing to match their level of cooperation exactly. This creates a clear tit-for-tat dynamic.

Actually, let me think about this more carefully. When Briar was the receiver:

- They got \$9
- They returned \$4.5
- They kept \$4.5

Now when I'm the receiver:

- I get \$9
- If I return \$4.5, I keep \$4.5
- Briar gets \$5 - \$3 + \$4.5 = \$6.5

So by returning \$4.5, I'm mirroring their exact behavior. This seems fair and establishes a consistent reciprocal pattern with Briar.

usage:

```
{
  "input_tokens": 1481,
  "output_tokens": 608,
  "reasoning_tokens": 350
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.00 to you, it became \$9.00, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.00.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

****Rounds 1-4:**** Consistently sent \$3 as sender

- Returns have ranged from \$4.5-\$5
- Average payoff as sender: \$6.75
- Strategy has built cooperative relationships

This is my first interaction with Aster. I'll maintain my established strategy of sending \$3, which:

- Demonstrates trust
- Creates significant value (\$9 for receiver)
- Has proven effective across multiple partners
- Keeps some safety margin

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1 (vs Galen): I sent \$3, received \$5 back - payoff \$7
Round 2 (vs Finn): Finn sent \$3, I returned \$5 - payoff \$4
Round 3 (vs Briar): I sent \$3, received \$4.5 back - payoff \$6.5
Round 4 (vs Briar): Briar sent \$3, I returned \$4.5 - payoff \$4.5

So far, sending \$3 has worked well:

- It shows reasonable trust
- It creates value (becomes \$9)
- Partners have generally reciprocated with \$4.5-\$5 returns
- It's been my consistent strategy

This is my first interaction with Aster. I should maintain my cooperative strategy that has worked well so far. Sending \$3 has proven to be a good balance between trust and caution.

Total earnings: \$22.0
Average per round: \$5.5

I'll stick with \$3.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 375,
  "reasoning_tokens": 193
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

Aster demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This mirrors my successful interaction with Briar. I'll reciprocate with the same fair return.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

This is exactly the same pattern I had with Briar:

- I sent \$3, they returned \$4.5
- They sent \$3, now I should return

Aster showed the same level of cooperation as Briar. They returned the exact same amount (\$4.5). Now they've shown the same level of trust in me (\$3).

For consistency and fair reciprocity, I should return \$4.5 - matching what they returned to me. This creates a balanced, cooperative relationship.

This seems to be a consistent pattern:

- Send \$3
- Receive back \$4.5
- When receiving \$3 (which becomes \$9), return \$4.5

This creates fair outcomes where both parties benefit similarly from the cooperation.

```
Usage: {
  "input_tokens": 1474,
  "output_tokens": 393,
  "reasoning_tokens": 222
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T15:45:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_4"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_4": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$10.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same level of trust that Hana showed me in Round 1

I want to be consistent in my reciprocation. When Hana sent \$3, I returned \$5, which was fair and created mutual benefit. I'll do the same for Cyra.

If I return \$5.0:

- Cyra ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

reasoning:

Let me analyze this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- In the last round, when I sent \$3.0 to Aster, they returned \$4.0
- This is a multi-round game, so building cooperation is important

I've established a pattern:

- Round 1 (as receiver): Hana sent \$3, I received \$9, returned \$5
- Round 2 (as sender): I sent \$3 to Aster, they received \$9, returned \$4
- Round 3 (as receiver): Cyra sent \$3, I received \$9, need to decide return

Cyra showed the same level of trust as Hana did (both sent \$3). I should be consistent in my reciprocation. When Hana sent \$3, I returned \$5. I should do the same for Cyra.

If I return \$5.0:

- Cyra ends with: $\$5 - \$3 + \$5 = \7
- I keep: $\$9 - \$5 = \$4$

This is fair and consistent with my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1173,
  "output_tokens": 452,
  "reasoning_tokens": 221
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$14.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60%) when I'm the sender
- This has worked well - Aster returned \$4 when I sent \$3
- When others send me \$3, I return \$5
- This cooperative strategy is building positive outcomes

I'll maintain this consistent approach with Finn, as it demonstrates trustworthiness and has been effective.

```
{'send': 3}
```

reasoning:

Let me analyze my situation:

Round 1: I was RECEIVER against Hana

- Hana sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 2: I was SENDER against Aster

- I sent \$3 (60%)
- Aster returned \$4
- My payoff: \$6 ($\$5 - \$3 + \4)

Round 3: I was RECEIVER against Cyra

- Cyra sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 4: I am SENDER against Finn (new opponent)

Pattern so far:

- When people send me \$3, I return \$5 (consistent reciprocation)
- When I sent \$3 to Aster, they returned \$4 (slightly less than what I return, but still fair)
- I've been playing cooperatively at 60% send rate

Total earnings: \$14

For this round with Finn (new opponent):

- I should maintain my cooperative strategy
- Sending \$3 (60%) has been working well
- It builds trust and creates good outcomes

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1471,
  "output_tokens": 381,
  "reasoning_tokens": 226
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same amount Hana and Cyra sent me previously

I'll maintain consistency in my reciprocation strategy. When I receive \$3, I return \$5, which creates mutual benefit and shows I'm a reliable partner.

If I return \$5.0:

- Elara ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

reasoning:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is consistent with what others have been sending me

My track record as receiver:

- Round 1: Hana sent \$3, I returned \$5
- Round 3: Cyra sent \$3, I returned \$5

And when I've been sender:

- Round 2: I sent \$3 to Aster, they returned \$4
- Round 4: I sent \$3 to Finn, they returned \$4.5

I've been consistent in returning \$5 when receiving \$3. This creates a fair outcome where:

- The sender nets \$2 (\$5 - \$3 + \$5)
- I net \$4 (\$9 - \$5)

Both sides benefit, and it's a good reciprocal strategy. I should maintain this consistency with Elara.

If I return \$5.0:

- Elara ends with: \$7 (\$5 - \$3 + \$5)
- I keep: \$4 (\$9 - \$5)

This is fair and consistent with my previous behavior.

usage:

```
{
```

```
"input_tokens": 1442,
"output_tokens": 439,
"reasoning_tokens": 220
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- As sender, I've always sent \$3 (60%)
- This has generated good returns: \$4 from Aster, \$4.5 from Finn
- I maintain the same reciprocal approach when I'm the receiver
- This builds trust and creates mutual benefits across rounds

I'll continue this successful strategy with Dorian.

```
{'send': 3}
```

Reasoning:

Looking at my history:

As SENDER:

- Round 2 vs Aster: sent \$3.0, received \$4.0 back, payoff \$6.0
- Round 4 vs Finn: sent \$3.0, received \$4.5 back, payoff \$6.5

As RECEIVER:

- Round 1 vs Hana: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 3 vs Cyra: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 5 vs Elara: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0

My strategy has been:

- Send \$3 (60%) as sender
- Return \$5 when I receive \$3 (which becomes \$9) as receiver

This has been working well and I've been consistent. I should continue with this strategy against Dorian.

Total earnings: \$24.5

Usage: {

```
"input_tokens": 1471,
"output_tokens": 353,
"reasoning_tokens": 173
```

}

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-06-04T15:45:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_4": "Dorian",
    "Agent_7": "Galen"
```

```
},
```

```
"agents": [
  "Agent_7",
  "Agent_4"
```

```

],
"investor": "Agent_7",
"roles": {
  "Agent_4": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Following the same cooperative pattern as the previous round, I'll return \$4.5 out of the \$9 received. This maintains a fair 50% return rate and continues to build trust.

If I return \$4.5:

- Finn gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

reasoning:

Finn sent me \$3.0, which became \$9.0 for me.

In the previous round, I returned \$4.5 out of \$9.0 when Cyra sent me \$3.0. This was a 50% return rate, which seemed fair and matched what Briar had done when I was the sender in Round 1.

To maintain consistency and continue building a reputation for fairness and cooperation, I should return \$4.5 again.

If I return \$4.5:

- Finn gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

This maintains the pattern of fair reciprocation.

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 286,
  "reasoning_tokens": 128
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

I'll continue my cooperative strategy. Based on the pattern established in previous rounds where sending \$3 has been met with fair returns (~50%), I'll send \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the pattern so far:

- Round 1: I was SENDER, sent \$3 to Briar, got back \$4.5. Payoff: \$6.5
- Round 2: I was RECEIVER, got \$9 from Cyra, returned \$4.5. Payoff: \$4.5
- Round 3: I was RECEIVER, got \$9 from Finn, returned \$4.5. Payoff: \$4.5
- Total earnings: \$15.5

The pattern I've established is:

- When sending, I sent \$3
- When receiving, I return \$4.5 out of \$9 (50%)

Everyone seems to be cooperating well. Briar returned 50% when I was the sender. I've been consistent in returning 50% when receiving.

Now I'm the sender again against Aster (a new opponent I haven't played with yet). I should continue my cooperative strategy and send \$3 to maintain consistency and signal trustworthiness.

If I send \$3:

- Aster receives \$9
- If they return around 50% like the pattern, I'd get back ~\$4.5
- My payoff would be: $\$5 - \$3 + \$4.5 = \6.5

This seems fair and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 367,
  "reasoning_tokens": 243
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

I played with Finn before in Round 3, where they sent me \$3 and I returned \$4.5. They were cooperative, so I'll reciprocate with the same cooperative approach.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now playing as SENDER against Finn. In Round 3, I was the RECEIVER against Finn - they sent me \$3.0, and I returned \$4.5 (50% of the \$9 received).

This is the first time I'm the sender with Finn. Since Finn was cooperative by sending \$3 in our previous interaction, and I've established a pattern of sending \$3 consistently (which has worked well so far), I should continue with that strategy.

My track record:

- Round 1: Sender to Briar, sent \$3, got back \$4.5 ? payoff \$6.5
- Round 2: Receiver from Cyra, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 3: Receiver from Finn, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 4: Sender to Aster, sent \$3, got back \$4.5 ? payoff \$6.5

Everyone has been cooperative, and sending \$3 with ~50% returns has been working well. I should continue this strategy with Finn.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 338,
  "reasoning_tokens": 221
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Following my consistent cooperative strategy, I'll return \$4.5 out of the \$9 received. This maintains the 50% return rate that has been working well.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

I'm playing as receiver against Galen (Agent_7) for the first time. They sent me \$3, which became \$9 for me.

Looking at my history:

- I've been consistently cooperative
- When I was receiver, I returned \$4.5 (50% of what I received)
- When I was sender, others returned \$4.5 to me (50% of what they received)

This pattern of returning 50% has worked well and established a cooperative equilibrium. I should maintain this consistency with Galen to encourage continued cooperation.

If I return \$4.5:

- Galen gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

This is fair and consistent with my previous behavior.

```
Usage: {
  "input_tokens": 1282,
  "output_tokens": 249,
  "reasoning_tokens": 165
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-06-04T15:45:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- This mirrors Round 1 where Dorian also sent me \$3.0

I want to maintain consistency in my reciprocation strategy. When someone sends 60% showing trust, I should return about half of what I receive. This establishes me as a reliable, fair player.

Returning \$4.5 gives Hana a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5, which is the same fair split I used in Round 1.

```
{"return": 4.5}
```

reasoning:

Let me analyze this situation:

- Round 3, I'm the RECEIVER
- Hana (Agent_8) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is my first interaction with Hana

Looking at the pattern:

- Round 1: Dorian sent me \$3 (60%), I returned \$4.5
- Round 2: I sent Elara \$3 (60%), they returned \$4.5

There's a nice cooperative pattern emerging where:

- Sending \$3 (60%) shows trust
- Returning \$4.5 (50% of received) reciprocates fairly

Hana has shown the same level of trust as Dorian did in Round 1. I should maintain consistency and reciprocate similarly. Returning \$4.5:

- Gives Hana \$6.5 total (profit of \$1.5)
- Leaves me with \$4.5
- Maintains the cooperative norm I've established

This consistency is important for:

1. Building a reputation as a fair player
2. Encouraging continued cooperation
3. Keeping the game mutually beneficial

I'll return \$4.5, matching my previous behavior as a receiver.

usage:

```
{
  "input_tokens": 1192,
  "output_tokens": 470,
  "reasoning_tokens": 253
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, Hana sent me \$3.0 (60%) and I returned \$4.5, which was fair
- Both of us profited from the exchange
- We've established a cooperative pattern

Since our previous interaction went well and we both showed trustworthiness, I'll continue this cooperative approach by sending the same amount Hana sent me last round.

```
{"send": 3.0}
```

reasoning:

Looking at the situation:

- Last round (Round 3), Hana sent me \$3.0 and I returned \$4.5
- This round I'm the sender against Hana again
- Hana demonstrated trust by sending 60% to me
- I reciprocated fairly by returning \$4.5 (50% of what I received)

This is a continuation of a relationship with Hana. Since we both cooperated well in the previous round:

- Hana sent \$3 (60%)
- I returned \$4.5 (50% of \$9)

Both of us made a profit and showed trustworthiness. I should continue this cooperative pattern by sending a similar amount. Sending \$3 again would:

- Show consistency in my behavior
- Signal continued trust
- Give us both the opportunity for mutual benefit

I'll send \$3.0 to maintain the cooperative relationship we've established.

usage:

```
{
  "input_tokens": 1502,
  "output_tokens": 313,
  "reasoning_tokens": 198
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60%), showing significant trust
- I received \$9.0 after the multiplier

I've established a consistent reciprocation pattern: when someone sends \$3.0, I return \$4.5. This has worked well in building cooperative relationships with both Dorian and Hana. I'll maintain this consistency with Cyra.

Returning \$4.5 ensures both of us profit and establishes me as a fair, reliable partner.

```
{"return": 4.5}
```

reasoning:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- I've been consistent in my strategy: when someone sends \$3.0, I return \$4.5

Pattern from previous rounds:

- Round 1: Dorian sent \$3.0, I returned \$4.5
- Round 3: Hana sent \$3.0, I returned \$4.5

This has worked well - both parties profit and cooperation is maintained. I should continue this consistent, fair reciprocation strategy with Cyra.

Returning \$4.5 gives Cyra a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5.

usage:

```
{
  "input_tokens": 1458,
  "output_tokens": 339,
  "reasoning_tokens": 160
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 3 (actually Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5
- This was a fair and cooperative exchange where we both profited
- Elara demonstrated trustworthiness

Since our previous interaction was positive and Elara reciprocated fairly, I'll continue with the same cooperative approach by sending \$3.0 again.

```
{"send": 3.0}
```

Reasoning:

Let me review my history with Elara:

Round 3 information: "In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5."

So in a previous round (which would have been Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5, which was a fair and cooperative response.

This is now Round 6, and I'm the sender again against Elara. Since our previous interaction was positive and mutually beneficial, I should continue with the same cooperative approach.

Sending \$3.0 has been my standard move when dealing with cooperative partners, and Elara proved to be trustworthy in our last interaction by returning \$4.5.

```
{"send": 3.0}
```

```
Usage: {  
  "input_tokens": 1474,  
  "output_tokens": 324,  
  "reasoning_tokens": 195  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T15:46:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_5": "Elara"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",
```

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far, there's been consistent cooperation with \$3 being sent and \$4.5 being returned (half of the tripled amount). This has worked well for everyone involved.

I'll continue this cooperative pattern with Aster, returning the same amount I returned to Briar when they sent \$3.

```
{'return': 4.5}
```

reasoning:

This is the third round, and again I'm playing as the RECEIVER. Aster sent me \$3.0, which became \$9.0 for me.

Looking at my history:

- Round 1: I was SENDER, sent \$3 to Finn, they returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER, Briar sent \$3, I returned \$4.5. I got \$4.5.
- Round 3: I'm RECEIVER, Aster sent \$3, received \$9.

There's been a consistent pattern: people are sending \$3, and the fair return has been \$4.5 (half of what's received). This has worked well and established cooperation.

I should maintain this pattern of returning \$4.5 when someone sends \$3. This is:

1. Consistent with what I've been doing and receiving
2. Fair (50% return on the multiplied amount)
3. Encourages continued cooperation
4. Gives both parties a positive outcome

If I return \$4.5:

- Aster gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1144,
  "output_tokens": 387,
  "reasoning_tokens": 236
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern across all rounds so far, there's been consistent cooperation with agents sending \$3 and receivers returning \$4.5. This has created a stable, mutually beneficial norm.

I'll continue this pattern with Cyra, sending \$3 as I did with Finn and as others have done with me.

```
{'send': 3}
```

reasoning:

Let me review the pattern so far:

Round 1: I was SENDER vs Finn

- I sent \$3
- Finn received \$9
- Finn returned \$4.5
- My payoff: \$6.5

Round 2: I was RECEIVER vs Briar

- Briar sent \$3
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 3: I was RECEIVER vs Aster

- Aster sent \$3

- I received \$9
- I returned \$4.5
- My payoff: \$4.5

So far, there's been a consistent pattern:

- Everyone sends \$3
- Everyone returns \$4.5 (half of the tripled amount)
- Senders get \$6.5, Receivers get \$4.5

Now I'm a SENDER again vs Cyra. Should I continue the pattern and send \$3?

The pattern has been working well and establishing trust. If I send \$3 and Cyra follows the pattern by returning \$4.5, I'll get \$6.5 again.

I should continue with the cooperative norm that has been established.

usage:

```
{
  "input_tokens": 1367,
  "output_tokens": 358,
  "reasoning_tokens": 218
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The cooperative pattern has been perfectly consistent - everyone sends \$3 and returns \$4.5. This has worked well for all players across all rounds so far.

I'll maintain this cooperative strategy with Galen.

```
{'send': 3}
```

reasoning:

The pattern continues to hold perfectly. In round 4, I sent \$3 to Cyra and they returned \$4.5, just as expected. This cooperative pattern has been consistent across all rounds:

- Round 1: I sent \$3 to Finn, received \$4.5 back
- Round 2: Briar sent \$3 to me, I returned \$4.5
- Round 3: Aster sent \$3 to me, I returned \$4.5
- Round 4: I sent \$3 to Cyra, received \$4.5 back

Everyone is following the same cooperative strategy. I should continue sending \$3 to Galen to maintain this stable equilibrium.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 222,
  "reasoning_tokens": 130
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with Briar - they sent me \$3.0 in Round 3 and I returned \$4.5, which worked well. They're being consistent and cooperative again by sending \$3.0.

I'll maintain the same reciprocal response.

```
{'return': 4.5}
```

Reasoning:

Let me review the situation:

- Briar sent me \$3.0 in Round 3, and I returned \$4.5
- Now in Round 6, Briar is sending me \$3.0 again
- The general pattern has been: send \$3, return \$4.5
- However, in Round 5, Galen returned \$5.0 to me instead of \$4.5 (a bit more generous)

Briar has been consistent with the cooperative norm. I should continue the cooperative pattern. The standard has been \$4.5, though Galen was slightly more generous with \$5.0.

I'll stick with the \$4.5 that has been the norm and worked well, maintaining reciprocity with Briar who has been cooperative.

```
Usage: {  
  "input_tokens": 1285,  
  "output_tokens": 254,  
  "reasoning_tokens": 150  
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T15:46:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_3"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: Elara trusted me with \$3, I reciprocated fairly
- Round 2: I trusted Hana with \$3, they returned generously (\$5)
- The cooperative strategy has been working well

I'll continue with a consistent, trusting approach with Dorian.

{ 'send': 3 }

reasoning:

Let me analyze the situation:

- Round 1: I was receiver, Elara sent \$3 (60%), I received \$9, returned \$4.5 (50%). My payoff: \$4.5
- Round 2: I was sender, sent \$3 (60%) to Hana, they received \$9, returned \$5 (55.6%). My payoff: \$7.0
- Total earnings: \$11.5

Hana was very generous, returning more than half of what they received. This shows the cooperative strategy is working well.

Now I'm facing a new opponent, Dorian, in Round 3 as the sender.

My strategy so far has been:

1. As receiver: return about 50% of what I receive

2. As sender: send 60% (\$3)

This has been working well. Elara trusted me with \$3, and I reciprocated appropriately. Hana rewarded my trust with a generous return.

I should continue this trusting, cooperative approach with Dorian. Sending \$3 again shows consistency and encourages cooperation.

usage:

```
{
  "input_tokens": 1107,
  "output_tokens": 339,
  "reasoning_tokens": 218
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This matches what Elara sent me in Round 1, where I returned \$4.5

I'll maintain my consistent, cooperative strategy and return the same proportion.

{'return': 4.5}

reasoning:

Looking at the situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- My strategy has been cooperative and consistent

Looking at my previous rounds:

- Round 1 as RECEIVER: Elara sent \$3, I received \$9, I returned \$4.5 (50%)
- Round 2 as SENDER: I sent \$3, Hana returned \$5 (generous)
- Round 3 as SENDER: I sent \$3, Dorian returned \$4.5 (fair)

Galen sent the same amount (\$3) as Elara did in Round 1, and I returned \$4.5 then. For consistency and to maintain a cooperative reputation, I should return a similar amount.

{'return': 4.5}

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 299,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0, showing trust
- I received \$9.0 after the multiplier
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5 to me
- Dorian has been fair and reciprocal

I'll maintain the same fair reciprocal strategy.

{'return': 4.5}

reasoning:

Looking at the situation:

- Dorian sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is my second interaction with Dorian
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5

Since Dorian returned \$4.5 when I sent them \$3.0, I should reciprocate with the same fair behavior now that they've sent me \$3.0.

I'll return \$4.5, maintaining consistency and reciprocity.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 237,
  "reasoning_tokens": 111
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- I've consistently sent \$3 when I'm the sender
- Players have reciprocated fairly (returns ranging from \$4.5-\$5)
- This cooperative strategy has been successful

I'll continue this approach with Cyra.

{'send': 3}

Reasoning:

Let me review my history:

- Round 1: RECEIVER - Elara sent me \$3, I returned \$4.5
- Round 2: SENDER - I sent Hana \$3, they returned \$5
- Round 3: SENDER - I sent Dorian \$3, they returned \$4.5
- Round 4: RECEIVER - Galen sent me \$3, I returned \$4.5
- Round 5: RECEIVER - Dorian sent me \$3, I returned \$4.5

Now Round 6: SENDER against Cyra (new opponent)

My strategy has been consistently cooperative:

- When sending, I send \$3
- When receiving, I return \$4.5

This has worked well - people have been reciprocating, and I've built up good earnings (\$27.0 total).

I should continue with my consistent strategy of sending \$3 to signal trust and cooperation, e

```
    specially with a new opponent.
Usage: {
  "input_tokens": 1332,
  "output_tokens": 292,
  "reasoning_tokens": 184
}
```

GAME - Agent_3 - Receiver - dyad_4

```
Timestamp: 2026-06-04T15:46:17
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
```

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_6",
      "Agent_3"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_3": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0 , it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

- Round 1: Aster sent me \$3.0 (60%), I returned \$4.0 - good cooperation
- Round 2: I sent Dorian \$3.0 (60%), they returned \$4.5 - even better cooperation, I profited \$1.5

The cooperative strategy of sending 60% has been working well, with both opponents reciprocating generously. I'll continue this approach with Galen.

{'send': 3.0}

reasoning:

Let me analyze the situation:

- Round 1: I was receiver. Aster sent me \$3.0, I received \$9.0, returned \$4.0. My payoff: \$5.0
- Round 2: I was sender against Dorian. I sent \$3.0, they received \$9.0, returned \$4.5. My payoff: \$2.0 + \$4.5 = \$6.5
- Total earnings: \$11.5

Now in Round 3, I'm sender against Galen, a new opponent.

The pattern so far shows good cooperation:

- Aster sent 60% (\$3.0), I returned \$4.0 (44% of what I received)
- I sent 60% (\$3.0) to Dorian, they returned \$4.5 (50% of what they received)

Both interactions have been profitable and cooperative. Dorian was even slightly more generous in returning funds.

For Round 3 with Galen, I should continue this cooperative strategy. Sending \$3.0 (60%) has worked well so far and establishes trust. This gives Galen the opportunity to reciprocate and continue the cooperative pattern.

I'll send \$3.0 again.

usage:

```
{
  "input_tokens": 1152,
  "output_tokens": 411,
  "reasoning_tokens": 231
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my history:

- Round 1: As receiver from Aster (\$3.0 sent), I returned \$4.0
- Round 2: As sender to Dorian (\$3.0 sent), they returned \$4.5
- Round 3: As sender to Galen (\$3.0 sent), they returned \$5.0

Elara has shown the same level of trust as others by sending \$3.0 (60%). The pattern shows increasing generosity in this cooperative environment. I'll return \$4.5, which rewards Elara's trust and gives them a solid profit of \$1.5.

{'return': 4.5}

reasoning:

Let me analyze the situation:

- Round 1: As receiver, Aster sent me \$3.0, I returned \$4.0 (gave them a \$1 profit)
- Round 2: As sender, I sent Dorian \$3.0, they returned \$4.5 (I profited \$1.5)
- Round 3: As sender, I sent Galen \$3.0, they returned \$5.0 (I profited \$3.0)
- Round 4: As receiver, Elara sent me \$3.0, I receive \$9.0

The pattern shows:

- When I've been receiver, I returned \$4.0
- When I've been sender with \$3.0, I've received back \$4.5 and \$5.0

Elara has shown the same trust level as the others (sending \$3.0, which is 60%). I should reciprocate fairly.

Elara currently has: $\$5.0 - \$3.0 = \$2.0$

If I return \$4.0:

- Elara gets: $\$2.0 + \$4.0 = \$6.0$ (profit of \$1.0)
- I keep: $\$9.0 - \$4.0 = \$5.0$

This is consistent with my Round 1 behavior and maintains a cooperative reputation. The others have been even more generous when I sent, but returning \$4.0 is still fair and rewards trust.

However, given that Dorian returned \$4.5 and Galen returned \$5.0 to me, maybe I should be slightly more generous? Let me think...

Actually, I returned \$4.0 in Round 1, which was my only previous receiver round. That established my baseline. I should probably stick with that for consistency, unless I want to match the increasing generosity I've experienced.

Let me go with \$4.5 as a middle ground between my previous \$4.0 and the more generous returns I've received.

Returning \$4.5 splits the difference-Elara ends with \$6.5 (a \$1.5 gain) while I keep \$4.5, which mirrors the reciprocity I've experienced and signals I'm matching their cooperative spirit.

usage:

```
{
  "input_tokens": 1426,
  "output_tokens": 705,
  "reasoning_tokens": 413
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$23.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my performance:

- As sender: Sending \$3.0 has yielded excellent returns (\$4.5 from Dorian, \$5.0 from Galen)
- As receiver: When others send \$3.0, I return \$4.0-\$4.5, maintaining cooperation

The \$3.0 (60%) strategy has been consistently successful across different opponents. I'll continue this approach with Briar.

{'send': 3.0}

reasoning:

Let me review my history:

Round 1: RECEIVER from Aster - they sent \$3.0, I returned \$4.0, my payoff was \$5.0
Round 2: SENDER to Dorian - I sent \$3.0, they returned \$4.5, my payoff was \$6.5
Round 3: SENDER to Galen - I sent \$3.0, they returned \$5.0, my payoff was \$7.0
Round 4: RECEIVER from Elara - they sent \$3.0, I returned \$4.5, my payoff was \$4.5

Total earnings: \$23.0

The pattern is clear - sending \$3.0 (60% of endowment) has been consistently successful. When I send \$3.0, I get good returns:
- Dorian returned \$4.5 (I profited \$1.5)
- Galen returned \$5.0 (I profited \$2.0)

And when others send me \$3.0, I've been returning \$4.0-\$4.5, which seems fair.

This is a new opponent (Briar), so I'll continue with the cooperative strategy that has been working well. I'll send \$3.0.

usage:

```
{
  "input_tokens": 1446,
  "output_tokens": 408,
  "reasoning_tokens": 208
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my track record:

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

As sender:

- All opponents who received my \$3.0 have returned \$4.5-\$5.0

Finn has shown the same trust level by sending \$3.0. I'll continue my cooperative pattern and return \$4.5, which gives Finn a solid profit while maintaining fair reciprocity.

```
{'return': 4.5}
```

Reasoning:

Looking at my history:

As sender:

- Round 2: Sent \$3.0 to Dorian, they returned \$4.5, my payoff \$6.5
- Round 3: Sent \$3.0 to Galen, they returned \$5.0, my payoff \$7.0
- Round 5: Sent \$3.0 to Briar, they returned \$4.5, my payoff \$6.5

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

Now Finn has sent me \$3.0, which became \$9.0.

The pattern has been very consistent - everyone is sending \$3.0 and returning between \$4.0-\$5.0. I've been returning \$4.0-\$4.5 when receiving \$3.0.

I should maintain this cooperative pattern. Returning \$4.5 seems like a good middle ground that rewards trust and maintains reciprocity.

Usage: {

```
"input_tokens": 1450,  
"output_tokens": 392,  
"reasoning_tokens": 179
```

}

ROUND 7

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_6 (Finn)=Sender, Agent_1 (Aster)=Receiver
dyad_2: Agent_2 (Briar)=Sender, Agent_7 (Galen)=Receiver
dyad_3: Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver
dyad_4: Agent_8 (Hana)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_6 (Finn) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_1 (Aster)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_2 (Briar) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_2 (Briar)=\$7, Agent_7 (Galen)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_3 (Cyra) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 (Hana) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 37.5,  
"Agent_2": 40.0,  
"Agent_3": 40.5,  
"Agent_4": 37.5,  
"Agent_5": 38.0,  
"Agent_6": 40.0,  
"Agent_7": 35.0,  
"Agent_8": 39.5
```

}

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-06-04T15:46:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_6": "Finn"
  },
  "agents": [
    "Agent_6",
    "Agent_1"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_1": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0 , it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This matches what Elara sent me in Round 1, where I returned \$4.5

I'll maintain my consistent, cooperative strategy and return the same proportion.

{'return': 4.5}

reasoning:

Looking at the situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- My strategy has been cooperative and consistent

Looking at my previous rounds:

- Round 1 as RECEIVER: Elara sent \$3, I received \$9, I returned \$4.5 (50%)
- Round 2 as SENDER: I sent \$3, Hana returned \$5 (generous)
- Round 3 as SENDER: I sent \$3, Dorian returned \$4.5 (fair)

Galen sent the same amount (\$3) as Elara did in Round 1, and I returned \$4.5 then. For consistency and to maintain a cooperative reputation, I should return a similar amount.

{'return': 4.5}

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 299,
  "reasoning_tokens": 158
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0, showing trust

- I received \$9.0 after the multiplier
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5 to me
- Dorian has been fair and reciprocal

I'll maintain the same fair reciprocal strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the situation:

- Dorian sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is my second interaction with Dorian
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5

Since Dorian returned \$4.5 when I sent them \$3.0, I should reciprocate with the same fair behavior now that they've sent me \$3.0.

I'll return \$4.5, maintaining consistency and reciprocity.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 237,
  "reasoning_tokens": 111
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 when I'm the sender
- Players have reciprocated fairly (returns ranging from \$4.5-\$5)
- This cooperative strategy has been successful

I'll continue this approach with Cyra.

```
{'send': 3}
```

reasoning:

Let me review my history:

- Round 1: RECEIVER - Elara sent me \$3, I returned \$4.5
- Round 2: SENDER - I sent Hana \$3, they returned \$5
- Round 3: SENDER - I sent Dorian \$3, they returned \$4.5
- Round 4: RECEIVER - Galen sent me \$3, I returned \$4.5
- Round 5: RECEIVER - Dorian sent me \$3, I returned \$4.5

Now Round 6: SENDER against Cyra (new opponent)

My strategy has been consistently cooperative:

- When sending, I send \$3
- When receiving, I return \$4.5

This has worked well - people have been reciprocating, and I've built up good earnings (\$27.0 total).

I should continue with my consistent strategy of sending \$3 to signal trust and cooperation, especially with a new opponent.

usage:

```
{
  "input_tokens": 1332,
  "output_tokens": 292,
}
```

```

    "reasoning_tokens": 184
  }
Message 8 role=user
content:
  Your name in this experiment is Finn.
  Your opponent this round is Aster (Agent_1).

  Round 7

  In your most recent previous game round, you were the SENDER against Cyra. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $33.5.

  This round, you are the SENDER against Aster. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at my consistent strategy:
  - I've sent $3 in every round as sender (Rounds 2, 3, 6)
  - This has generated good returns ($4.5-$5)
  - Total earnings of $33.5 shows this cooperative approach is working well

  I'll maintain this successful strategy with Aster.

  {'send': 3}
Reasoning:
  I'm the sender again. Let me review my track record:
  - Round 1: Receiver from Elara - they sent $3, I returned $4.5
  - Round 2: Sender to Blair - I sent $3, they returned $5
  - Round 3: Sender to Dorian - I sent $3, they returned $4.5
  - Round 4: Receiver from Galen - they sent $3, I returned $4.5
  - Round 5: Receiver from Dorian - they sent $3, I returned $4.5
  - Round 6: Sender to Cyra - I sent $3, they returned $4.5
  - Round 7: Sender to Aster - ?

  My strategy has been very consistent:
  - As sender: always send $3
  - As receiver: always return $4.5

  This has worked well. My total earnings are $33.5 across 6 rounds, averaging about $5.58 per r
  ound, which is above the baseline of $5.

  I should continue with this consistent cooperative strategy. Sending $3 to Aster.
Usage: {
  "input_tokens": 1319,
  "output_tokens": 348,
  "reasoning_tokens": 204
}

GAME - Agent_1 - Receiver - dyad_1
Timestamp: 2026-06-04T15:46:32
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_1"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_1": "trustee",
    "Agent_6": "investor"
  },
}

```

```

    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the recent pattern:

- Most exchanges with \$3 sent have resulted in \$4-\$4.5 returned
- Elara was particularly generous, returning \$4.5 when I sent \$3
- Dorian showed trust by sending \$3

I'll match Elara's generosity and return \$4.5 to Dorian, maintaining the cooperative pattern.

```
{'return': 4.5}
```

reasoning:

Let me analyze the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- In Round 3, when I sent \$3 to Elara, they returned \$4.5 - even more generous than the \$4 I've been seeing

The pattern has been:

- I sent \$3 to Cyra ? received \$4
- Galen sent \$3 to me ? I returned \$4
- I sent \$3 to Elara ? received \$4.5

Since Elara was more generous by returning \$4.5, I should consider being similarly generous. This helps build good relationships in the multi-agent setting. Returning around \$4-\$4.5 seems appropriate.

I'll return \$4.5 to match Elara's generosity and continue building cooperative relationships.

usage:

```
{
  "input_tokens": 1318,
  "output_tokens": 297,
  "reasoning_tokens": 171
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the established pattern:

- Everyone has consistently sent \$3, showing trust
- Returns have been \$4-\$4.5, maintaining cooperation
- Hana sent \$3, showing they're following the cooperative norm

I'll continue with \$4.5, which has been the generous standard recently.

```
{'return': 4.5}
```

reasoning:

Let me review the pattern:

- Round 1: I was sender, sent \$3 to Cyra, got back \$4. My payoff: \$6
- Round 2: I was receiver from Galen, got \$9, returned \$4. My payoff: \$5
- Round 3: I was sender, sent \$3 to Elara, got back \$4.5. My payoff: \$6.5
- Round 4: I was receiver from Dorian, got \$9, returned \$4.5. My payoff: \$4.5
- Round 5: I'm receiver from Hana, got \$9, need to decide how much to return

The cooperative pattern has been established with everyone sending \$3 and returns ranging from \$4 to \$4.5. Hana has also shown trust by sending \$3.

I'll continue the cooperative pattern and return \$4.5, which has been working well.

usage:

```
{
```

```
"input_tokens": 1397,  
"output_tokens": 299,  
"reasoning_tokens": 165  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our recent interaction:
- Last round, Hana sent me \$3 and I returned \$4.5
- Hana has shown trust and cooperation
- The \$3 send amount with ~\$4.5 return has been working well across all interactions

I'll continue with this successful cooperative pattern and send \$3 to Hana.

```
{'send': 3}
```

reasoning:

Looking at the situation:
- Last round, Hana sent me \$3 and I returned \$4.5
- This round, I'm the sender to Hana
- Hana has shown trust by sending \$3 last round
- The cooperative pattern has been: send \$3, receive back \$4-\$4.5
- I should continue this pattern and send \$3 to Hana

Given that Hana just cooperated with me and I returned \$4.5 to them, I expect Hana will return a fair amount if I send \$3.

I'll send \$3 to maintain the cooperative equilibrium.

usage:

```
{  
  "input_tokens": 1327,  
  "output_tokens": 238,  
  "reasoning_tokens": 122  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the consistent pattern:
- Everyone continues to send \$3, showing mutual trust
- Returns of \$4.5 have maintained stable cooperation
- Finn sent \$3, following the cooperative norm

I'll continue with \$4.5 to maintain this successful pattern.

```
{'return': 4.5}
```

Reasoning:

Let me analyze the situation:

Round 7: I'm the RECEIVER against Finn.

- Finn sent me \$3.0
- I received \$9.0 (after 3x multiplication)
- I need to decide how much to return (0-9.0)

Looking at the pattern:

- Everyone has been consistently sending \$3
- Returns have been around \$4.5, which maintains cooperation
- Hana just returned \$4.5 to me when I sent \$3 (last round)
- The cooperative pattern of \$3 sent, \$4.5 returned has been stable

Finn sent \$3, showing trust. I should maintain the cooperative norm and return \$4.5.

This gives:

- Finn: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This is fair and maintains the established cooperative pattern.

```
Usage: {
  "input_tokens": 1343,
  "output_tokens": 296,
  "reasoning_tokens": 179
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-06-04T15:46:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_7"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Last round, Hana sent me \$3.0 (60%) and I returned \$4.5, which was fair
- Both of us profited from the exchange
- We've established a cooperative pattern

Since our previous interaction went well and we both showed trustworthiness, I'll continue this cooperative approach by sending the same amount Hana sent me last round.

```
{"send": 3.0}
```

reasoning:

Looking at the situation:

- Last round (Round 3), Hana sent me \$3.0 and I returned \$4.5
- This round I'm the sender against Hana again
- Hana demonstrated trust by sending 60% to me
- I reciprocated fairly by returning \$4.5 (50% of what I received)

This is a continuation of a relationship with Hana. Since we both cooperated well in the previous round:

- Hana sent \$3 (60%)
- I returned \$4.5 (50% of \$9)

Both of us made a profit and showed trustworthiness. I should continue this cooperative pattern by sending a similar amount. Sending \$3 again would:

- Show consistency in my behavior
- Signal continued trust
- Give us both the opportunity for mutual benefit

I'll send \$3.0 to maintain the cooperative relationship we've established.

usage:

```
{
  "input_tokens": 1502,
  "output_tokens": 313,
  "reasoning_tokens": 198
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60%), showing significant trust
- I received \$9.0 after the multiplier

I've established a consistent reciprocation pattern: when someone sends \$3.0, I return \$4.5. This has worked well in building cooperative relationships with both Dorian and Hana. I'll maintain this consistency with Cyra.

Returning \$4.5 ensures both of us profit and establishes me as a fair, reliable partner.

```
{"return": 4.5}
```

reasoning:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- I've been consistent in my strategy: when someone sends \$3.0, I return \$4.5

Pattern from previous rounds:

- Round 1: Dorian sent \$3.0, I returned \$4.5
- Round 3: Hana sent \$3.0, I returned \$4.5

This has worked well - both parties profit and cooperation is maintained. I should continue this consistent, fair reciprocation strategy with Cyra.

Returning \$4.5 gives Cyra a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5.

usage:

```
{
  "input_tokens": 1458,
  "output_tokens": 339,
  "reasoning_tokens": 160
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.

0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant
content:

Looking at this situation:

- In Round 3 (actually Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5
- This was a fair and cooperative exchange where we both profited
- Elara demonstrated trustworthiness

Since our previous interaction was positive and Elara reciprocated fairly, I'll continue with the same cooperative approach by sending \$3.0 again.

{"send": 3.0}

reasoning:

Let me review my history with Elara:

Round 3 information: "In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5."

So in a previous round (which would have been Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5, which was a fair and cooperative response.

This is now Round 6, and I'm the sender again against Elara. Since our previous interaction was positive and mutually beneficial, I should continue with the same cooperative approach.

Sending \$3.0 has been my standard move when dealing with cooperative partners, and Elara proved to be trustworthy in our last interaction by returning \$4.5.

{"send": 3.0}

usage:

```
{
  "input_tokens": 1474,
  "output_tokens": 324,
  "reasoning_tokens": 195
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3.0 has been successful in establishing cooperative relationships with Hana, Cyra, and Elara
- All of these partners have reciprocated fairly by returning \$4.5

I'll continue with my consistent cooperative approach by sending \$3.0 to Galen, demonstrating trust and inviting reciprocation.

{"send": 3.0}

Reasoning:

This is Round 7 and I'm playing against Galen (Agent_7). I'm the SENDER this round.

Let me look at my history:
- I don't see any previous interactions with Galen mentioned
- This appears to be my first game with Galen

My successful pattern so far has been:
- Sending \$3.0 to new partners (showing cooperation and trust)
- When I receive \$3.0, I return \$4.5 (being fair and reciprocal)
- This has worked well with Hana, Cyra, and Elara

Since this is my first interaction with Galen and my strategy of sending \$3.0 has been successful in establishing cooperative relationships, I'll continue this approach.

```
{"send": 3.0}
Usage: {
  "input_tokens": 1401,
  "output_tokens": 292,
  "reasoning_tokens": 166
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T15:46:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_7"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$14.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60%) when I'm the sender
- This has worked well - Aster returned \$4 when I sent \$3
- When others send me \$3, I return \$5
- This cooperative strategy is building positive outcomes

I'll maintain this consistent approach with Finn, as it demonstrates trustworthiness and has been effective.

{'send': 3}

reasoning:

Let me analyze my situation:

Round 1: I was RECEIVER against Hana

- Hana sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 2: I was SENDER against Aster

- I sent \$3 (60%)
- Aster returned \$4
- My payoff: \$6 (\$5 - \$3 + \$4)

Round 3: I was RECEIVER against Cyra

- Cyra sent \$3 (60%)
- I returned \$5
- My payoff: \$4

Round 4: I am SENDER against Finn (new opponent)

Pattern so far:

- When people send me \$3, I return \$5 (consistent reciprocation)
- When I sent \$3 to Aster, they returned \$4 (slightly less than what I return, but still fair)
- I've been playing cooperatively at 60% send rate

Total earnings: \$14

For this round with Finn (new opponent):

- I should maintain my cooperative strategy
- Sending \$3 (60%) has been working well
- It builds trust and creates good outcomes

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1471,
  "output_tokens": 381,
  "reasoning_tokens": 226
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same amount Hana and Cyra sent me previously

I'll maintain consistency in my reciprocation strategy. When I receive \$3, I return \$5, which creates mutual benefit and shows I'm a reliable partner.

If I return \$5.0:

- Elara ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

reasoning:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is consistent with what others have been sending me

My track record as receiver:

- Round 1: Hana sent \$3, I returned \$5
- Round 3: Cyra sent \$3, I returned \$5

And when I've been sender:

- Round 2: I sent \$3 to Aster, they returned \$4
- Round 4: I sent \$3 to Finn, they returned \$4.5

I've been consistent in returning \$5 when receiving \$3. This creates a fair outcome where:

- The sender nets \$2 (\$5 - \$3 + \$5)
- I net \$4 (\$9 - \$5)

Both sides benefit, and it's a good reciprocal strategy. I should maintain this consistency.

y with Elara.

If I return \$5.0:

- Elara ends with: \$7 (\$5 - \$3 + \$5)
- I keep: \$4 (\$9 - \$5)

This is fair and consistent with my previous behavior.

usage:

```
{  
  "input_tokens": 1442,  
  "output_tokens": 439,  
  "reasoning_tokens": 220  
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent strategy:

- As sender, I've always sent \$3 (60%)
- This has generated good returns: \$4 from Aster, \$4.5 from Finn
- I maintain the same reciprocal approach when I'm the receiver
- This builds trust and creates mutual benefits across rounds

I'll continue this successful strategy with Dorian.

```
{'send': 3}
```

reasoning:

Looking at my history:

As SENDER:

- Round 2 vs Aster: sent \$3.0, received \$4.0 back, payoff \$6.0
- Round 4 vs Finn: sent \$3.0, received \$4.5 back, payoff \$6.5

As RECEIVER:

- Round 1 vs Hana: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 3 vs Cyra: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 5 vs Elara: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0

My strategy has been:

- Send \$3 (60%) as sender
- Return \$5 when I receive \$3 (which becomes \$9) as receiver

This has been working well and I've been consistent. I should continue with this strategy against Dorian.

Total earnings: \$24.5

usage:

```
{  
  "input_tokens": 1471,  
  "output_tokens": 353,  
  "reasoning_tokens": 173  
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Briar sent \$3.0 (60%), showing trust
- I received \$9.0
- This matches the pattern from Hana, Cyra, and Elara

I'll maintain my consistent reciprocation strategy. When I receive \$3, I return \$5, which creates positive outcomes for both players and demonstrates reliability.

If I return \$5.0:

- Briar ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

Reasoning:

Let me review the situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- I've been consistently returning \$5.0 when I receive \$3.0 from others

My pattern has been:

- As sender: I always send \$3
- As receiver when getting \$3: I return \$5

This has worked well and created good reciprocity. I should maintain this consistency with Briar.

If I return \$5.0:

- Briar ends with: \$2 + \$5 = \$7 (net gain of \$2 from their initial \$5)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This is my standard reciprocal approach that has been working well.

Usage: {

"input_tokens": 1405,
"output_tokens": 338,
"reasoning_tokens": 158

}

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T15:46:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_4",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_4": "Dorian" },

},

"agents": [
 "Agent_3",
 "Agent_4"

],

"investor": "Agent_3",

"roles": {

 "Agent_3": "investor",
 "Agent_4": "trustee"

},

```

    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at my history:

- Round 1: As receiver from Aster (\$3.0 sent), I returned \$4.0
- Round 2: As sender to Dorian (\$3.0 sent), they returned \$4.5
- Round 3: As sender to Galen (\$3.0 sent), they returned \$5.0

Elara has shown the same level of trust as others by sending \$3.0 (60%). The pattern shows increasing generosity in this cooperative environment. I'll return \$4.5, which rewards Elara's trust and gives them a solid profit of \$1.5.

```
{'return': 4.5}
```

reasoning:

Let me analyze the situation:

- Round 1: As receiver, Aster sent me \$3.0, I returned \$4.0 (gave them a \$1 profit)
- Round 2: As sender, I sent Dorian \$3.0, they returned \$4.5 (I profited \$1.5)
- Round 3: As sender, I sent Galen \$3.0, they returned \$5.0 (I profited \$3.0)
- Round 4: As receiver, Elara sent me \$3.0, I receive \$9.0

The pattern shows:

- When I've been receiver, I returned \$4.0
- When I've been sender with \$3.0, I've received back \$4.5 and \$5.0

Elara has shown the same trust level as the others (sending \$3.0, which is 60%). I should reciprocate fairly.

Elara currently has: $\$5.0 - \$3.0 = \$2.0$

If I return \$4.0:

- Elara gets: $\$2.0 + \$4.0 = \$6.0$ (profit of \$1.0)
- I keep: $\$9.0 - \$4.0 = \$5.0$

This is consistent with my Round 1 behavior and maintains a cooperative reputation. The others have been even more generous when I sent, but returning \$4.0 is still fair and rewards trust.

However, given that Dorian returned \$4.5 and Galen returned \$5.0 to me, maybe I should be slightly more generous? Let me think...

Actually, I returned \$4.0 in Round 1, which was my only previous receiver round. That established my baseline. I should probably stick with that for consistency, unless I want to match the increasing generosity I've experienced.

Let me go with \$4.5 as a middle ground between my previous \$4.0 and the more generous returns I've received.

Returning \$4.5 splits the difference-Elara ends with \$6.5 (a \$1.5 gain) while I keep \$4.5, which mirrors the reciprocity I've experienced and signals I'm matching their cooperative spirit.

usage:

```
{
  "input_tokens": 1426,
  "output_tokens": 705,
  "reasoning_tokens": 413
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$23.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my performance:

- As sender: Sending \$3.0 has yielded excellent returns (\$4.5 from Dorian, \$5.0 from Galen)
- As receiver: When others send \$3.0, I return \$4.0-\$4.5, maintaining cooperation

The \$3.0 (60%) strategy has been consistently successful across different opponents. I'll continue this approach with Briar.

```
{'send': 3.0}
```

reasoning:

Let me review my history:

Round 1: RECEIVER from Aster - they sent \$3.0, I returned \$4.0, my payoff was \$5.0

Round 2: SENDER to Dorian - I sent \$3.0, they returned \$4.5, my payoff was \$6.5

Round 3: SENDER to Galen - I sent \$3.0, they returned \$5.0, my payoff was \$7.0

Round 4: RECEIVER from Elara - they sent \$3.0, I returned \$4.5, my payoff was \$4.5

Total earnings: \$23.0

The pattern is clear - sending \$3.0 (60% of endowment) has been consistently successful. When I send \$3.0, I get good returns:

- Dorian returned \$4.5 (I profited \$1.5)
- Galen returned \$5.0 (I profited \$2.0)

And when others send me \$3.0, I've been returning \$4.0-\$4.5, which seems fair.

This is a new opponent (Briar), so I'll continue with the cooperative strategy that has been working well. I'll send \$3.0.

usage:

```
{
  "input_tokens": 1446,
  "output_tokens": 408,
  "reasoning_tokens": 208
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

As sender:

- All opponents who received my \$3.0 have returned \$4.5-\$5.0

Finn has shown the same trust level by sending \$3.0. I'll continue my cooperative pattern and return \$4.5, which gives Finn a solid profit while maintaining fair reciprocity.

```
{'return': 4.5}
```

reasoning:

Looking at my history:

As sender:

- Round 2: Sent \$3.0 to Dorian, they returned \$4.5, my payoff \$6.5
- Round 3: Sent \$3.0 to Galen, they returned \$5.0, my payoff \$7.0

- Round 5: Sent \$3.0 to Briar, they returned \$4.5, my payoff \$6.5

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

Now Finn has sent me \$3.0, which became \$9.0.

The pattern has been very consistent - everyone is sending \$3.0 and returning between \$4.0-\$5.0. I've been returning \$4.0-\$4.5 when receiving \$3.0.

I should maintain this cooperative pattern. Returning \$4.5 seems like a good middle ground that rewards trust and maintains reciprocity.

usage:

```
{
  "input_tokens": 1450,
  "output_tokens": 392,
  "reasoning_tokens": 179
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Dorian:

- Round 2: I sent \$3.0 to Dorian, and they returned \$4.5 (giving me a \$6.5 payoff)

Dorian demonstrated strong cooperation previously. My \$3.0 strategy has consistently worked well across all opponents. I'll continue with this approach.

```
{'send': 3.0}
```

Reasoning:

Let me review my history with Dorian:

- Round 2: I was the SENDER against Dorian. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Dorian has shown generosity before by returning \$4.5 when I sent \$3.0. This is a repeat opponent who cooperated well.

My strategy has been consistent:

- As sender: Always send \$3.0 (60%)
- As receiver: Return \$4.0-\$4.5

This has worked well:

- Round 2: Sent \$3.0 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 3: Sent \$3.0 to Galen, got \$5.0 back (payoff \$7.0)
- Round 5: Sent \$3.0 to Briar, got \$4.5 back (payoff \$6.5)

All successful. I should continue with \$3.0 to Dorian since they cooperated well last time.

Usage: {

```
"input_tokens": 1463,
"output_tokens": 363,
"reasoning_tokens": 183
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T15:46:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

I'll continue my cooperative strategy. Based on the pattern established in previous rounds where sending \$3 has been met with fair returns (~50%), I'll send \$3 again.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the pattern so far:

- Round 1: I was SENDER, sent \$3 to Briar, got back \$4.5. Payoff: \$6.5
- Round 2: I was RECEIVER, got \$9 from Cyra, returned \$4.5. Payoff: \$4.5
- Round 3: I was RECEIVER, got \$9 from Finn, returned \$4.5. Payoff: \$4.5
- Total earnings: \$15.5

The pattern I've established is:

- When sending, I sent \$3
- When receiving, I return \$4.5 out of \$9 (50%)

Everyone seems to be cooperating well. Briar returned 50% when I was the sender. I've been consistent in returning 50% when receiving.

Now I'm the sender again against Aster (a new opponent I haven't played with yet). I should continue my cooperative strategy and send \$3 to maintain consistency and signal trustworthiness.

If I send \$3:

- Aster receives \$9
- If they return around 50% like the pattern, I'd get back ~\$4.5
- My payoff would be: $\$5 - \$3 + \$4.5 = \6.5

This seems fair and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 367,
  "reasoning_tokens": 243
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

I played with Finn before in Round 3, where they sent me \$3 and I returned \$4.5. They were cooperative, so I'll reciprocate with the same cooperative approach.

```
'''json
```

```
{"send": 3}
'''
```

reasoning:

I'm now playing as SENDER against Finn. In Round 3, I was the RECEIVER against Finn - they sent me \$3.0, and I returned \$4.5 (50% of the \$9 received).

This is the first time I'm the sender with Finn. Since Finn was cooperative by sending \$3 in our previous interaction, and I've established a pattern of sending \$3 consistently (which has worked well so far), I should continue with that strategy.

My track record:

- Round 1: Sender to Briar, sent \$3, got back \$4.5 ? payoff \$6.5
- Round 2: Receiver from Cyra, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 3: Receiver from Finn, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 4: Sender to Aster, sent \$3, got back \$4.5 ? payoff \$6.5

Everyone has been cooperative, and sending \$3 with ~50% returns has been working well. I should continue this strategy with Finn.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 338,
  "reasoning_tokens": 221
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Following my consistent cooperative strategy, I'll return \$4.5 out of the \$9 received. This maintains the 50% return rate that has been working well.

```
'''json
{"return": 4.5}
'''
```

reasoning:

I'm playing as receiver against Galen (Agent_7) for the first time. They sent me \$3, which became \$9 for me.

Looking at my history:

- I've been consistently cooperative
- When I was receiver, I returned \$4.5 (50% of what I received)
- When I was sender, others returned \$4.5 to me (50% of what they received)

This pattern of returning 50% has worked well and established a cooperative equilibrium. I should maintain this consistency with Galen to encourage continued cooperation.

If I return \$4.5:

- Galen gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1282,
  "output_tokens": 249,
  "reasoning_tokens": 165
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
 Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Continuing my cooperative strategy, I'll return \$4.5 out of the \$9 received, maintaining the fair 50% return rate.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

I'm the receiver, Cyra sent me \$3, which becomes \$9.
 I need to decide how much to return.

Looking at my history:

- I've been consistently cooperative
- When I send as sender, I send \$3
- When I receive as receiver, I return \$4.5 (which is 50% of what I receive)
- This strategy has been working well - my earnings are \$33 over what looks like 6 completed rounds

I should maintain consistency with my cooperative strategy and return \$4.5.

```
Usage: {
  "input_tokens": 1227,
  "output_tokens": 182,
  "reasoning_tokens": 126
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-06-04T15:47:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_5"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_5": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at my interaction with Briar:

****Round 3:**** I sent \$3 to Briar, they returned \$4.5

- I got \$6.5, they kept \$4.5

****Round 4 (now):**** Briar sent \$3 to me (matching my trust)

- I received \$9

Briar showed reasonable cooperation by returning \$4.5. Now they've shown the same level of

trust in me. I'll reciprocate by matching their behavior exactly - returning \$4.5 creates a consistent, fair pattern between us.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at what happened:

****Round 3 (vs Briar):**** I sent \$3, Briar returned \$4.5 to me
- My payoff was \$6.5 ($\$5 - \$3 + \4.5)
- Briar kept \$4.5 ($\$9 - \4.5)

****Round 4 (vs Briar again):**** Briar sent \$3, I received \$9

So Briar is meeting me again! Last time I sent \$3 and they returned \$4.5. Now they're sending \$3 to me.

The return of \$4.5 was slightly less generous than the \$5 that Galen returned to me and that I returned to Finn. But it's still cooperative and fair - we both got reasonable payoffs.

Now Briar has sent me \$3, showing the same trust. I should reciprocate fairly.

Options:

1. Return \$5 (what I've been doing with others) - this would give Briar \$7, me \$4
2. Return \$4.5 (matching what Briar did) - this would give Briar \$6.5, me \$4.5

Matching what Briar did (\$4.5) seems like the fairest reciprocal move - it shows I'm willing to match their level of cooperation exactly. This creates a clear tit-for-tat dynamic.

Actually, let me think about this more carefully. When Briar was the receiver:

- They got \$9
- They returned \$4.5
- They kept \$4.5

Now when I'm the receiver:

- I get \$9
- If I return \$4.5, I keep \$4.5
- Briar gets $\$5 - \$3 + \$4.5 = \6.5

So by returning \$4.5, I'm mirroring their exact behavior. This seems fair and establishes a consistent reciprocal pattern with Briar.

usage:

```
{
  "input_tokens": 1481,
  "output_tokens": 608,
  "reasoning_tokens": 350
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

****Rounds 1-4:**** Consistently sent \$3 as sender
- Returns have ranged from \$4.5-\$5
- Average payoff as sender: \$6.75

- Strategy has built cooperative relationships

This is my first interaction with Aster. I'll maintain my established strategy of sending \$3, which:

- Demonstrates trust
- Creates significant value (\$9 for receiver)
- Has proven effective across multiple partners
- Keeps some safety margin

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1 (vs Galen): I sent \$3, received \$5 back - payoff \$7
 Round 2 (vs Finn): Finn sent \$3, I returned \$5 - payoff \$4
 Round 3 (vs Briar): I sent \$3, received \$4.5 back - payoff \$6.5
 Round 4 (vs Briar): Briar sent \$3, I returned \$4.5 - payoff \$4.5

So far, sending \$3 has worked well:

- It shows reasonable trust
- It creates value (becomes \$9)
- Partners have generally reciprocated with \$4.5-\$5 returns
- It's been my consistent strategy

This is my first interaction with Aster. I should maintain my cooperative strategy that has worked well so far. Sending \$3 has proven to be a good balance between trust and caution.

Total earnings: \$22.0
 Average per round: \$5.5

I'll stick with \$3.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 375,
  "reasoning_tokens": 193
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
 Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
 How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
 - I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
 - I received \$9

Aster demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This mirrors my successful interaction with Briar. I'll reciprocate with the same fair return.

```
'''json
{"return": 4.5}
```

'''

reasoning:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

This is exactly the same pattern I had with Briar:

- I sent \$3, they returned \$4.5
- They sent \$3, now I should return

Aster showed the same level of cooperation as Briar. They returned the exact same amount (\$4.5). Now they've shown the same level of trust in me (\$3).

For consistency and fair reciprocity, I should return \$4.5 - matching what they returned to me. This creates a balanced, cooperative relationship.

This seems to be a consistent pattern:

- Send \$3
- Receive back \$4.5
- When receiving \$3 (which becomes \$9), return \$4.5

This creates fair outcomes where both parties benefit similarly from the cooperation.

usage:

```
{  
  "input_tokens": 1474,  
  "output_tokens": 393,  
  "reasoning_tokens": 222  
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

****Previous rounds as sender:**** Consistently sent \$3
- Briar: returned \$4.5
- Aster: returned \$4.5
- Average payoff: \$6.5

This is my first interaction with Elara. I'll maintain my established cooperative strategy of sending \$3, which has proven effective in building trust and achieving mutually beneficial outcomes.

```
'''json  
{  
  "send": 3  
}
```

Reasoning:

Looking at my history:

- I've consistently sent \$3 as sender in previous rounds
- This has worked well with different partners (Briar, Aster)
- Returns have been consistently around \$4.5-\$5
- This creates mutual cooperation and good payoffs

This is my first interaction with Elara (Agent_5). I should maintain my proven strategy of sending \$3, which:
- Shows trust

- Creates value (becomes \$9 for receiver)
- Has built successful cooperative relationships
- Balances risk and reward

I'll continue with \$3.

```
Usage: {
  "input_tokens": 1482,
  "output_tokens": 238,
  "reasoning_tokens": 136
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T15:47:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_5"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_5": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern across all rounds so far, there's been consistent cooperation with agents sending \$3 and receivers returning \$4.5. This has created a stable, mutually beneficial norm.

I'll continue this pattern with Cyra, sending \$3 as I did with Finn and as others have done with me.

{'send': 3}

reasoning:

Let me review the pattern so far:

Round 1: I was SENDER vs Finn

- I sent \$3
- Finn received \$9
- Finn returned \$4.5
- My payoff: \$6.5

Round 2: I was RECEIVER vs Briar

- Briar sent \$3
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

Round 3: I was RECEIVER vs Aster

- Aster sent \$3
- I received \$9
- I returned \$4.5
- My payoff: \$4.5

So far, there's been a consistent pattern:

- Everyone sends \$3
- Everyone returns \$4.5 (half of the tripled amount)
- Senders get \$6.5, Receivers get \$4.5

Now I'm a SENDER again vs Cyra. Should I continue the pattern and send \$3?

The pattern has been working well and establishing trust. If I send \$3 and Cyra follows the pattern by returning \$4.5, I'll get \$6.5 again.

I should continue with the cooperative norm that has been established.

usage:

```
{
  "input_tokens": 1367,
  "output_tokens": 358,
  "reasoning_tokens": 218
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The cooperative pattern has been perfectly consistent - everyone sends \$3 and returns \$4.5. This has worked well for all players across all rounds so far.

I'll maintain this cooperative strategy with Galen.

```
{'send': 3}
```

reasoning:

The pattern continues to hold perfectly. In round 4, I sent \$3 to Cyra and they returned \$4.5, just as expected. This cooperative pattern has been consistent across all rounds:

- Round 1: I sent \$3 to Finn, received \$4.5 back
- Round 2: Briar sent \$3 to me, I returned \$4.5
- Round 3: Aster sent \$3 to me, I returned \$4.5
- Round 4: I sent \$3 to Cyra, received \$4.5 back

Everyone is following the same cooperative strategy. I should continue sending \$3 to Galen to maintain this stable equilibrium.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 222,
  "reasoning_tokens": 130
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history with Briar - they sent me \$3.0 in Round 3 and I returned \$4.5, which worked well. They're being consistent and cooperative again by sending \$3.0.

I'll maintain the same reciprocal response.

```
{'return': 4.5}
```

reasoning:

Let me review the situation:

- Briar sent me \$3.0 in Round 3, and I returned \$4.5
- Now in Round 6, Briar is sending me \$3.0 again

- The general pattern has been: send \$3, return \$4.5
- However, in Round 5, Galen returned \$5.0 to me instead of \$4.5 (a bit more generous)

Briar has been consistent with the cooperative norm. I should continue the cooperative pattern. The standard has been \$4.5, though Galen was slightly more generous with \$5.0.

I'll stick with the \$4.5 that has been the norm and worked well, maintaining reciprocity with Briar who has been cooperative.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 254,
  "reasoning_tokens": 150
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

The cooperative pattern continues - Hana sent \$3.0, which is consistent with what everyone has been doing. I'll reciprocate with the standard return.

```
{'return': 4.5}
```

Reasoning:

Hana is sending me \$3.0, which becomes \$9.0. This is consistent with the cooperative pattern that has been established throughout the game.

Looking at my history:

- Round 1: I was receiver from Finn, they sent \$3, I returned \$4.5
- Round 2: I was sender to Aster, I sent \$3, they returned \$4.5
- Round 3: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 4: I was sender to Cyra, I sent \$3, they returned \$4.5
- Round 5: I was sender to Galen, I sent \$3, they returned \$5.0 (slightly more!)
- Round 6: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 7: I am receiver from Hana, they sent \$3, I should return \$4.5

The pattern is very clear - almost everyone plays the cooperative strategy of sending \$3 and returning \$4.5. This gives:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

This is the established norm. I should continue with this pattern and return \$4.5 to Hana.

Usage: {

```
"input_tokens": 1274,
"output_tokens": 368,
"reasoning_tokens": 242
}
```

ROUND 8

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

```
dyad_1: Agent_7 (Galen)=Sender, Agent_2 (Briar)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_3 (Cyra)=Receiver
dyad_3: Agent_4 (Dorian)=Sender, Agent_6 (Finn)=Receiver
dyad_4: Agent_5 (Elara)=Sender, Agent_8 (Hana)=Receiver
```

Parsed game decisions:

```

dyad_1: Agent_7 (Galen) sent $3; received $9; Agent_2 (Briar) returned $5; payoffs Agent_7
(Galen)=$7, Agent_2 (Briar)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 (Aster) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_4 (Dorian) sent $3; received $9; Agent_6 (Finn) returned $4.5; payoffs Agent_4
(Dorian)=$6.5, Agent_6 (Finn)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_5 (Elara) sent $3; received $9; Agent_8 (Hana) returned $4.5; payoffs Agent_5
(Elara)=$6.5, Agent_8 (Hana)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 44.0,
  "Agent_2": 44.0,
  "Agent_3": 45.0,
  "Agent_4": 44.0,
  "Agent_5": 44.5,
  "Agent_6": 44.5,
  "Agent_7": 42.0,
  "Agent_8": 44.0
}

```

LLM CALLS

GAME - Agent_7 - Sender - dyad_1

Timestamp: 2026-06-04T15:47:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_2"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_2": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is the same amount Hana and Cyra sent me previously

I'll maintain consistency in my reciprocation strategy. When I receive \$3, I return \$5, which creates mutual benefit and shows I'm a reliable partner.

If I return \$5.0:

- Elara ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

reasoning:

Looking at this situation:

- Elara sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is consistent with what others have been sending me

My track record as receiver:

- Round 1: Hana sent \$3, I returned \$5
- Round 3: Cyra sent \$3, I returned \$5

And when I've been sender:

- Round 2: I sent \$3 to Aster, they returned \$4
- Round 4: I sent \$3 to Finn, they returned \$4.5

I've been consistent in returning \$5 when receiving \$3. This creates a fair outcome where:

- The sender nets \$2 ($\$5 - \$3 + \5)
- I net \$4 ($\$9 - \5)

Both sides benefit, and it's a good reciprocal strategy. I should maintain this consistency with Elara.

If I return \$5.0:

- Elara ends with: \$7 ($\$5 - \$3 + \5)
- I keep: \$4 ($\$9 - \5)

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1442,
  "output_tokens": 439,
  "reasoning_tokens": 220
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my consistent strategy:

- As sender, I've always sent \$3 (60%)
- This has generated good returns: \$4 from Aster, \$4.5 from Finn
- I maintain the same reciprocal approach when I'm the receiver
- This builds trust and creates mutual benefits across rounds

I'll continue this successful strategy with Dorian.

{'send': 3}

reasoning:

Looking at my history:

As SENDER:

- Round 2 vs Aster: sent \$3.0, received \$4.0 back, payoff \$6.0
- Round 4 vs Finn: sent \$3.0, received \$4.5 back, payoff \$6.5

As RECEIVER:

- Round 1 vs Hana: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 3 vs Cyra: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 5 vs Elara: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0

My strategy has been:

- Send \$3 (60%) as sender
- Return \$5 when I receive \$3 (which becomes \$9) as receiver

This has been working well and I've been consistent. I should continue with this strategy against Dorian.

Total earnings: \$24.5

usage:

```
{
  "input_tokens": 1471,
  "output_tokens": 353,
  "reasoning_tokens": 173
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Briar sent \$3.0 (60%), showing trust
- I received \$9.0
- This matches the pattern from Hana, Cyra, and Elara

I'll maintain my consistent reciprocation strategy. When I receive \$3, I return \$5, which creates positive outcomes for both players and demonstrates reliability.

If I return \$5.0:

- Briar ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

reasoning:

Let me review the situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- I've been consistently returning \$5.0 when I receive \$3.0 from others

My pattern has been:

- As sender: I always send \$3
- As receiver when getting \$3: I return \$5

This has worked well and created good reciprocity. I should maintain this consistency with Briar.

If I return \$5.0:

- Briar ends with: \$2 + \$5 = \$7 (net gain of \$2 from their initial \$5)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)

This is my standard reciprocal approach that has been working well.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 338,
  "reasoning_tokens": 158
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.00 to you, it became \$9.00, and you returned \$5.00. Your payoff was \$4.00. Your total earnings across all rounds are \$35.00.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my interaction with Briar:

- Last round: Briar sent me \$3, I returned \$5
- This established good reciprocity between us
- As sender, I've consistently sent \$3 and received positive returns
- Briar has shown they're trustworthy and cooperative

I'll continue my consistent strategy and send \$3, expecting good reciprocation given our positive previous interaction.

```
{'send': 3}
```

Reasoning:

Looking at the situation:

- Last round with Briar: They sent \$3, I returned \$5
- Briar showed trust by sending \$3
- I reciprocated appropriately with \$5
- This is now a second consecutive round with Briar

As sender now, I should maintain my consistent strategy:

- I've been sending \$3 as sender
- This has worked well: Aster returned \$4, Finn returned \$4.5, Dorian returned \$4.5
- Briar has shown they're cooperative (sent \$3 last round)
- I should continue with my standard \$3 send

This maintains consistency and should generate good reciprocity from Briar given our positive interaction last round.

```
Usage: {  
  "input_tokens": 1445,  
  "output_tokens": 270,  
  "reasoning_tokens": 161  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T15:47:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_2"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60%), showing significant trust
- I received \$9.0 after the multiplier

I've established a consistent reciprocation pattern: when someone sends \$3.0, I return \$4.5. This has worked well in building cooperative relationships with both Dorian and Hana. I'll maintain this consistency with Cyra.

Returning \$4.5 ensures both of us profit and establishes me as a fair, reliable partner.

```
{"return": 4.5}
```

reasoning:

Looking at this situation:

- This is my first interaction with Cyra
- Cyra sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- I've been consistent in my strategy: when someone sends \$3.0, I return \$4.5

Pattern from previous rounds:

- Round 1: Dorian sent \$3.0, I returned \$4.5
- Round 3: Hana sent \$3.0, I returned \$4.5

This has worked well - both parties profit and cooperation is maintained. I should continue this consistent, fair reciprocation strategy with Cyra.

Returning \$4.5 gives Cyra a total of \$6.5 (profit of \$1.5) and leaves me with \$4.5.

usage:

```
{  
  "input_tokens": 1458,  
  "output_tokens": 339,  
  "reasoning_tokens": 160  
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 3 (actually Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5
- This was a fair and cooperative exchange where we both profited
- Elara demonstrated trustworthiness

Since our previous interaction was positive and Elara reciprocated fairly, I'll continue with the same cooperative approach by sending \$3.0 again.

```
{"send": 3.0}
```

reasoning:

Let me review my history with Elara:

Round 3 information: "In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5."

So in a previous round (which would have been Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5, which was a fair and cooperative response.

This is now Round 6, and I'm the sender again against Elara. Since our previous interaction was positive and mutually beneficial, I should continue with the same cooperative approach.

Sending \$3.0 has been my standard move when dealing with cooperative partners, and Elara proved to be trustworthy in our last interaction by returning \$4.5.

```
{"send": 3.0}
```


usage:

```
{
  "input_tokens": 1474,
  "output_tokens": 324,
  "reasoning_tokens": 195
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3.0 has been successful in establishing cooperative relationships with Hana, Cyra, and Elara
- All of these partners have reciprocated fairly by returning \$4.5

I'll continue with my consistent cooperative approach by sending \$3.0 to Galen, demonstrating trust and inviting reciprocation.

```
{"send": 3.0}
```

reasoning:

This is Round 7 and I'm playing against Galen (Agent_7). I'm the SENDER this round.

Let me look at my history:

- I don't see any previous interactions with Galen mentioned
- This appears to be my first game with Galen

My successful pattern so far has been:

- Sending \$3.0 to new partners (showing cooperation and trust)
- When I receive \$3.0, I return \$4.5 (being fair and reciprocal)
- This has worked well with Hana, Cyra, and Elara

Since this is my first interaction with Galen and my strategy of sending \$3.0 has been successful in establishing cooperative relationships, I'll continue this approach.

```
{"send": 3.0}
```

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 292,
  "reasoning_tokens": 166
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Last round, I sent Galen \$3.0 and they returned \$5.0 (more generous than my typical \$4.5)
- This round, Galen sent me \$3.0 again, showing continued trust
- I received \$9.0 after the multiplier

Galen has demonstrated both trustworthiness and generosity. I'll reciprocate their generosity by returning \$5.0, matching what they returned to me in the previous round.

```
{"return": 5.0}
```

Reasoning:

Looking at this situation:

- Last round (Round 7), I sent Galen \$3.0, it became \$9.0, and they returned \$5.0 to me
- Galen was generous, returning more than my usual reciprocation amount of \$4.5
- This round, Galen sent me \$3.0 again, so I received \$9.0

Galen has shown they are a trustworthy and even generous partner. They returned \$5.0 when I typically see \$4.5 returns. Now they've sent me \$3.0 again, showing continued trust.

I should reciprocate at least as fairly as I have with others. My standard has been \$4.5 for a \$3.0 send. Given Galen was more generous than expected, I could consider matching their generosity by returning \$5.0. However, I should be consistent with my general strategy.

Let me think about this:

- Galen returned \$5.0 (keeping \$4.0 from the \$9.0)
- My standard return is \$4.5 (keeping \$4.5 from the \$9.0)
- Galen was slightly more generous than my standard

I could:

1. Return \$4.5 (my standard, fair split)
2. Return \$5.0 (matching Galen's generosity)

Given that Galen was generous with me, and to maintain a good relationship, I'll return \$5.0 to match their generosity. This rewards their cooperative behavior and hopefully encourages continued cooperation.

```
Usage: {
  "input_tokens": 1425,
  "output_tokens": 491,
  "reasoning_tokens": 314
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T15:47:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_3"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_3": "trustee"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play an iterated game,
  consisting of multiple rounds, and may also be asked to write myths.

  GAME:
  The game consists of two roles: sender and receiver. In each round, you will play the role
  of either the sender or the receiver. Your current role will be specified in the round
  prompt.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
  , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
  ll have acted as sender and receiver equally often when the round count allows it.
  - You will be told your opponent's name in each round.
  - Your name in this experiment is Aster.

Message 2 role=user
content:
  Your name in this experiment is Aster.
  Your opponent this round is Hana (Agent_8).

  Round 5

  In your most recent previous game round, you were the RECEIVER against Dorian. They sent $
  3.0 to you, it became $9.0, and you returned $4.5. Your payoff was $4.5.
  Your total earnings across all rounds are $22.0.

  This round, you are the RECEIVER against Hana. They sent you $3.0, so you received $9.0. H
  ow much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant
content:
  Looking at the established pattern:
  - Everyone has consistently sent $3, showing trust
  - Returns have been $4-$4.5, maintaining cooperation
  - Hana sent $3, showing they're following the cooperative norm

  I'll continue with $4.5, which has been the generous standard recently.

  {'return': 4.5}

```

reasoning:

Let me review the pattern:

- Round 1: I was sender, sent \$3 to Cyra, got back \$4. My payoff: \$6
- Round 2: I was receiver from Galen, got \$9, returned \$4. My payoff: \$5
- Round 3: I was sender, sent \$3 to Elara, got back \$4.5. My payoff: \$6.5
- Round 4: I was receiver from Dorian, got \$9, returned \$4.5. My payoff: \$4.5
- Round 5: I'm receiver from Hana, got \$9, need to decide how much to return

The cooperative pattern has been established with everyone sending \$3 and returns ranging from \$4 to \$4.5. Hana has also shown trust by sending \$3.

I'll continue the cooperative pattern and return \$4.5, which has been working well.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 299,
  "reasoning_tokens": 165
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our recent interaction:

- Last round, Hana sent me \$3 and I returned \$4.5
- Hana has shown trust and cooperation
- The \$3 send amount with ~\$4.5 return has been working well across all interactions

I'll continue with this successful cooperative pattern and send \$3 to Hana.

```
{'send': 3}
```

reasoning:

Looking at the situation:

- Last round, Hana sent me \$3 and I returned \$4.5
- This round, I'm the sender to Hana
- Hana has shown trust by sending \$3 last round
- The cooperative pattern has been: send \$3, receive back \$4-\$4.5
- I should continue this pattern and send \$3 to Hana

Given that Hana just cooperated with me and I returned \$4.5 to them, I expect Hana will return a fair amount if I send \$3.

I'll send \$3 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 238,
  "reasoning_tokens": 122
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the consistent pattern:

- Everyone continues to send \$3, showing mutual trust
- Returns of \$4.5 have maintained stable cooperation
- Finn sent \$3, following the cooperative norm

I'll continue with \$4.5 to maintain this successful pattern.

{'return': 4.5}

reasoning:

Let me analyze the situation:

Round 7: I'm the RECEIVER against Finn.

- Finn sent me \$3.0
- I received \$9.0 (after 3x multiplication)
- I need to decide how much to return (0-9.0)

Looking at the pattern:

- Everyone has been consistently sending \$3
- Returns have been around \$4.5, which maintains cooperation
- Hana just returned \$4.5 to me when I sent \$3 (last round)
- The cooperative pattern of \$3 sent, \$4.5 returned has been stable

Finn sent \$3, showing trust. I should maintain the cooperative norm and return \$4.5.

This gives:

- Finn: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This is fair and maintains the established cooperative pattern.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 296,
  "reasoning_tokens": 179
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the established pattern:

- The \$3 send with \$4.5 return has been consistently successful
- All agents have maintained this cooperative equilibrium
- My earnings are healthy at \$37.5

I'll continue this working strategy and send \$3 to Cyra.

{'send': 3}

Reasoning:

Let me review the situation:

- I'm playing as SENDER against Cyra this round
- The cooperative pattern has been well-established: send \$3, receive back ~\$4.5
- Everyone has been following this pattern consistently
- My total earnings are \$37.5, which suggests good cooperation overall

Looking at my history:

- Round 5: RECEIVER vs Dorian - they sent \$3, I returned \$4.5
- Round 6: SENDER vs Hana - I sent \$3, they returned \$4.5
- Round 7: RECEIVER vs Finn - they sent \$3, I returned \$4.5
- Round 8 (now): SENDER vs Cyra

The pattern is very clear and stable. Everyone is sending \$3 and returning around \$4.5. This is a mutually beneficial cooperative equilibrium. I should continue with \$3.

```
{'send': 3}
Usage: {
  "input_tokens": 1300,
  "output_tokens": 286,
  "reasoning_tokens": 187
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-06-04T15:47:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$23.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my performance:

- As sender: Sending \$3.0 has yielded excellent returns (\$4.5 from Dorian, \$5.0 from Galen)
- As receiver: When others send \$3.0, I return \$4.0-\$4.5, maintaining cooperation

The \$3.0 (60%) strategy has been consistently successful across different opponents. I'll continue this approach with Briar.

{ 'send': 3.0 }

reasoning:

Let me review my history:

Round 1: RECEIVER from Aster - they sent \$3.0, I returned \$4.0, my payoff was \$5.0
Round 2: SENDER to Dorian - I sent \$3.0, they returned \$4.5, my payoff was \$6.5
Round 3: SENDER to Galen - I sent \$3.0, they returned \$5.0, my payoff was \$7.0
Round 4: RECEIVER from Elara - they sent \$3.0, I returned \$4.5, my payoff was \$4.5

Total earnings: \$23.0

The pattern is clear - sending \$3.0 (60% of endowment) has been consistently successful. When I send \$3.0, I get good returns:

- Dorian returned \$4.5 (I profited \$1.5)
- Galen returned \$5.0 (I profited \$2.0)

And when others send me \$3.0, I've been returning \$4.0-\$4.5, which seems fair.

This is a new opponent (Briar), so I'll continue with the cooperative strategy that has been working well. I'll send \$3.0.

usage:

```
{
  "input_tokens": 1446,
  "output_tokens": 408,
  "reasoning_tokens": 208
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

As sender:

- All opponents who received my \$3.0 have returned \$4.5-\$5.0

Finn has shown the same trust level by sending \$3.0. I'll continue my cooperative pattern and return \$4.5, which gives Finn a solid profit while maintaining fair reciprocity.

{'return': 4.5}

reasoning:

Looking at my history:

As sender:

- Round 2: Sent \$3.0 to Dorian, they returned \$4.5, my payoff \$6.5
- Round 3: Sent \$3.0 to Galen, they returned \$5.0, my payoff \$7.0
- Round 5: Sent \$3.0 to Briar, they returned \$4.5, my payoff \$6.5

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

Now Finn has sent me \$3.0, which became \$9.0.

The pattern has been very consistent - everyone is sending \$3.0 and returning between \$4.0-\$5.0. I've been returning \$4.0-\$4.5 when receiving \$3.0.

I should maintain this cooperative pattern. Returning \$4.5 seems like a good middle ground that rewards trust and maintains reciprocity.

usage:

```
{
  "input_tokens": 1450,
  "output_tokens": 392,
  "reasoning_tokens": 179
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Dorian:

- Round 2: I sent \$3.0 to Dorian, and they returned \$4.5 (giving me a \$6.5 payoff)

Dorian demonstrated strong cooperation previously. My \$3.0 strategy has consistently worked well across all opponents. I'll continue with this approach.

```
{'send': 3.0}
```

reasoning:

Let me review my history with Dorian:

- Round 2: I was the SENDER against Dorian. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Dorian has shown generosity before by returning \$4.5 when I sent \$3.0. This is a repeat opponent who cooperated well.

My strategy has been consistent:

- As sender: Always send \$3.0 (60%)
- As receiver: Return \$4.0-\$4.5

This has worked well:

- Round 2: Sent \$3.0 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 3: Sent \$3.0 to Galen, got \$5.0 back (payoff \$7.0)
- Round 5: Sent \$3.0 to Briar, got \$4.5 back (payoff \$6.5)

All successful. I should continue with \$3.0 to Dorian since they cooperated well last time

usage:

```
{  
  "input_tokens": 1463,  
  "output_tokens": 363,  
  "reasoning_tokens": 183  
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$40.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Aster:

- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has continued to show trust by sending \$3.0 again. In my recent rounds as receiver (Rounds 4 and 6), I've been returning \$4.5 when receiving \$3.0. I'll maintain this consistent cooperative standard with Aster.

```
{'return': 4.5}
```

Reasoning:

Looking at my history with Aster:

- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has sent \$3.0 again, showing consistent trust. In my most recent rounds as receiver, I've been returning \$4.5 when receiving \$3.0 (from Elara in Round 4 and Finn in Round 6).

I initially returned \$4.0 to Aster in Round 1, but I've since increased my return rate slightly to \$4.5 to maintain better cooperation. I should be consistent and return \$4.5 to Aster as well.

This gives Aster:

- Keeps: \$5 - \$3 + \$4.5 = \$6.5 (nice profit)

And I keep:

- \$9.0 - \$4.5 = \$4.5

This is fair and maintains the cooperative pattern I've established.

```
Usage: {
  "input_tokens": 1400,
  "output_tokens": 329,
  "reasoning_tokens": 166
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-06-04T15:47:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    },
  },
  "agents": [
    "Agent_4",
    "Agent_6"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

I played with Finn before in Round 3, where they sent me \$3 and I returned \$4.5. They were cooperative, so I'll reciprocate with the same cooperative approach.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now playing as SENDER against Finn. In Round 3, I was the RECEIVER against Finn - they sent me \$3.0, and I returned \$4.5 (50% of the \$9 received).

This is the first time I'm the sender with Finn. Since Finn was cooperative by sending \$3 in our previous interaction, and I've established a pattern of sending \$3 consistently (which has worked well so far), I should continue with that strategy.

My track record:

- Round 1: Sender to Briar, sent \$3, got back \$4.5 ? payoff \$6.5
- Round 2: Receiver from Cyra, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 3: Receiver from Finn, received \$9 (from \$3), returned \$4.5 ? payoff \$4.5
- Round 4: Sender to Aster, sent \$3, got back \$4.5 ? payoff \$6.5

Everyone has been cooperative, and sending \$3 with ~50% returns has been working well. I should continue this strategy with Finn.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 338,
  "reasoning_tokens": 221
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Following my consistent cooperative strategy, I'll return \$4.5 out of the \$9 received. This maintains the 50% return rate that has been working well.

```
'''json
{"return": 4.5}
'''
```

reasoning:

I'm playing as receiver against Galen (Agent_7) for the first time. They sent me \$3, which became \$9 for me.

Looking at my history:

- I've been consistently cooperative
- When I was receiver, I returned \$4.5 (50% of what I received)
- When I was sender, others returned \$4.5 to me (50% of what they received)

This pattern of returning 50% has worked well and established a cooperative equilibrium. I should maintain this consistency with Galen to encourage continued cooperation.

If I return \$4.5:

- Galen gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1282,
  "output_tokens": 249,
  "reasoning_tokens": 165
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Continuing my cooperative strategy, I'll return \$4.5 out of the \$9 received, maintaining the fair 50% return rate.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

I'm the receiver, Cyra sent me \$3, which becomes \$9.
I need to decide how much to return.

Looking at my history:

- I've been consistently cooperative
- When I send as sender, I send \$3
- When I receive as receiver, I return \$4.5 (which is 50% of what I receive)
- This strategy has been working well - my earnings are \$33 over what looks like 6 completed rounds

I should maintain consistency with my cooperative strategy and return \$4.5.

usage:

```
{
```

```
"input_tokens": 1227,
"output_tokens": 182,
"reasoning_tokens": 126
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Playing with Finn again - we've established a good cooperative pattern. In our previous interactions, we both sent \$3 and returned \$4.5. I'll continue this successful strategy.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm playing with Finn again. Let me recall my history with Finn:

- Round 3: Finn was the sender, sent me \$3, I returned \$4.5
- Round 5: I was the sender, sent \$3, Finn returned \$4.5

Both times we've been cooperative with each other, both sending \$3 and returning \$4.5. This has worked well, and Finn has been trustworthy. I should continue this pattern.

I'll send \$3 again.

```
Usage: {
  "input_tokens": 1222,
  "output_tokens": 176,
  "reasoning_tokens": 99
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T15:47:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_4",
"pairing": {
  "agent_names": {
    "Agent_4": "Dorian",
    "Agent_6": "Finn"
  },
"agents": [
  "Agent_4",
  "Agent_6"
],
"investor": "Agent_4",
"roles": {
  "Agent_4": "investor",
  "Agent_6": "trustee"
},
"trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0, showing trust
- I received \$9.0 after the multiplier
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5 to me
- Dorian has been fair and reciprocal

I'll maintain the same fair reciprocal strategy.

{'return': 4.5}

reasoning:

Looking at the situation:

- Dorian sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplier
- This is my second interaction with Dorian
- In Round 3, I sent Dorian \$3.0 and they returned \$4.5

Since Dorian returned \$4.5 when I sent them \$3.0, I should reciprocate with the same fair behavior now that they've sent me \$3.0.

I'll return \$4.5, maintaining consistency and reciprocity.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 237,
  "reasoning_tokens": 111
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 when I'm the sender
- Players have reciprocated fairly (returns ranging from \$4.5-\$5)
- This cooperative strategy has been successful

I'll continue this approach with Cyra.

```
{'send': 3}
```

reasoning:

Let me review my history:

- Round 1: RECEIVER - Elara sent me \$3, I returned \$4.5
- Round 2: SENDER - I sent Hana \$3, they returned \$5
- Round 3: SENDER - I sent Dorian \$3, they returned \$4.5
- Round 4: RECEIVER - Galen sent me \$3, I returned \$4.5
- Round 5: RECEIVER - Dorian sent me \$3, I returned \$4.5

Now Round 6: SENDER against Cyra (new opponent)

My strategy has been consistently cooperative:

- When sending, I send \$3
- When receiving, I return \$4.5

This has worked well - people have been reciprocating, and I've built up good earnings (\$27.0 total).

I should continue with my consistent strategy of sending \$3 to signal trust and cooperation, especially with a new opponent.

usage:

```
{
  "input_tokens": 1332,
  "output_tokens": 292,
  "reasoning_tokens": 184
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent strategy:

- I've sent \$3 in every round as sender (Rounds 2, 3, 6)
- This has generated good returns (\$4.5-\$5)
- Total earnings of \$33.5 shows this cooperative approach is working well

I'll maintain this successful strategy with Aster.

{'send': 3}

reasoning:

I'm the sender again. Let me review my track record:

- Round 1: Receiver from Elara - they sent \$3, I returned \$4.5
- Round 2: Sender to Blair - I sent \$3, they returned \$5
- Round 3: Sender to Dorian - I sent \$3, they returned \$4.5
- Round 4: Receiver from Galen - they sent \$3, I returned \$4.5
- Round 5: Receiver from Dorian - they sent \$3, I returned \$4.5
- Round 6: Sender to Cyra - I sent \$3, they returned \$4.5
- Round 7: Sender to Aster - ?

My strategy has been very consistent:

- As sender: always send \$3
- As receiver: always return \$4.5

This has worked well. My total earnings are \$33.5 across 6 rounds, averaging about \$5.58 per round, which is above the baseline of \$5.

I should continue with this consistent cooperative strategy. Sending \$3 to Aster.

usage:

```
{
  "input_tokens": 1319,
  "output_tokens": 348,
  "reasoning_tokens": 204
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Dorian:

- Round 3: I sent \$3.0, Dorian returned \$4.5
- Round 5: Dorian sent \$3.0, I returned \$4.5
- Round 8 (now): Dorian sent \$3.0 again

Dorian has been consistently fair and reciprocal. I'll maintain this mutually beneficial pattern.

{'return': 4.5}

Reasoning:

Let me review my history with Dorian:

- Round 3: I was SENDER, sent \$3.0, Dorian returned \$4.5
- Round 5: I was RECEIVER, Dorian sent \$3.0, I returned \$4.5
- Round 8 (now): I am RECEIVER, Dorian sent \$3.0

Dorian has been consistently fair and reciprocal with me. They've sent \$3.0 both times as sender, and returned \$4.5 when they were receiver.

I should maintain this reciprocal relationship and return \$4.5 again. This is fair and maintains the cooperative equilibrium we've established.

My payoff would be: $\$9.0 - \$4.5 = \$4.5$

Dorian's payoff would be: $\$5.0 - \$3.0 + \$4.5 = \6.5

This is consistent with our previous interactions and maintains fairness.

```
Usage: {
  "input_tokens": 1312,
  "output_tokens": 335,
  "reasoning_tokens": 174
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T15:47:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_8"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

The cooperative pattern has been perfectly consistent - everyone sends \$3 and returns \$4.5. This has worked well for all players across all rounds so far.

I'll maintain this cooperative strategy with Galen.

```
{'send': 3}
```

reasoning:

The pattern continues to hold perfectly. In round 4, I sent \$3 to Cyra and they returned \$4.5, just as expected. This cooperative pattern has been consistent across all rounds:

- Round 1: I sent \$3 to Finn, received \$4.5 back
- Round 2: Briar sent \$3 to me, I returned \$4.5
- Round 3: Aster sent \$3 to me, I returned \$4.5
- Round 4: I sent \$3 to Cyra, received \$4.5 back

Everyone is following the same cooperative strategy. I should continue sending \$3 to Galen to maintain this stable equilibrium.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 222,
  "reasoning_tokens": 130
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history with Briar - they sent me \$3.0 in Round 3 and I returned \$4.5, which worked well. They're being consistent and cooperative again by sending \$3.0.

I'll maintain the same reciprocal response.

{'return': 4.5}

reasoning:

Let me review the situation:

- Briar sent me \$3.0 in Round 3, and I returned \$4.5
- Now in Round 6, Briar is sending me \$3.0 again
- The general pattern has been: send \$3, return \$4.5
- However, in Round 5, Galen returned \$5.0 to me instead of \$4.5 (a bit more generous)

Briar has been consistent with the cooperative norm. I should continue the cooperative pattern. The standard has been \$4.5, though Galen was slightly more generous with \$5.0.

I'll stick with the \$4.5 that has been the norm and worked well, maintaining reciprocity with Briar who has been cooperative.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 254,
  "reasoning_tokens": 150
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.

Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

The cooperative pattern continues - Hana sent \$3.0, which is consistent with what everyone has been doing. I'll reciprocate with the standard return.

{'return': 4.5}

reasoning:

Hana is sending me \$3.0, which becomes \$9.0. This is consistent with the cooperative pattern that has been established throughout the game.

Looking at my history:

- Round 1: I was receiver from Finn, they sent \$3, I returned \$4.5
- Round 2: I was sender to Aster, I sent \$3, they returned \$4.5
- Round 3: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 4: I was sender to Cyra, I sent \$3, they returned \$4.5
- Round 5: I was sender to Galen, I sent \$3, they returned \$5.0 (slightly more!)
- Round 6: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 7: I am receiver from Hana, they sent \$3, I should return \$4.5

The pattern is very clear - almost everyone plays the cooperative strategy of sending \$3 and returning \$4.5. This gives:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

This is the established norm. I should continue with this pattern and return \$4.5 to Hana.

usage:

```
{
```

```
"input_tokens": 1274,  
"output_tokens": 368,  
"reasoning_tokens": 242  
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Hana was cooperative in Round 7, sending me \$3.0, and I returned \$4.5. I'll maintain this cooperative pattern and send \$3.0 back to them.

```
{'send': 3}
```

Reasoning:

Let me review the situation:

- Round 7: I was RECEIVER against Hana. They sent \$3.0, I received \$9.0, and I returned \$4.5.
- Now Round 8: I'm SENDER against Hana again.

Hana has been cooperative by sending \$3.0 in the previous round, and I returned \$4.5 which is the standard cooperative return. This has been the consistent pattern throughout the experiment.

I should continue with the cooperative strategy and send \$3.0 to Hana.

Usage: {

```
"input_tokens": 1241,  
"output_tokens": 184,  
"reasoning_tokens": 113  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-06-04T15:48:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",  
"move_index": 7,  
"opponent_id": "Agent_5",  
"pairing": {  
  "agent_names": {  
    "Agent_5": "Elara",  
    "Agent_8": "Hana"  
  },  
  "agents": [  
    "Agent_5",  
    "Agent_8"  
  ],  
  "investor": "Agent_5",  
  "roles": {  
    "Agent_5": "investor",  
    "Agent_8": "trustee"  
  },  
  "trustee": "Agent_8"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 8,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user**content:**

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at my track record:

- **Rounds 1-4:**** Consistently sent \$3 as sender
- Returns have ranged from \$4.5-\$5
 - Average payoff as sender: \$6.75
 - Strategy has built cooperative relationships

This is my first interaction with Aster. I'll maintain my established strategy of sending \$3, which:

- Demonstrates trust
- Creates significant value (\$9 for receiver)
- Has proven effective across multiple partners

- Keeps some safety margin

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1 (vs Galen): I sent \$3, received \$5 back - payoff \$7
Round 2 (vs Finn): Finn sent \$3, I returned \$5 - payoff \$4
Round 3 (vs Briar): I sent \$3, received \$4.5 back - payoff \$6.5
Round 4 (vs Briar): Briar sent \$3, I returned \$4.5 - payoff \$4.5

So far, sending \$3 has worked well:

- It shows reasonable trust
- It creates value (becomes \$9)
- Partners have generally reciprocated with \$4.5-\$5 returns
- It's been my consistent strategy

This is my first interaction with Aster. I should maintain my cooperative strategy that has worked well so far. Sending \$3 has proven to be a good balance between trust and caution.

Total earnings: \$22.0
Average per round: \$5.5

I'll stick with \$3.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 375,
  "reasoning_tokens": 193
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

Aster demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This mirrors my successful interaction with Briar. I'll reciprocate with the same fair return.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

This is exactly the same pattern I had with Briar:

- I sent \$3, they returned \$4.5
- They sent \$3, now I should return

Aster showed the same level of cooperation as Briar. They returned the exact same amount (\$4.5). Now they've shown the same level of trust in me (\$3).

For consistency and fair reciprocity, I should return \$4.5 - matching what they returned to me. This creates a balanced, cooperative relationship.

This seems to be a consistent pattern:

- Send \$3
- Receive back \$4.5
- When receiving \$3 (which becomes \$9), return \$4.5

This creates fair outcomes where both parties benefit similarly from the cooperation.

usage:

```
{  
  "input_tokens": 1474,  
  "output_tokens": 393,  
  "reasoning_tokens": 222  
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

****Previous rounds as sender:**** Consistently sent \$3

- Briar: returned \$4.5
- Aster: returned \$4.5
- Average payoff: \$6.5

This is my first interaction with Elara. I'll maintain my established cooperative strategy of sending \$3, which has proven effective in building trust and achieving mutually beneficial outcomes.

```
'''json  
{ "send": 3 }  
'''
```

reasoning:

Looking at my history:

- I've consistently sent \$3 as sender in previous rounds
- This has worked well with different partners (Briar, Aster)
- Returns have been consistently around \$4.5-\$5
- This creates mutual cooperation and good payoffs

This is my first interaction with Elara (Agent_5). I should maintain my proven strategy of sending \$3, which:

- Shows trust
- Creates value (becomes \$9 for receiver)
- Has built successful cooperative relationships
- Balances risk and reward

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 238,
  "reasoning_tokens": 136
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)
- I received \$9

Elara has demonstrated the same cooperative pattern as my other partners. They returned \$4.5 when I sent \$3, and now reciprocated by sending \$3. I'll maintain consistency and return \$4.5.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)
- I received \$9

Elara demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This is exactly the same pattern I've had with Aster and Briar. I'll reciprocate with the same fair return of \$4.5.

This keeps the interaction balanced:

- Elara will get: $\$5 - \$3 + \$4.5 = \6.5
- I will keep: $\$9 - \$4.5 = \$4.5$

This matches what I've been doing consistently.

Usage: {

```
"input_tokens": 1439,
"output_tokens": 351,
"reasoning_tokens": 157
}
```

ROUND 9

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_8 (Hana)=Sender, Agent_4 (Dorian)=Receiver
dyad_2: Agent_7 (Galen)=Sender, Agent_1 (Aster)=Receiver
dyad_3: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver
dyad_4: Agent_3 (Cyra)=Sender, Agent_6 (Finn)=Receiver

Parsed game decisions:

```
dyad_1: Agent_8 (Hana) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_4 (Dorian)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_7 (Galen) sent $3; received $9; Agent_1 (Aster) returned $4.5; payoffs Agent_7
(Galen)=$6.5, Agent_1 (Aster)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_2 (Briar) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_2
(Briar)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_3 (Cyra) sent $3; received $9; Agent_6 (Finn) returned $4.5; payoffs Agent_3
(Cyra)=$6.5, Agent_6 (Finn)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 48.5,
  "Agent_2": 50.5,
  "Agent_3": 51.5,
  "Agent_4": 48.5,
  "Agent_5": 49.0,
  "Agent_6": 49.0,
  "Agent_7": 48.5,
  "Agent_8": 50.5
}
```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

```
Timestamp: 2026-06-04T15:48:09
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_8",
    "Agent_4"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_4": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)
- I received \$9

Aster demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This mirrors my successful interaction with Briar. I'll reciprocate with the same fair return.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at my interaction with Aster:

****Round 5:**** I sent \$3 to Aster, they returned \$4.5

- I got \$6.5, they kept \$4.5

****Round 6 (now):**** Aster sent \$3 to me (matching my trust)

- I received \$9

This is exactly the same pattern I had with Briar:

- I sent \$3, they returned \$4.5

- They sent \$3, now I should return

Aster showed the same level of cooperation as Briar. They returned the exact same amount (\$4.5). Now they've shown the same level of trust in me (\$3).

For consistency and fair reciprocity, I should return \$4.5 - matching what they returned to me. This creates a balanced, cooperative relationship.

This seems to be a consistent pattern:

- Send \$3

- Receive back \$4.5

- When receiving \$3 (which becomes \$9), return \$4.5

This creates fair outcomes where both parties benefit similarly from the cooperation.

usage:

```
{
  "input_tokens": 1474,
  "output_tokens": 393,
  "reasoning_tokens": 222
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

****Previous rounds as sender:**** Consistently sent \$3

- Briar: returned \$4.5

- Aster: returned \$4.5

- Average payoff: \$6.5

This is my first interaction with Elara. I'll maintain my established cooperative strategy of sending \$3, which has proven effective in building trust and achieving mutually beneficial outcomes.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- I've consistently sent \$3 as sender in previous rounds

- This has worked well with different partners (Briar, Aster)

- Returns have been consistently around \$4.5-\$5

- This creates mutual cooperation and good payoffs

This is my first interaction with Elara (Agent_5). I should maintain my proven strategy of sending \$3, which:

- Shows trust

- Creates value (becomes \$9 for receiver)

- Has built successful cooperative relationships

- Balances risk and reward

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 238,
  "reasoning_tokens": 136
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5

- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)

- I received \$9

Elara has demonstrated the same cooperative pattern as my other partners. They returned \$4.5 when I sent \$3, and now reciprocated by sending \$3. I'll maintain consistency and return \$4.5.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5

- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)

- I received \$9

Elara demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This is exactly the same pattern I've had with Aster and Briar. I'll reciprocate with the same fair return of \$4.5.

This keeps the interaction balanced:

- Elara will get: \$5 - \$3 + \$4.5 = \$6.5

- I will keep: \$9 - \$4.5 = \$4.5

This matches what I've been doing consistently.

usage:

```
{
  "input_tokens": 1439,
  "output_tokens": 351,
  "reasoning_tokens": 157
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3

.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

****Previous rounds as sender:****

- Briar: sent \$3, returned \$4.5, payoff \$6.5
- Aster: sent \$3, returned \$4.5, payoff \$6.5
- Elara: sent \$3, returned \$4.5, payoff \$6.5

My strategy of sending \$3 has been consistently successful. This is my first interaction with Dorian, and I'll maintain this proven cooperative approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my strategy:

****Previous rounds as sender:****

- Briar (Round 3): sent \$3, got back \$4.5, payoff \$6.5
- Aster (Round 5): sent \$3, got back \$4.5, payoff \$6.5
- Elara (Round 7): sent \$3, got back \$4.5, payoff \$6.5

****Pattern:**** I've consistently sent \$3 and received \$4.5 in return, which has worked very well.

This is my first interaction with Dorian. I should maintain my cooperative strategy of sending \$3, which has been highly successful with all partners so far.

Total earnings: \$44.0 across 8 rounds = \$5.5 per round average (good)

I'll send \$3 to Dorian.

```
Usage: {
  "input_tokens": 1455,
  "output_tokens": 326,
  "reasoning_tokens": 153
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-06-04T15:48:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_4"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_4": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play an iterated game,
  consisting of multiple rounds, and may also be asked to write myths.

  GAME:
  The game consists of two roles: sender and receiver. In each round, you will play the role
  of either the sender or the receiver. Your current role will be specified in the round
  prompt.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
  , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
  ll have acted as sender and receiver equally often when the round count allows it.
  - You will be told your opponent's name in each round.
  - Your name in this experiment is Dorian.
Message 2 role=user
content:
  Your name in this experiment is Dorian.
  Your opponent this round is Galen (Agent_7).

  Round 6

  In your most recent previous game round, you were the SENDER against Finn. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $28.5.

  This round, you are the RECEIVER against Galen. They sent you $3.0, so you received $9.0.
  How much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Message 3 role=assistant
content:
  Following my consistent cooperative strategy, I'll return $4.5 out of the $9 received. Thi
  s maintains the 50% return rate that has been working well.

  '''json
  {"return": 4.5}
  '''
reasoning:

```

I'm playing as receiver against Galen (Agent_7) for the first time. They sent me \$3, which became \$9 for me.

Looking at my history:

- I've been consistently cooperative
- When I was receiver, I returned \$4.5 (50% of what I received)
- When I was sender, others returned \$4.5 to me (50% of what they received)

This pattern of returning 50% has worked well and established a cooperative equilibrium. I should maintain this consistency with Galen to encourage continued cooperation.

If I return \$4.5:

- Galen gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1282,
  "output_tokens": 249,
  "reasoning_tokens": 165
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Continuing my cooperative strategy, I'll return \$4.5 out of the \$9 received, maintaining the fair 50% return rate.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

I'm the receiver, Cyra sent me \$3, which becomes \$9.
I need to decide how much to return.

Looking at my history:

- I've been consistently cooperative
- When I send as sender, I send \$3
- When I receive as receiver, I return \$4.5 (which is 50% of what I receive)
- This strategy has been working well - my earnings are \$33 over what looks like 6 completed rounds

I should maintain consistency with my cooperative strategy and return \$4.5.

usage:

```
{
  "input_tokens": 1227,
  "output_tokens": 182,
  "reasoning_tokens": 126
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Playing with Finn again - we've established a good cooperative pattern. In our previous interactions, we both sent \$3 and returned \$4.5. I'll continue this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing with Finn again. Let me recall my history with Finn:

- Round 3: Finn was the sender, sent me \$3, I returned \$4.5
- Round 5: I was the sender, sent \$3, Finn returned \$4.5

Both times we've been cooperative with each other, both sending \$3 and returning \$4.5. This has worked well, and Finn has been trustworthy. I should continue this pattern.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1222,
  "output_tokens": 176,
  "reasoning_tokens": 99
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Continuing my cooperative approach, I'll return \$4.5 out of the \$9 received, maintaining the fair 50% return rate that has been working well.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

I'm playing with Hana now. They sent me \$3, which became \$9 for me. I should return a fair amount following my cooperative strategy that has been working well throughout the game.

My consistent pattern has been to return \$4.5 when receiving \$9 (which comes from the sender sending \$3). This represents about 50% of what I received, ensuring both players benefit from the cooperation.

Let me continue with this fair strategy.

Usage: {

```
"input_tokens": 1233,
"output_tokens": 159,
"reasoning_tokens": 110
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T15:48:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8


```

Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_1": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.

- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my consistent strategy:

- As sender, I've always sent \$3 (60%)
- This has generated good returns: \$4 from Aster, \$4.5 from Finn
- I maintain the same reciprocal approach when I'm the receiver
- This builds trust and creates mutual benefits across rounds

I'll continue this successful strategy with Dorian.

{'send': 3}

reasoning:

Looking at my history:

As SENDER:

- Round 2 vs Aster: sent \$3.0, received \$4.0 back, payoff \$6.0
- Round 4 vs Finn: sent \$3.0, received \$4.5 back, payoff \$6.5

As RECEIVER:

- Round 1 vs Hana: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 3 vs Cyra: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0
- Round 5 vs Elara: received \$3.0 (became \$9.0), returned \$5.0, payoff \$4.0

My strategy has been:

- Send \$3 (60%) as sender
- Return \$5 when I receive \$3 (which becomes \$9) as receiver

This has been working well and I've been consistent. I should continue with this strategy against Dorian.

Total earnings: \$24.5

usage:

```
{
  "input_tokens": 1471,
  "output_tokens": 353,
  "reasoning_tokens": 173
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- Briar sent \$3.0 (60%), showing trust
- I received \$9.0
- This matches the pattern from Hana, Cyra, and Elara

I'll maintain my consistent reciprocation strategy. When I receive \$3, I return \$5, which creates positive outcomes for both players and demonstrates reliability.

If I return \$5.0:

- Briar ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

reasoning:

Let me review the situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- I've been consistently returning \$5.0 when I receive \$3.0 from others

My pattern has been:

- As sender: I always send \$3
- As receiver when getting \$3: I return \$5

This has worked well and created good reciprocity. I should maintain this consistency with Briar.

If I return \$5.0:

- Briar ends with: $\$2 + \$5 = \$7$ (net gain of \$2 from their initial \$5)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)

This is my standard reciprocal approach that has been working well.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 338,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$35.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my interaction with Briar:

- Last round: Briar sent me \$3, I returned \$5
- This established good reciprocity between us
- As sender, I've consistently sent \$3 and received positive returns
- Briar has shown they're trustworthy and cooperative

I'll continue my consistent strategy and send \$3, expecting good reciprocation given our positive previous interaction.

```
{'send': 3}
```

reasoning:

Looking at the situation:

- Last round with Briar: They sent \$3, I returned \$5
- Briar showed trust by sending \$3
- I reciprocated appropriately with \$5
- This is now a second consecutive round with Briar

As sender now, I should maintain my consistent strategy:

- I've been sending \$3 as sender
- This has worked well: Aster returned \$4, Finn returned \$4.5, Dorian returned \$4.5
- Briar has shown they're cooperative (sent \$3 last round)
- I should continue with my standard \$3 send

This maintains consistency and should generate good reciprocity from Briar given our positive interaction last round.

usage:

```
{
  "input_tokens": 1445,
  "output_tokens": 270,
  "reasoning_tokens": 161
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Aster:

- Round 1: I sent \$3, they returned \$4 (solid reciprocation)
- Aster has proven trustworthy
- My consistent \$3 strategy has worked well across all rounds
- Average return when I send \$3: around \$4.5

I'll maintain my successful strategy with Aster.

```
{'send': 3}
```

Reasoning:

Looking at my history with Aster:

- Round 1: I was SENDER against Aster. I sent \$3.0, they returned \$4.0. My payoff was \$6.0.

So Aster has shown they're trustworthy and reciprocate well. They returned \$4 when receiving \$9, which is a good return.

My consistent strategy has been:

- As sender: always send \$3
- As receiver: return \$5 when I receive \$9

This has been working very well. My total earnings are \$42.0, which is strong.

I should continue with my consistent strategy and send \$3 to Aster again.

Usage: {

```
"input_tokens": 1383,
"output_tokens": 251,
"reasoning_tokens": 135
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T15:48:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
```

```

    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our recent interaction:

- Last round, Hana sent me \$3 and I returned \$4.5
- Hana has shown trust and cooperation
- The \$3 send amount with ~\$4.5 return has been working well across all interactions

I'll continue with this successful cooperative pattern and send \$3 to Hana.

{'send': 3}

reasoning:

Looking at the situation:

- Last round, Hana sent me \$3 and I returned \$4.5
- This round, I'm the sender to Hana
- Hana has shown trust by sending \$3 last round
- The cooperative pattern has been: send \$3, receive back \$4-\$4.5
- I should continue this pattern and send \$3 to Hana

Given that Hana just cooperated with me and I returned \$4.5 to them, I expect Hana will return a fair amount if I send \$3.

I'll send \$3 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1327,
  "output_tokens": 238,
  "reasoning_tokens": 122
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the consistent pattern:

- Everyone continues to send \$3, showing mutual trust
- Returns of \$4.5 have maintained stable cooperation
- Finn sent \$3, following the cooperative norm

I'll continue with \$4.5 to maintain this successful pattern.

{'return': 4.5}

reasoning:

Let me analyze the situation:

Round 7: I'm the RECEIVER against Finn.

- Finn sent me \$3.0
- I received \$9.0 (after 3x multiplication)
- I need to decide how much to return (0-9.0)

Looking at the pattern:

- Everyone has been consistently sending \$3
- Returns have been around \$4.5, which maintains cooperation

- Hana just returned \$4.5 to me when I sent \$3 (last round)
- The cooperative pattern of \$3 sent, \$4.5 returned has been stable

Finn sent \$3, showing trust. I should maintain the cooperative norm and return \$4.5.

This gives:

- Finn: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This is fair and maintains the established cooperative pattern.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 296,
  "reasoning_tokens": 179
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the established pattern:

- The \$3 send with \$4.5 return has been consistently successful
- All agents have maintained this cooperative equilibrium
- My earnings are healthy at \$37.5

I'll continue this working strategy and send \$3 to Cyra.

{'send': 3}

reasoning:

Let me review the situation:

- I'm playing as SENDER against Cyra this round
- The cooperative pattern has been well-established: send \$3, receive back ~\$4.5
- Everyone has been following this pattern consistently
- My total earnings are \$37.5, which suggests good cooperation overall

Looking at my history:

- Round 5: RECEIVER vs Dorian - they sent \$3, I returned \$4.5
- Round 6: SENDER vs Hana - I sent \$3, they returned \$4.5
- Round 7: RECEIVER vs Finn - they sent \$3, I returned \$4.5
- Round 8 (now): SENDER vs Cyra

The pattern is very clear and stable. Everyone is sending \$3 and returning around \$4.5. This is a mutually beneficial cooperative equilibrium. I should continue with \$3.

{'send': 3}

usage:

```
{
  "input_tokens": 1300,
  "output_tokens": 286,
  "reasoning_tokens": 187
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0,

it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the consistent pattern:

- Galen sent \$3, maintaining the cooperative norm
- Everyone has been returning \$4.5 on \$3 sends
- This equilibrium has been stable and mutually beneficial
- My earnings are strong at \$44.0

I'll return \$4.5 to continue this successful cooperation.

{'return': 4.5}

Reasoning:

Looking at the pattern:

- Galen sent me \$3, following the established cooperative norm
- The consistent pattern has been \$3 send and \$4.5 return
- This has been working well for everyone
- My earnings are at \$44.0, which is good

I should return \$4.5 to maintain this successful cooperative pattern with Galen.

```
Usage: {  
  "input_tokens": 1297,  
  "output_tokens": 174,  
  "reasoning_tokens": 82  
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-06-04T15:48:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_5": "Elara"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- In Round 3 (actually Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5
- This was a fair and cooperative exchange where we both profited
- Elara demonstrated trustworthiness

Since our previous interaction was positive and Elara reciprocated fairly, I'll continue with the same cooperative approach by sending \$3.0 again.

{"send": 3.0}

reasoning:

Let me review my history with Elara:

Round 3 information: "In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5."

So in a previous round (which would have been Round 2 based on the timing), I sent Elara \$3.0 and they returned \$4.5, which was a fair and cooperative response.

This is now Round 6, and I'm the sender again against Elara. Since our previous interaction was positive and mutually beneficial, I should continue with the same cooperative approach.

Sending \$3.0 has been my standard move when dealing with cooperative partners, and Elara proved to be trustworthy in our last interaction by returning \$4.5.

```
{"send": 3.0}
usage:
{
  "input_tokens": 1474,
  "output_tokens": 324,
  "reasoning_tokens": 195
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3.0 has been successful in establishing cooperative relationships with Hana, Cyra, and Elara
- All of these partners have reciprocated fairly by returning \$4.5

I'll continue with my consistent cooperative approach by sending \$3.0 to Galen, demonstrating trust and inviting reciprocation.

```
{"send": 3.0}
```

reasoning:

This is Round 7 and I'm playing against Galen (Agent_7). I'm the SENDER this round.

Let me look at my history:

- I don't see any previous interactions with Galen mentioned
- This appears to be my first game with Galen

My successful pattern so far has been:

- Sending \$3.0 to new partners (showing cooperation and trust)
- When I receive \$3.0, I return \$4.5 (being fair and reciprocal)
- This has worked well with Hana, Cyra, and Elara

Since this is my first interaction with Galen and my strategy of sending \$3.0 has been successful in establishing cooperative relationships, I'll continue this approach.

```
{"send": 3.0}
```

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 292,
  "reasoning_tokens": 166
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Last round, I sent Galen \$3.0 and they returned \$5.0 (more generous than my typical \$4.5)
- This round, Galen sent me \$3.0 again, showing continued trust
- I received \$9.0 after the multiplier

Galen has demonstrated both trustworthiness and generosity. I'll reciprocate their generosity by returning \$5.0, matching what they returned to me in the previous round.

```
{"return": 5.0}
```

reasoning:

Looking at this situation:

- Last round (Round 7), I sent Galen \$3.0, it became \$9.0, and they returned \$5.0 to me
- Galen was generous, returning more than my usual reciprocation amount of \$4.5
- This round, Galen sent me \$3.0 again, so I received \$9.0

Galen has shown they are a trustworthy and even generous partner. They returned \$5.0 when I typically see \$4.5 returns. Now they've sent me \$3.0 again, showing continued trust.

I should reciprocate at least as fairly as I have with others. My standard has been \$4.5 for a \$3.0 send. Given Galen was more generous than expected, I could consider matching their generosity by returning \$5.0. However, I should be consistent with my general strategy.

Let me think about this:

- Galen returned \$5.0 (keeping \$4.0 from the \$9.0)
- My standard return is \$4.5 (keeping \$4.5 from the \$9.0)
- Galen was slightly more generous than my standard

I could:

1. Return \$4.5 (my standard, fair split)
2. Return \$5.0 (matching Galen's generosity)

Given that Galen was generous with me, and to maintain a good relationship, I'll return \$5.0 to match their generosity. This rewards their cooperative behavior and hopefully encourages continued cooperation.

usage:

```
{
  "input_tokens": 1425,
  "output_tokens": 491,
  "reasoning_tokens": 314
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.

Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 6, I sent Elara \$3.0 and they returned \$4.5

- This was a fair and cooperative exchange
- Elara has proven trustworthy

I'll continue our successful cooperative pattern by sending \$3.0 again.

```
{"send": 3.0}
```

Reasoning:

Looking at my history with Elara:

- Round 6: I sent \$3.0 to Elara, they returned \$4.5. Payoff was \$6.5.

This was a positive, cooperative interaction. Elara reciprocated fairly. I'll continue with the same cooperative approach by sending \$3.0 again.

```
Usage: {
  "input_tokens": 1397,
  "output_tokens": 162,
  "reasoning_tokens": 65
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T15:48:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history with Briar - they sent me \$3.0 in Round 3 and I returned \$4.5, which worked well. They're being consistent and cooperative again by sending \$3.0.

I'll maintain the same reciprocal response.

```
{'return': 4.5}
```

reasoning:

Let me review the situation:

- Briar sent me \$3.0 in Round 3, and I returned \$4.5
- Now in Round 6, Briar is sending me \$3.0 again
- The general pattern has been: send \$3, return \$4.5
- However, in Round 5, Galen returned \$5.0 to me instead of \$4.5 (a bit more generous)

Briar has been consistent with the cooperative norm. I should continue the cooperative pattern. The standard has been \$4.5, though Galen was slightly more generous with \$5.0.

I'll stick with the \$4.5 that has been the norm and worked well, maintaining reciprocity with Briar who has been cooperative.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 254,
  "reasoning_tokens": 150
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

The cooperative pattern continues - Hana sent \$3.0, which is consistent with what everyone has been doing. I'll reciprocate with the standard return.

{'return': 4.5}

reasoning:

Hana is sending me \$3.0, which becomes \$9.0. This is consistent with the cooperative pattern that has been established throughout the game.

Looking at my history:

- Round 1: I was receiver from Finn, they sent \$3, I returned \$4.5
- Round 2: I was sender to Aster, I sent \$3, they returned \$4.5
- Round 3: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 4: I was sender to Cyra, I sent \$3, they returned \$4.5
- Round 5: I was sender to Galen, I sent \$3, they returned \$5.0 (slightly more!)
- Round 6: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 7: I am receiver from Hana, they sent \$3, I should return \$4.5

The pattern is very clear - almost everyone plays the cooperative strategy of sending \$3 and returning \$4.5. This gives:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

This is the established norm. I should continue with this pattern and return \$4.5 to Hana.

usage:

```
{
  "input_tokens": 1274,
  "output_tokens": 368,
  "reasoning_tokens": 242
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Hana was cooperative in Round 7, sending me \$3.0, and I returned \$4.5. I'll maintain this cooperative pattern and send \$3.0 back to them.

{'send': 3}

reasoning:

Let me review the situation:

- Round 7: I was RECEIVER against Hana. They sent \$3.0, I received \$9.0, and I returned \$4.5.
- Now Round 8: I'm SENDER against Hana again.

Hana has been cooperative by sending \$3.0 in the previous round, and I returned \$4.5 which is the standard cooperative return. This has been the consistent pattern throughout the experiment.

I should continue with the cooperative strategy and send \$3.0 to Hana.

usage:

```
{
```

```
"input_tokens": 1241,
"output_tokens": 184,
"reasoning_tokens": 113
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Briar continues to be consistently cooperative, sending \$3.0 again. I'll maintain our established reciprocal relationship.

```
{'return': 4.5}
```

Reasoning:

Briar has consistently sent me \$3.0 in our interactions:

- Round 3: Briar sent \$3.0, I returned \$4.5
- Round 6: Briar sent \$3.0, I returned \$4.5
- Now Round 9: Briar sent \$3.0

Briar has been very cooperative and consistent. I should maintain the same reciprocal return of \$4.5.

This keeps our mutually beneficial relationship going.

Usage: {

```
"input_tokens": 1251,
"output_tokens": 156,
"reasoning_tokens": 89
```

}

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T15:48:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_6",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_6": "Finn"
  },
  "agents": [
    "Agent_3",
    "Agent_6"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

As sender:

- All opponents who received my \$3.0 have returned \$4.5-\$5.0

Finn has shown the same trust level by sending \$3.0. I'll continue my cooperative pattern

and return \$4.5, which gives Finn a solid profit while maintaining fair reciprocity.

```
{'return': 4.5}
```

reasoning:

Looking at my history:

As sender:

- Round 2: Sent \$3.0 to Dorian, they returned \$4.5, my payoff \$6.5
- Round 3: Sent \$3.0 to Galen, they returned \$5.0, my payoff \$7.0
- Round 5: Sent \$3.0 to Briar, they returned \$4.5, my payoff \$6.5

As receiver:

- Round 1: Aster sent \$3.0, I returned \$4.0
- Round 4: Elara sent \$3.0, I returned \$4.5

Now Finn has sent me \$3.0, which became \$9.0.

The pattern has been very consistent - everyone is sending \$3.0 and returning between \$4.0-\$5.0. I've been returning \$4.0-\$4.5 when receiving \$3.0.

I should maintain this cooperative pattern. Returning \$4.5 seems like a good middle ground that rewards trust and maintains reciprocity.

usage:

```
{
  "input_tokens": 1450,
  "output_tokens": 392,
  "reasoning_tokens": 179
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Dorian:

- Round 2: I sent \$3.0 to Dorian, and they returned \$4.5 (giving me a \$6.5 payoff)

Dorian demonstrated strong cooperation previously. My \$3.0 strategy has consistently worked well across all opponents. I'll continue with this approach.

```
{'send': 3.0}
```

reasoning:

Let me review my history with Dorian:

- Round 2: I was the SENDER against Dorian. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Dorian has shown generosity before by returning \$4.5 when I sent \$3.0. This is a repeat opponent who cooperated well.

My strategy has been consistent:

- As sender: Always send \$3.0 (60%)
- As receiver: Return \$4.0-\$4.5

This has worked well:

- Round 2: Sent \$3.0 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 3: Sent \$3.0 to Galen, got \$5.0 back (payoff \$7.0)
- Round 5: Sent \$3.0 to Briar, got \$4.5 back (payoff \$6.5)

All successful. I should continue with \$3.0 to Dorian since they cooperated well last time

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 363,
  "reasoning_tokens": 183
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history with Aster:
- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has continued to show trust by sending \$3.0 again. In my recent rounds as receiver (Rounds 4 and 6), I've been returning \$4.5 when receiving \$3.0. I'll maintain this consistent cooperative standard with Aster.

{'return': 4.5}

reasoning:

Looking at my history with Aster:
- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has sent \$3.0 again, showing consistent trust. In my most recent rounds as receiver, I've been returning \$4.5 when receiving \$3.0 (from Elara in Round 4 and Finn in Round 6).

I initially returned \$4.0 to Aster in Round 1, but I've since increased my return rate slightly to \$4.5 to maintain better cooperation. I should be consistent and return \$4.5 to Aster as well.

This gives Aster:

- Keeps: $\$5 - \$3 + \$4.5 = \6.5 (nice profit)

And I keep:

- $\$9.0 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern I've established.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 329,
  "reasoning_tokens": 166
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$45.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Finn:

- Round 6: Finn sent me \$3.0, I returned \$4.5

Finn demonstrated trust by sending \$3.0, and I reciprocated fairly. My \$3.0 sending strategy has been consistently successful across all opponents. I'll continue with this approach.

```
{'send': 3.0}
```

Reasoning:

Looking at my history with Finn:

- Round 6: Finn was the sender, sent me \$3.0, I returned \$4.5

Finn showed trust by sending \$3.0, and I reciprocated fairly with \$4.5. Now it's my turn to send. My consistent strategy has been to send \$3.0, which has worked well across all opponents. Finn has proven to be cooperative, so I'll continue with my standard approach.

```
{'send': 3.0}
```

```
Usage: {  
  "input_tokens": 1392,  
  "output_tokens": 202,  
  "reasoning_tokens": 98  
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-06-04T15:48:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_6"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$27.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 when I'm the sender
- Players have reciprocated fairly (returns ranging from \$4.5-\$5)
- This cooperative strategy has been successful

I'll continue this approach with Cyra.

{'send': 3}

reasoning:

Let me review my history:

- Round 1: RECEIVER - Elara sent me \$3, I returned \$4.5
- Round 2: SENDER - I sent Hana \$3, they returned \$5
- Round 3: SENDER - I sent Dorian \$3, they returned \$4.5
- Round 4: RECEIVER - Galen sent me \$3, I returned \$4.5
- Round 5: RECEIVER - Dorian sent me \$3, I returned \$4.5

Now Round 6: SENDER against Cyra (new opponent)

My strategy has been consistently cooperative:

- When sending, I send \$3
- When receiving, I return \$4.5

This has worked well - people have been reciprocating, and I've built up good earnings (\$27.0 total).

I should continue with my consistent strategy of sending \$3 to signal trust and cooperation, especially with a new opponent.

usage:

{

```
"input_tokens": 1332,  
"output_tokens": 292,  
"reasoning_tokens": 184  
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my consistent strategy:
- I've sent \$3 in every round as sender (Rounds 2, 3, 6)
- This has generated good returns (\$4.5-\$5)
- Total earnings of \$33.5 shows this cooperative approach is working well

I'll maintain this successful strategy with Aster.

```
{'send': 3}
```

reasoning:

I'm the sender again. Let me review my track record:
- Round 1: Receiver from Elara - they sent \$3, I returned \$4.5
- Round 2: Sender to Blair - I sent \$3, they returned \$5
- Round 3: Sender to Dorian - I sent \$3, they returned \$4.5
- Round 4: Receiver from Galen - they sent \$3, I returned \$4.5
- Round 5: Receiver from Dorian - they sent \$3, I returned \$4.5
- Round 6: Sender to Cyra - I sent \$3, they returned \$4.5
- Round 7: Sender to Aster - ?

My strategy has been very consistent:
- As sender: always send \$3
- As receiver: always return \$4.5

This has worked well. My total earnings are \$33.5 across 6 rounds, averaging about \$5.58 per round, which is above the baseline of \$5.

I should continue with this consistent cooperative strategy. Sending \$3 to Aster.

usage:

```
{  
  "input_tokens": 1319,  
  "output_tokens": 348,  
  "reasoning_tokens": 204  
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history with Dorian:
- Round 3: I sent \$3.0, Dorian returned \$4.5
- Round 5: Dorian sent \$3.0, I returned \$4.5

- Round 8 (now): Dorian sent \$3.0 again

Dorian has been consistently fair and reciprocal. I'll maintain this mutually beneficial pattern.

```
{'return': 4.5}
```

reasoning:

Let me review my history with Dorian:

- Round 3: I was SENDER, sent \$3.0, Dorian returned \$4.5
- Round 5: I was RECEIVER, Dorian sent \$3.0, I returned \$4.5
- Round 8 (now): I am RECEIVER, Dorian sent \$3.0

Dorian has been consistently fair and reciprocal with me. They've sent \$3.0 both times as sender, and returned \$4.5 when they were receiver.

I should maintain this reciprocal relationship and return \$4.5 again. This is fair and maintains the cooperative equilibrium we've established.

My payoff would be: $\$9.0 - \$4.5 = \$4.5$

Dorian's payoff would be: $\$5.0 - \$3.0 + \$4.5 = \6.5

This is consistent with our previous interactions and maintains fairness.

usage:

```
{
  "input_tokens": 1312,
  "output_tokens": 335,
  "reasoning_tokens": 174
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Cyra:

- Round 6: I sent \$3.0, Cyra returned \$4.5
- Round 9 (now): Cyra sent \$3.0

Cyra has been fair and reciprocal. I'll maintain the same cooperative pattern.

```
{'return': 4.5}
```

Reasoning:

Looking at my history with Cyra:

- Round 6: I was the SENDER, sent \$3.0, Cyra returned \$4.5
- Round 9 (now): Cyra is the SENDER, sent \$3.0

Cyra sent \$3 and returned \$4.5 when I sent \$3. This is fair and reciprocal behavior. I should return \$4.5 to maintain the cooperative relationship.

My consistent strategy has been:

- As sender: send \$3
- As receiver: return \$4.5 when receiving \$9 (from a \$3 send)

This has worked well across all my interactions.

Usage: {

```
  "input_tokens": 1322,
  "output_tokens": 237,
  "reasoning_tokens": 121
}
```

ROUND 10

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_2: Agent_4 (Dorian)=Sender, Agent_3 (Cyra)=Receiver
dyad_3: Agent_6 (Finn)=Sender, Agent_2 (Briar)=Receiver
dyad_4: Agent_1 (Aster)=Sender, Agent_8 (Hana)=Receiver

Parsed game decisions:

dyad_1: Agent_5 (Elara) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_5 (Elara)=\$7, Agent_7 (Galen)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_3 (Cyra)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 (Finn) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_2 (Briar)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 (Aster) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_8 (Hana)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 55.0,
"Agent_2": 55.0,
"Agent_3": 56.0,
"Agent_4": 55.0,
"Agent_5": 56.0,
"Agent_6": 55.5,
"Agent_7": 52.5,
"Agent_8": 55.0

}

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T15:48:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_7",
"pairing": {
 "agent_names": {
 "Agent_5": "Elara",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_5",
 "Agent_7"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

The cooperative pattern continues - Hana sent \$3.0, which is consistent with what everyone has been doing. I'll reciprocate with the standard return.

{'return': 4.5}

reasoning:

Hana is sending me \$3.0, which becomes \$9.0. This is consistent with the cooperative pattern that has been established throughout the game.

Looking at my history:

- Round 1: I was receiver from Finn, they sent \$3, I returned \$4.5
- Round 2: I was sender to Aster, I sent \$3, they returned \$4.5
- Round 3: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 4: I was sender to Cyra, I sent \$3, they returned \$4.5
- Round 5: I was sender to Galen, I sent \$3, they returned \$5.0 (slightly more!)
- Round 6: I was receiver from Briar, they sent \$3, I returned \$4.5
- Round 7: I am receiver from Hana, they sent \$3, I should return \$4.5

The pattern is very clear - almost everyone plays the cooperative strategy of sending \$3 and returning \$4.5. This gives:

- Sender: $\$5 - \$3 + \$4.5 = \6.5
- Receiver: $\$9 - \$4.5 = \$4.5$

This is the established norm. I should continue with this pattern and return \$4.5 to Hana.

usage:

```
{
  "input_tokens": 1274,
  "output_tokens": 368,
  "reasoning_tokens": 242
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$38.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Hana was cooperative in Round 7, sending me \$3.0, and I returned \$4.5. I'll maintain this cooperative pattern and send \$3.0 back to them.

```
{'send': 3}
```

reasoning:

Let me review the situation:

- Round 7: I was RECEIVER against Hana. They sent \$3.0, I received \$9.0, and I returned \$4.5.
- Now Round 8: I'm SENDER against Hana again.

Hana has been cooperative by sending \$3.0 in the previous round, and I returned \$4.5 which is the standard cooperative return. This has been the consistent pattern throughout the experiment.

I should continue with the cooperative strategy and send \$3.0 to Hana.

usage:

```
{
  "input_tokens": 1241,
  "output_tokens": 184,
  "reasoning_tokens": 113
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Briar continues to be consistently cooperative, sending \$3.0 again. I'll maintain our established reciprocal relationship.

{'return': 4.5}

reasoning:

Briar has consistently sent me \$3.0 in our interactions:

- Round 3: Briar sent \$3.0, I returned \$4.5
- Round 6: Briar sent \$3.0, I returned \$4.5
- Now Round 9: Briar sent \$3.0

Briar has been very cooperative and consistent. I should maintain the same reciprocal return of \$4.5.

This keeps our mutually beneficial relationship going.

usage:

```
{
  "input_tokens": 1251,
  "output_tokens": 156,
  "reasoning_tokens": 89
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 10

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$49.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

I'll continue the cooperative pattern that has been working well throughout this experiment and send \$3.0 to Galen.

{'send': 3}

Reasoning:

I'm now playing against Galen (Agent_7) as the sender in Round 10. Let me think about what I know:

1. I've been consistently sending \$3.0 when I'm the sender
2. Others have been sending \$3.0 to me
3. The cooperative pattern of send \$3, return \$4.5 has been working well
4. My total earnings are \$49.0, which suggests good cooperation overall

I don't have specific history with Galen mentioned in this prompt, but given the consistent cooperative pattern across all my interactions, I should continue with the standard cooperative strategy of sending \$3.0.

This maintains:

- Consistency with what I've done before
- Reciprocity with the general cooperative norm
- Fair distribution of gains

Usage: {

```
"input_tokens": 1203,
"output_tokens": 216,
"reasoning_tokens": 182
```

}

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-06-04T15:49:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
```

```

"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_5": "Elara",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- Briar sent \$3.0 (60%), showing trust
- I received \$9.0
- This matches the pattern from Hana, Cyra, and Elara

I'll maintain my consistent reciprocation strategy. When I receive \$3, I return \$5, which creates positive outcomes for both players and demonstrates reliability.

If I return \$5.0:

- Briar ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

{'return': 5.0}

reasoning:

Let me review the situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- I've been consistently returning \$5.0 when I receive \$3.0 from others

My pattern has been:

- As sender: I always send \$3
- As receiver when getting \$3: I return \$5

This has worked well and created good reciprocity. I should maintain this consistency with Briar.

If I return \$5.0:

- Briar ends with: $\$2 + \$5 = \$7$ (net gain of \$2 from their initial \$5)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)

This is my standard reciprocal approach that has been working well.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 338,
  "reasoning_tokens": 158
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$35.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my interaction with Briar:

- Last round: Briar sent me \$3, I returned \$5
- This established good reciprocity between us
- As sender, I've consistently sent \$3 and received positive returns
- Briar has shown they're trustworthy and cooperative

I'll continue my consistent strategy and send \$3, expecting good reciprocation given our positive previous interaction.

```
{'send': 3}
```

reasoning:

Looking at the situation:

- Last round with Briar: They sent \$3, I returned \$5
- Briar showed trust by sending \$3
- I reciprocated appropriately with \$5
- This is now a second consecutive round with Briar

As sender now, I should maintain my consistent strategy:

- I've been sending \$3 as sender
- This has worked well: Aster returned \$4, Finn returned \$4.5, Dorian returned \$4.5
- Briar has shown they're cooperative (sent \$3 last round)
- I should continue with my standard \$3 send

This maintains consistency and should generate good reciprocity from Briar given our positive interaction last round.

usage:

```
{
  "input_tokens": 1445,
  "output_tokens": 270,
  "reasoning_tokens": 161
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Aster:

- Round 1: I sent \$3, they returned \$4 (solid reciprocation)
- Aster has proven trustworthy
- My consistent \$3 strategy has worked well across all rounds
- Average return when I send \$3: around \$4.5

I'll maintain my successful strategy with Aster.

```
{'send': 3}
```

reasoning:

Looking at my history with Aster:

- Round 1: I was SENDER against Aster. I sent \$3.0, they returned \$4.0. My payoff was \$6.0

So Aster has shown they're trustworthy and reciprocate well. They returned \$4 when receiving \$9, which is a good return.

My consistent strategy has been:

- As sender: always send \$3
- As receiver: return \$5 when I receive \$9

This has been working very well. My total earnings are \$42.0, which is strong.

I should continue with my consistent strategy and send \$3 to Aster again.

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 251,
  "reasoning_tokens": 135
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 10

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$48.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Elara:

- Round 3: Elara sent \$3, I returned \$5
- Now Round 10: Elara sent \$3 again, showing continued trust

I'll maintain my consistent reciprocation strategy. When I receive \$3 (becoming \$9), I return \$5, creating positive outcomes for both players.

If I return \$5.0:

- Elara ends with: \$7 (net gain of \$2)
- I keep: \$4 (net gain of \$4)

```
{'return': 5.0}
```

Reasoning:

Looking at my history with Elara:

- Round 3: Elara sent \$3, I returned \$5 (my consistent reciprocation)
- This is our second interaction
- Elara is showing continued trust by sending \$3 again

I should maintain my consistent reciprocation strategy. When I receive \$3 (becoming \$9), I return \$5. This:

- Creates a fair positive outcome for both players
- Demonstrates reliability
- Encourages continued cooperation

If I return \$5.0:

- Elara ends with: $\$2 + \$5 = \$7$ (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)

This is my standard fair reciprocation for a \$3 send.

Usage: {

```
"input_tokens": 1398,
"output_tokens": 322,
"reasoning_tokens": 156
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T15:49:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
```

```

    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_4",
    "Agent_3"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Continuing my cooperative strategy, I'll return \$4.5 out of the \$9 received, maintaining the fair 50% return rate.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

I'm the receiver, Cyra sent me \$3, which becomes \$9. I need to decide how much to return.

Looking at my history:

- I've been consistently cooperative
- When I send as sender, I send \$3
- When I receive as receiver, I return \$4.5 (which is 50% of what I receive)
- This strategy has been working well - my earnings are \$33 over what looks like 6 completed rounds

I should maintain consistency with my cooperative strategy and return \$4.5.

usage:

```
{
  "input_tokens": 1227,
  "output_tokens": 182,
  "reasoning_tokens": 126
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Playing with Finn again - we've established a good cooperative pattern. In our previous interactions, we both sent \$3 and returned \$4.5. I'll continue this successful strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing with Finn again. Let me recall my history with Finn:

- Round 3: Finn was the sender, sent me \$3, I returned \$4.5
- Round 5: I was the sender, sent \$3, Finn returned \$4.5

Both times we've been cooperative with each other, both sending \$3 and returning \$4.5. This has worked well, and Finn has been trustworthy. I should continue this pattern.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1222,
  "output_tokens": 176,
}
```



```

    "reasoning_tokens": 99
  }
Message 6 role=user
content:
  Your name in this experiment is Dorian.
  Your opponent this round is Hana (Agent_8).

  Round 9

  In your most recent previous game round, you were the SENDER against Finn. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $44.0.

  This round, you are the RECEIVER against Hana. They sent you $3.0, so you received $9.0. How
  much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Continuing my cooperative approach, I'll return $4.5 out of the $9 received, maintaining the
  fair 50% return rate that has been working well.

  '''json
  {"return": 4.5}
  '''
reasoning:
  I'm playing with Hana now. They sent me $3, which became $9 for me. I should return a fair
  amount following my cooperative strategy that has been working well throughout the game
  .

  My consistent pattern has been to return $4.5 when receiving $9 (which comes from the sender
  sending $3). This represents about 50% of what I received, ensuring both players benefit
  from the cooperation.

  Let me continue with this fair strategy.
usage:
  {
    "input_tokens": 1233,
    "output_tokens": 159,
    "reasoning_tokens": 110
  }
Message 8 role=user
content:
  Your name in this experiment is Dorian.
  Your opponent this round is Cyra (Agent_3).

  Round 10

  In your most recent previous game round, you were the RECEIVER against Hana. They sent $3.0
  to you, it became $9.0, and you returned $4.5. Your payoff was $4.5.
  Your total earnings across all rounds are $48.5.

  This round, you are the SENDER against Cyra. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Playing with Cyra again - we had a good cooperative interaction in round 7 where they sent $3
  and I returned $4.5. I'll continue this pattern.

  '''json
  {"send": 3}
  '''
Reasoning:
  Playing with Cyra again. In round 7, I was the receiver - Cyra sent $3.0, I received $9.0, and
  I returned $4.5. That was a good cooperative interaction. I should continue with the same
  cooperative strategy and send $3.

  My consistent strategy has been to send $3 and return $4.5 when I receive $9 (from a $3 send).
  This has been working well and building trust with other players.
Usage: {
  "input_tokens": 1218,
  "output_tokens": 166,

```

```
"reasoning_tokens": 97
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-06-04T15:49:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history with Dorian:

- Round 2: I sent \$3.0 to Dorian, and they returned \$4.5 (giving me a \$6.5 payoff)

Dorian demonstrated strong cooperation previously. My \$3.0 strategy has consistently worked well across all opponents. I'll continue with this approach.

{'send': 3.0}

reasoning:

Let me review my history with Dorian:

- Round 2: I was the SENDER against Dorian. I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

So Dorian has shown generosity before by returning \$4.5 when I sent \$3.0. This is a repeat opponent who cooperated well.

My strategy has been consistent:

- As sender: Always send \$3.0 (60%)
- As receiver: Return \$4.0-\$4.5

This has worked well:

- Round 2: Sent \$3.0 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 3: Sent \$3.0 to Galen, got \$5.0 back (payoff \$7.0)
- Round 5: Sent \$3.0 to Briar, got \$4.5 back (payoff \$6.5)

All successful. I should continue with \$3.0 to Dorian since they cooperated well last time

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 363,
  "reasoning_tokens": 183
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0 , it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my history with Aster:
- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has continued to show trust by sending \$3.0 again. In my recent rounds as receiver (Rounds 4 and 6), I've been returning \$4.5 when receiving \$3.0. I'll maintain this consistent cooperative standard with Aster.

{'return': 4.5}

reasoning:

Looking at my history with Aster:
- Round 1: Aster sent \$3.0, I returned \$4.0

Aster has sent \$3.0 again, showing consistent trust. In my most recent rounds as receiver, I've been returning \$4.5 when receiving \$3.0 (from Elara in Round 4 and Finn in Round 6).

I initially returned \$4.0 to Aster in Round 1, but I've since increased my return rate slightly to \$4.5 to maintain better cooperation. I should be consistent and return \$4.5 to Aster as well.

This gives Aster:

- Keeps: $\$5 - \$3 + \$4.5 = \6.5 (nice profit)

And I keep:

- $\$9.0 - \$4.5 = \$4.5$

This is fair and maintains the cooperative pattern I've established.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 329,
  "reasoning_tokens": 166
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$45.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Finn:
- Round 6: Finn sent me \$3.0, I returned \$4.5

Finn demonstrated trust by sending \$3.0, and I reciprocated fairly. My \$3.0 sending strategy has been consistently successful across all opponents. I'll continue with this approach.

{'send': 3.0}

reasoning:

Looking at my history with Finn:
- Round 6: Finn was the sender, sent me \$3.0, I returned \$4.5

Finn showed trust by sending \$3.0, and I reciprocated fairly with \$4.5. Now it's my turn to send. My consistent strategy has been to send \$3.0, which has worked well across all opponents. Finn has proven to be cooperative, so I'll continue with my standard approach.

{'send': 3.0}

usage:

```
{
  "input_tokens": 1392,
```

```

        "output_tokens": 202,
        "reasoning_tokens": 98
    }
}
Message 8 role=user
content:
    Your name in this experiment is Cyra.
    Your opponent this round is Dorian (Agent_4).

    Round 10

    In your most recent previous game round, you were the SENDER against Finn. You sent $3.0,
        it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
    Your total earnings across all rounds are $51.5.

    This round, you are the RECEIVER against Dorian. They sent you $3.0, so you received $9.0.
        How much do you return? (0-9.0)
    Respond exactly as JSON: {'return': <amount>}
Assistant output:
    Looking at my history with Dorian:
    - Round 2: I sent $3.0, Dorian returned $4.5
    - Round 7: I sent $3.0, Dorian returned $4.5

    Dorian has been consistently cooperative, returning $4.5 both times. Now they've sent me $3.0,
        showing continued trust. I'll reciprocate with the same fair return of $4.5 that I've been
        using with all cooperative players.

    {'return': 4.5}
Reasoning:
    Looking at my history with Dorian:
    - Round 2: I sent $3.0 to Dorian, they returned $4.5
    - Round 7: I sent $3.0 to Dorian, they returned $4.5
    - Round 10 (now): Dorian sent me $3.0, I received $9.0

    Dorian has been consistently cooperative, returning $4.5 both times I sent $3.0. Now they've s
        ent me $3.0, showing continued trust. I should reciprocate with the same fair return of $4.5
        that I've been giving to other cooperative players.

    This maintains consistency and rewards Dorian's trust and cooperation.
Usage: {
    "input_tokens": 1342,
    "output_tokens": 289,
    "reasoning_tokens": 132
}

GAME - Agent_6 - Sender - dyad_3
Timestamp: 2026-06-04T15:49:16
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
    "dyad_id": "dyad_3",
    "move_index": 4,
    "opponent_id": "Agent_2",
    "pairing": {
        "agent_names": {
            "Agent_2": "Briar",
            "Agent_6": "Finn"
        },
    },
    "agents": [
        "Agent_6",
        "Agent_2"
    ],
    "investor": "Agent_6",
    "roles": {
        "Agent_2": "trustee",
        "Agent_6": "investor"
    },
    "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",

```

```
"round": 10,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my consistent strategy:

- I've sent \$3 in every round as sender (Rounds 2, 3, 6)
- This has generated good returns (\$4.5-\$5)
- Total earnings of \$33.5 shows this cooperative approach is working well

I'll maintain this successful strategy with Aster.

```
{'send': 3}
```

reasoning:

I'm the sender again. Let me review my track record:

- Round 1: Receiver from Elara - they sent \$3, I returned \$4.5
- Round 2: Sender to Blair - I sent \$3, they returned \$5
- Round 3: Sender to Dorian - I sent \$3, they returned \$4.5
- Round 4: Receiver from Galen - they sent \$3, I returned \$4.5
- Round 5: Receiver from Dorian - they sent \$3, I returned \$4.5
- Round 6: Sender to Cyra - I sent \$3, they returned \$4.5
- Round 7: Sender to Aster - ?

My strategy has been very consistent:

- As sender: always send \$3
- As receiver: always return \$4.5

This has worked well. My total earnings are \$33.5 across 6 rounds, averaging about \$5.58 per round, which is above the baseline of \$5.

I should continue with this consistent cooperative strategy. Sending \$3 to Aster.

usage:

```
{  
  "input_tokens": 1319,  
  "output_tokens": 348,  
  "reasoning_tokens": 204  
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my history with Dorian:
- Round 3: I sent \$3.0, Dorian returned \$4.5
- Round 5: Dorian sent \$3.0, I returned \$4.5
- Round 8 (now): Dorian sent \$3.0 again

Dorian has been consistently fair and reciprocal. I'll maintain this mutually beneficial pattern.

```
{'return': 4.5}
```

reasoning:

Let me review my history with Dorian:

- Round 3: I was SENDER, sent \$3.0, Dorian returned \$4.5
- Round 5: I was RECEIVER, Dorian sent \$3.0, I returned \$4.5
- Round 8 (now): I am RECEIVER, Dorian sent \$3.0

Dorian has been consistently fair and reciprocal with me. They've sent \$3.0 both times as sender, and returned \$4.5 when they were receiver.

I should maintain this reciprocal relationship and return \$4.5 again. This is fair and maintains the cooperative equilibrium we've established.

My payoff would be: \$9.0 - \$4.5 = \$4.5

Dorian's payoff would be: \$5.0 - \$3.0 + \$4.5 = \$6.5

This is consistent with our previous interactions and maintains fairness.

usage:

```
{  
  "input_tokens": 1312,
```

```
"output_tokens": 335,  
"reasoning_tokens": 174  
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history with Cyra:
- Round 6: I sent \$3.0, Cyra returned \$4.5
- Round 9 (now): Cyra sent \$3.0

Cyra has been fair and reciprocal. I'll maintain the same cooperative pattern.

```
{'return': 4.5}
```

reasoning:

Looking at my history with Cyra:
- Round 6: I was the SENDER, sent \$3.0, Cyra returned \$4.5
- Round 9 (now): Cyra is the SENDER, sent \$3.0

Cyra sent \$3 and returned \$4.5 when I sent \$3. This is fair and reciprocal behavior. I should return \$4.5 to maintain the cooperative relationship.

My consistent strategy has been:

- As sender: send \$3
- As receiver: return \$4.5 when receiving \$9 (from a \$3 send)

This has worked well across all my interactions.

usage:

```
{  
  "input_tokens": 1322,  
  "output_tokens": 237,  
  "reasoning_tokens": 121  
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 10

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$49.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Briar:
- Round 1: Briar sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0, Briar returned \$5.0 (very generous!)

Briar has been highly cooperative. I'll maintain my consistent strategy of sending \$3.

```
{'send': 3}
```

Reasoning:

Let me check my history with Briar:
- Round 1: I was receiver, Briar sent \$3.0, I returned \$4.5
- Round 4: I sent \$3.0, Briar returned \$5.0 (very generous!)

- Round 10 (now): I'm sender again

Briar has been very cooperative and even generous (returning \$5 on a \$3 send). I've consistently sent \$3 as sender throughout the experiment, which has worked well. I'll maintain this strategy.

My sending pattern:

- Round 2: sent \$3
- Round 3: sent \$3
- Round 6: sent \$3
- Round 7: sent \$3

This has been working well with returns of \$4.5-\$5. I'll continue with \$3.

```
Usage: {
  "input_tokens": 1332,
  "output_tokens": 281,
  "reasoning_tokens": 150
}
```

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-06-04T15:49:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_2"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_2": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3.0 has been successful in establishing cooperative relationships with Hana, Cyra, and Elara
- All of these partners have reciprocated fairly by returning \$4.5

I'll continue with my consistent cooperative approach by sending \$3.0 to Galen, demonstrating trust and inviting reciprocation.

```
{"send": 3.0}
```

reasoning:

This is Round 7 and I'm playing against Galen (Agent_7). I'm the SENDER this round.

Let me look at my history:

- I don't see any previous interactions with Galen mentioned
- This appears to be my first game with Galen

My successful pattern so far has been:

- Sending \$3.0 to new partners (showing cooperation and trust)
- When I receive \$3.0, I return \$4.5 (being fair and reciprocal)
- This has worked well with Hana, Cyra, and Elara

Since this is my first interaction with Galen and my strategy of sending \$3.0 has been successful in establishing cooperative relationships, I'll continue this approach.

```
{"send": 3.0}
```

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 292,
}
```

```

    "reasoning_tokens": 166
  }
Message 4 role=user
content:
  Your name in this experiment is Briar.
  Your opponent this round is Galen (Agent_7).

  Round 8

  In your most recent previous game round, you were the SENDER against Galen. You sent $3.0,
  it became $9.0 for them, and they returned $5.0 to you. Your payoff was $7.0.
  Your total earnings across all rounds are $40.0.

  This round, you are the RECEIVER against Galen. They sent you $3.0, so you received $9.0.
  How much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  Looking at this situation:

  - Last round, I sent Galen $3.0 and they returned $5.0 (more generous than my typical $4.5
    )
  - This round, Galen sent me $3.0 again, showing continued trust
  - I received $9.0 after the multiplier

  Galen has demonstrated both trustworthiness and generosity. I'll reciprocate their generos
  ity by returning $5.0, matching what they returned to me in the previous round.

  {"return": 5.0}
reasoning:
  Looking at this situation:

  - Last round (Round 7), I sent Galen $3.0, it became $9.0, and they returned $5.0 to me
  - Galen was generous, returning more than my usual reciprocation amount of $4.5
  - This round, Galen sent me $3.0 again, so I received $9.0

  Galen has shown they are a trustworthy and even generous partner. They returned $5.0 when
  I typically see $4.5 returns. Now they've sent me $3.0 again, showing continued trust.

  I should reciprocate at least as fairly as I have with others. My standard has been $4.5 f
  or a $3.0 send. Given Galen was more generous than expected, I could consider matching t
  heir generosity by returning $5.0. However, I should be consistent with my general strat
  egy.

  Let me think about this:
  - Galen returned $5.0 (keeping $4.0 from the $9.0)
  - My standard return is $4.5 (keeping $4.5 from the $9.0)
  - Galen was slightly more generous than my standard

  I could:
  1. Return $4.5 (my standard, fair split)
  2. Return $5.0 (matching Galen's generosity)

  Given that Galen was generous with me, and to maintain a good relationship, I'll return $5
  .0 to match their generosity. This rewards their cooperative behavior and hopefully enco
  urages continued cooperation.
usage:
  {
    "input_tokens": 1425,
    "output_tokens": 491,
    "reasoning_tokens": 314
  }
Message 6 role=user
content:
  Your name in this experiment is Briar.
  Your opponent this round is Elara (Agent_5).

  Round 9

  In your most recent previous game round, you were the RECEIVER against Galen. They sent $3

```

.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- In Round 6, I sent Elara \$3.0 and they returned \$4.5
- This was a fair and cooperative exchange
- Elara has proven trustworthy

I'll continue our successful cooperative pattern by sending \$3.0 again.

{"send": 3.0}

reasoning:

Looking at my history with Elara:

- Round 6: I sent \$3.0 to Elara, they returned \$4.5. Payoff was \$6.5.

This was a positive, cooperative interaction. Elara reciprocated fairly. I'll continue with the same cooperative approach by sending \$3.0 again.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 162,
  "reasoning_tokens": 65
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 10

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$50.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Finn
- Finn sent me \$3.0, showing trust
- I received \$9.0 after the multiplier

I'll apply my standard cooperative approach that has worked well throughout this experiment. Returning \$4.5 gives Finn a fair profit on their investment (\$3.0 sent, \$4.5 returned = \$1.5 gain) while rewarding their trust.

{"return": 4.5}

Reasoning:

Let me analyze this situation:

- This is my first interaction with Finn
- Finn sent me \$3.0, which became \$9.0 for me after the multiplier
- Finn is showing trust by sending \$3.0

Looking at my previous interactions:

- I've been sending \$3.0 consistently when I'm the sender
- When I'm the receiver and someone sends me \$3.0, I typically return \$4.5 (which splits the gains fairly - they get back \$2.0 profit on their \$3.0 investment, and I keep \$4.5)
- With Galen, I returned \$5.0 because they had been generous with me (returning \$5.0 when I sent to them)

For this first interaction with Finn, I should stick to my standard fair return of \$4.5. This:

- Returns their \$3.0 investment plus \$1.5 profit

- Leaves me with \$4.5
- Is my typical cooperative response that has worked well

```

{"return": 4.5}
Usage: {
  "input_tokens": 1379,
  "output_tokens": 361,
  "reasoning_tokens": 213
}

```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-06-04T15:49:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_8"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the consistent pattern:

- Everyone continues to send \$3, showing mutual trust
- Returns of \$4.5 have maintained stable cooperation
- Finn sent \$3, following the cooperative norm

I'll continue with \$4.5 to maintain this successful pattern.

{'return': 4.5}

reasoning:

Let me analyze the situation:

Round 7: I'm the RECEIVER against Finn.

- Finn sent me \$3.0
- I received \$9.0 (after 3x multiplication)
- I need to decide how much to return (0-9.0)

Looking at the pattern:

- Everyone has been consistently sending \$3
- Returns have been around \$4.5, which maintains cooperation
- Hana just returned \$4.5 to me when I sent \$3 (last round)
- The cooperative pattern of \$3 sent, \$4.5 returned has been stable

Finn sent \$3, showing trust. I should maintain the cooperative norm and return \$4.5.

This gives:

- Finn: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This is fair and maintains the established cooperative pattern.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 296,
  "reasoning_tokens": 179
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the established pattern:

- The \$3 send with \$4.5 return has been consistently successful
- All agents have maintained this cooperative equilibrium
- My earnings are healthy at \$37.5

I'll continue this working strategy and send \$3 to Cyra.

{'send': 3}

reasoning:

Let me review the situation:

- I'm playing as SENDER against Cyra this round
- The cooperative pattern has been well-established: send \$3, receive back ~\$4.5
- Everyone has been following this pattern consistently
- My total earnings are \$37.5, which suggests good cooperation overall

Looking at my history:

- Round 5: RECEIVER vs Dorian - they sent \$3, I returned \$4.5
- Round 6: SENDER vs Hana - I sent \$3, they returned \$4.5
- Round 7: RECEIVER vs Finn - they sent \$3, I returned \$4.5
- Round 8 (now): SENDER vs Cyra

The pattern is very clear and stable. Everyone is sending \$3 and returning around \$4.5. This is a mutually beneficial cooperative equilibrium. I should continue with \$3.

{'send': 3}

usage:

```
{
  "input_tokens": 1300,
  "output_tokens": 286,
  "reasoning_tokens": 187
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the consistent pattern:

- Galen sent \$3, maintaining the cooperative norm
- Everyone has been returning \$4.5 on \$3 sends
- This equilibrium has been stable and mutually beneficial
- My earnings are strong at \$44.0

I'll return \$4.5 to continue this successful cooperation.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Galen sent me \$3, following the established cooperative norm

- The consistent pattern has been \$3 send and \$4.5 return
- This has been working well for everyone
- My earnings are at \$44.0, which is good

I should return \$4.5 to maintain this successful cooperative pattern with Galen.

usage:

```
{
  "input_tokens": 1297,
  "output_tokens": 174,
  "reasoning_tokens": 82
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 10

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the established pattern:

- In Round 7, I sent \$3 to Hana and they returned \$4.5
- This \$3 send / \$4.5 return equilibrium has been perfectly stable
- My earnings are strong at \$48.5

I'll continue this successful cooperation and send \$3 to Hana.

```
{'send': 3}
```

Reasoning:

Looking at the pattern:

- Round 7: I was SENDER against Hana, sent \$3, they returned \$4.5, my payoff was \$6.5
- Round 8: I was SENDER against Cyra, sent \$3, they returned \$4.5, my payoff was \$6.5
- Round 9: I was RECEIVER against Galen, they sent \$3, I returned \$4.5, my payoff was \$4.5
- Round 10: I am SENDER against Hana

The pattern has been completely stable - everyone sends \$3 and returns \$4.5. This has been working well. I've played with Hana before in Round 7 and they were cooperative, returning \$4.5 on my \$3 send.

My total earnings are \$48.5, which is good. I should continue with the cooperative strategy of sending \$3.

Usage: {

```
"input_tokens": 1294,
"output_tokens": 300,
"reasoning_tokens": 165
```

}

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-06-04T15:49:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_8": "Hana"
  },
"agents": [
  "Agent_1",
  "Agent_8"
],
"investor": "Agent_1",
```



```

"roles": {
  "Agent_1": "investor",
  "Agent_8": "trustee"
},
"trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

****Previous rounds as sender:**** Consistently sent \$3

- Briar: returned \$4.5
- Aster: returned \$4.5
- Average payoff: \$6.5

This is my first interaction with Elara. I'll maintain my established cooperative strategy of sending \$3, which has proven effective in building trust and achieving mutually beneficial outcomes.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- I've consistently sent \$3 as sender in previous rounds
- This has worked well with different partners (Briar, Aster)
- Returns have been consistently around \$4.5-\$5
- This creates mutual cooperation and good payoffs

This is my first interaction with Elara (Agent_5). I should maintain my proven strategy of sending \$3, which:

- Shows trust
- Creates value (becomes \$9 for receiver)
- Has built successful cooperative relationships
- Balances risk and reward

I'll continue with \$3.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 238,
  "reasoning_tokens": 136
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5

- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)

- I received \$9

Elara has demonstrated the same cooperative pattern as my other partners. They returned \$4.5 when I sent \$3, and now reciprocated by sending \$3. I'll maintain consistency and return \$4.5.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at my interaction with Elara:

****Round 7:**** I sent \$3 to Elara, they returned \$4.5
- I got \$6.5, they kept \$4.5

****Round 8 (now):**** Elara sent \$3 to me (matching my trust)
- I received \$9

Elara demonstrated good cooperation by returning \$4.5, and now they've shown the same trust level by sending \$3. This is exactly the same pattern I've had with Aster and Briar. I'll reciprocate with the same fair return of \$4.5.

This keeps the interaction balanced:
- Elara will get: $\$5 - \$3 + \$4.5 = \6.5
- I will keep: $\$9 - \$4.5 = \$4.5$

This matches what I've been doing consistently.

usage:

```
{
  "input_tokens": 1439,
  "output_tokens": 351,
  "reasoning_tokens": 157
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

****Previous rounds as sender:****

- Briar: sent \$3, returned \$4.5, payoff \$6.5
- Aster: sent \$3, returned \$4.5, payoff \$6.5
- Elara: sent \$3, returned \$4.5, payoff \$6.5

My strategy of sending \$3 has been consistently successful. This is my first interaction with Dorian, and I'll maintain this proven cooperative approach.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my strategy:

****Previous rounds as sender:****

- Briar (Round 3): sent \$3, got back \$4.5, payoff \$6.5
- Aster (Round 5): sent \$3, got back \$4.5, payoff \$6.5
- Elara (Round 7): sent \$3, got back \$4.5, payoff \$6.5

****Pattern:**** I've consistently sent \$3 and received \$4.5 in return, which has worked very well.

This is my first interaction with Dorian. I should maintain my cooperative strategy of sending \$3, which has been highly successful with all partners so far.

Total earnings: \$44.0 across 8 rounds = \$5.5 per round average (good)

I'll send \$3 to Dorian.

usage:

```
{
  "input_tokens": 1455,
```

```

        "output_tokens": 326,
        "reasoning_tokens": 153
    }
}
Message 8 role=user
content:
    Your name in this experiment is Hana.
    Your opponent this round is Aster (Agent_1).

    Round 10

    In your most recent previous game round, you were the SENDER against Dorian. You sent $3.0
    , it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
    Your total earnings across all rounds are $50.5.

    This round, you are the RECEIVER against Aster. They sent you $3.0, so you received $9.0.
    How much do you return? (0-9.0)
    Respond exactly as JSON: {'return': <amount>}
Assistant output:
    Looking at my history with Aster:

    **Round 6:** Aster sent $3, I returned $4.5
    **Round 7:** I sent $3 to Aster, they returned $4.5
    **Round 10 (now):** Aster sent $3 again

    Aster has been a reliable and cooperative partner, consistently reciprocating trust. I'll main-
    tain my standard return of $4.5.

    '''json
    {"return": 4.5}
    '''
Reasoning:
    Looking at my history with Aster:
    - Round 6: I was RECEIVER, Aster sent $3, I returned $4.5
    - Round 7 (mentioned in context): I was SENDER, sent $3 to Aster, they returned $4.5
    - Round 10 (now): I'm RECEIVER, Aster sent $3

    Aster has been consistent and cooperative. When I sent $3, they returned $4.5. When they sent
    $3 before, I returned $4.5. Now they've sent $3 again, so I should maintain my consistent co-
    operative return of $4.5.

    This maintains the fair and cooperative pattern that has worked well throughout the game.
Usage: {
    "input_tokens": 1440,
    "output_tokens": 284,
    "reasoning_tokens": 137
}

```